Introduction to Complex Networks

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Consejo Superior de Investigaciones Científicas
@svalver

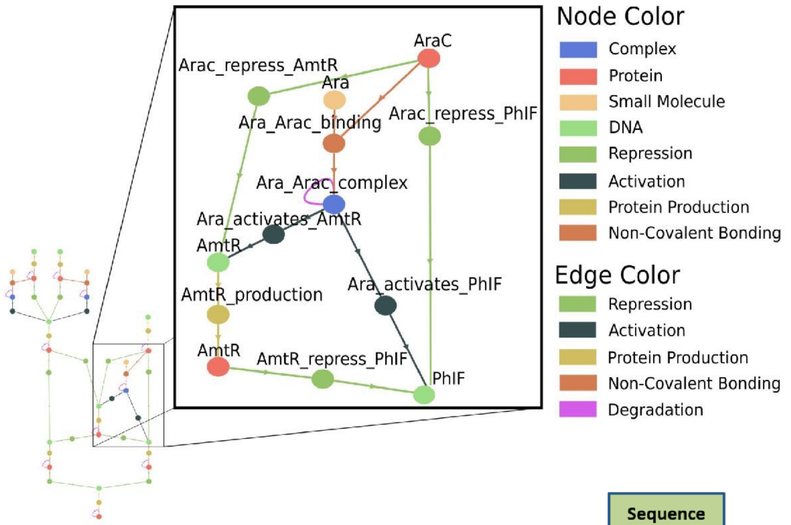
A Visual Language for Biology

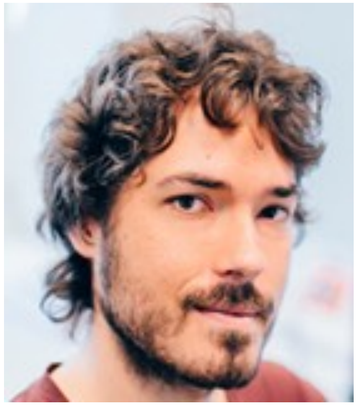
Can we find a good notation for biological complexity?

Bertrand Russell

"*A good notation has a subtlety and suggestiveness which at times make it seem almost like a live teacher ... and a perfect notation would be a substitute for thought*"

quoted by Woodger (1937) The Axiomatic Method in Biology, pp. 18





Angel Goñi

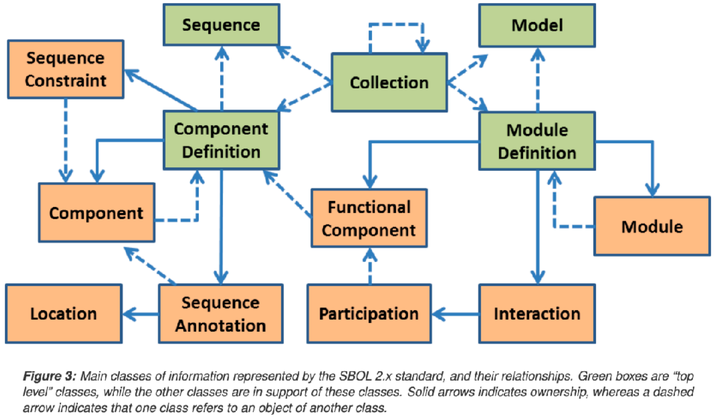


Figure 3: *Main classes of information represented by the SBOL 2.x standard, and their relationships. Green boxes are "top level" classes, while the other classes are in support of these classes. Solid arrows indicates ownership, whereas a dashed arrow indicates that one class refers to an object of another class.*

Madsen et al. (2019) *Synthetic Biology Open Language (SBOL) v 2.3*

A Visual Language for Technology

Valverde et al. (2002) *Scale-Free Networks from Optimal Design*

Alan Kay

Hierarchical Small-Worlds in Software Architecture

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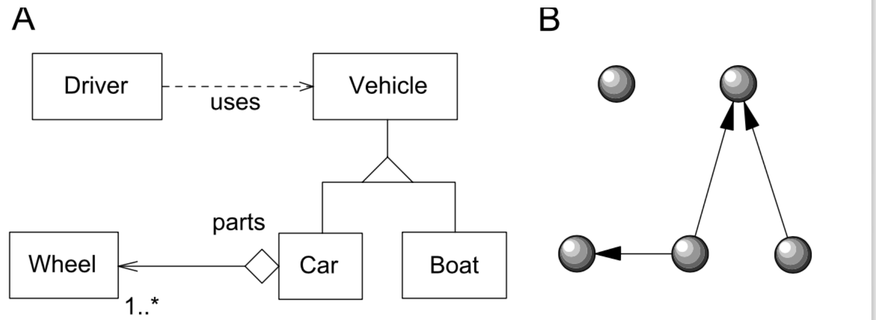
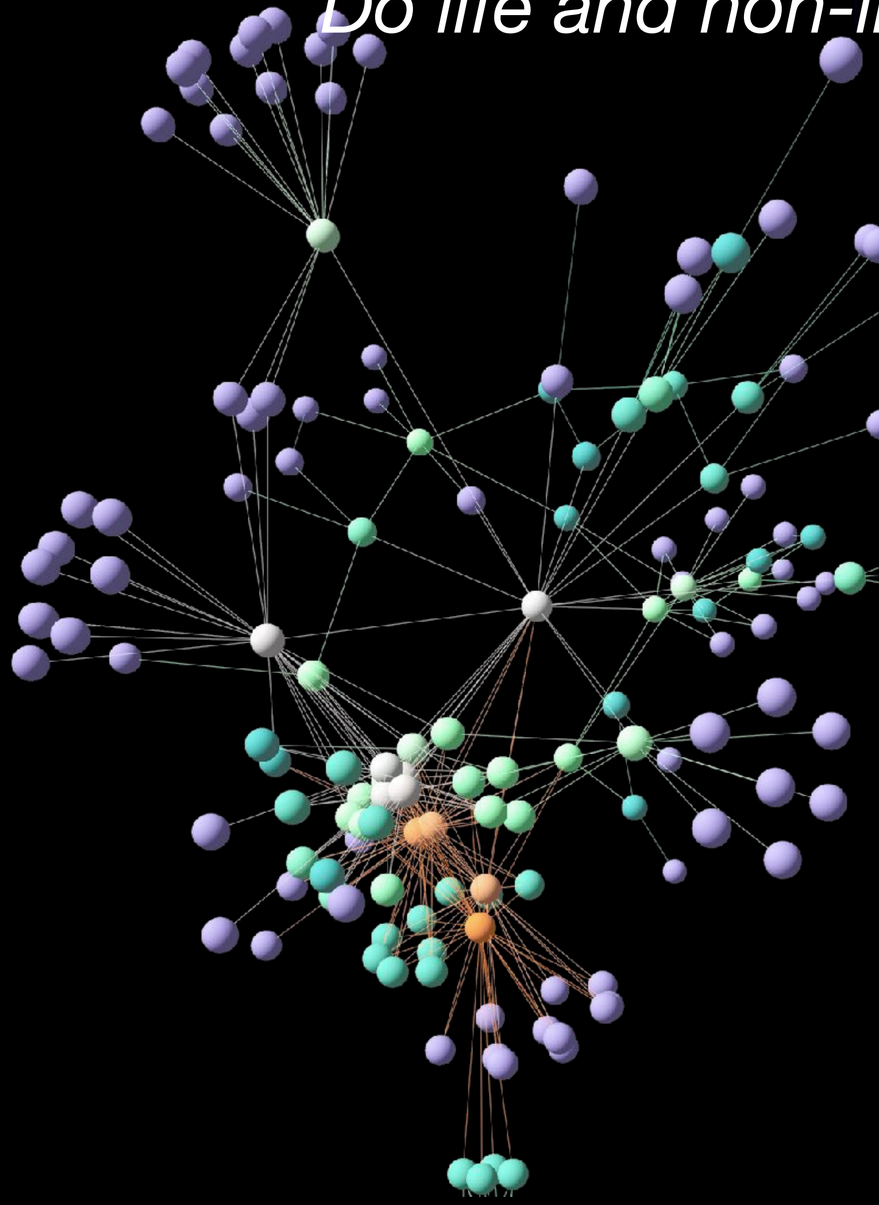


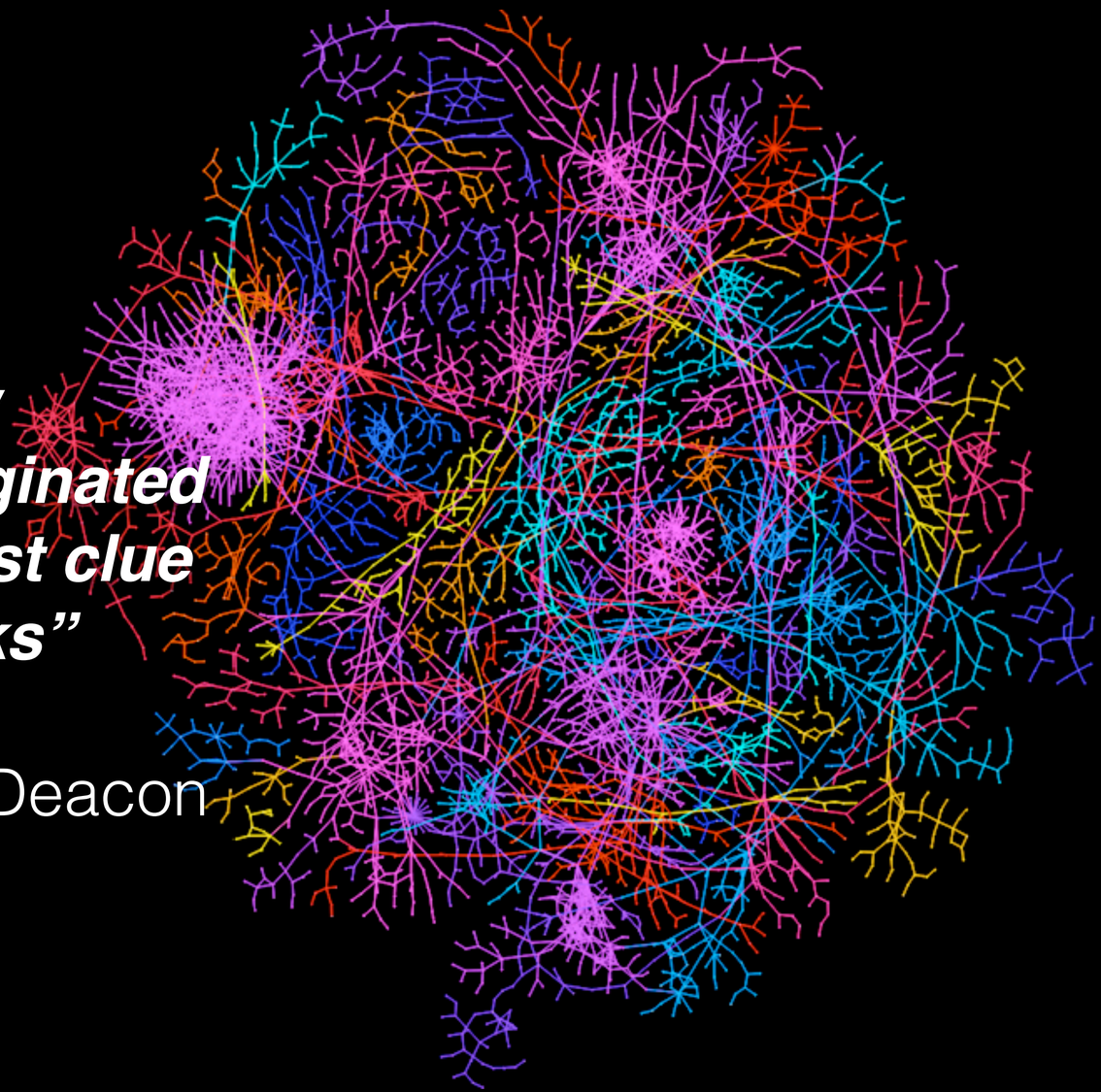
Universality

Do life and non-life share the same basic architecture?

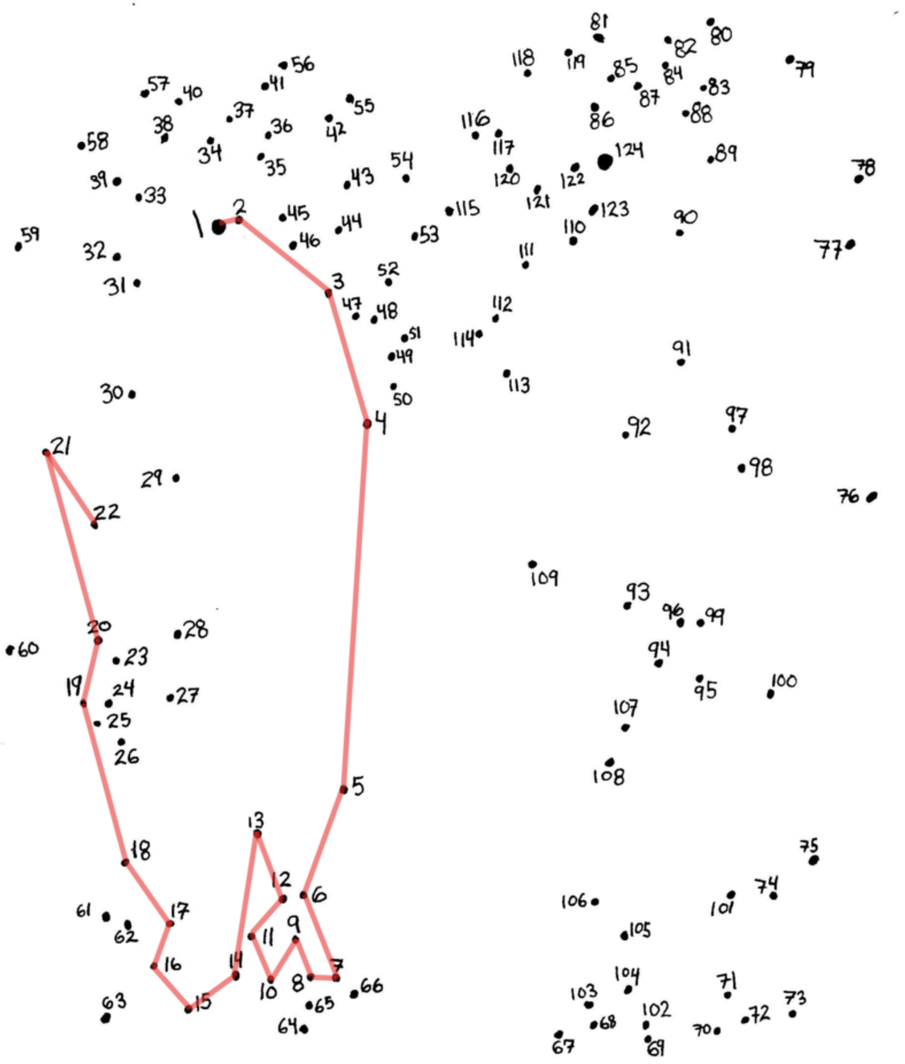


"Knowing how something originated often is the best clue for how it works"

- Terrence Deacon



Index



Basic Properties

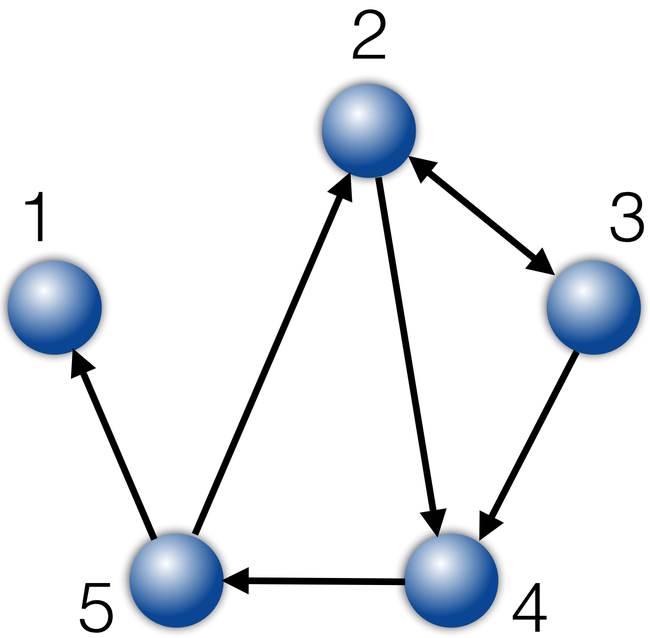
Robustness and Fragility

Hubs, Connectors and Paths

Evolution of Networks

Community Structure

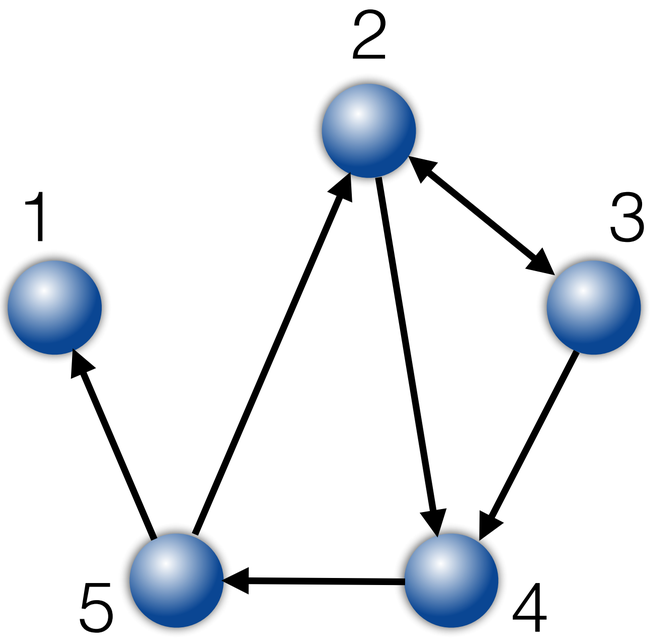
Adjacency Matrix



$$A = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \end{pmatrix}$$

Network Representation

Edge List



2 3

3 2

2 4

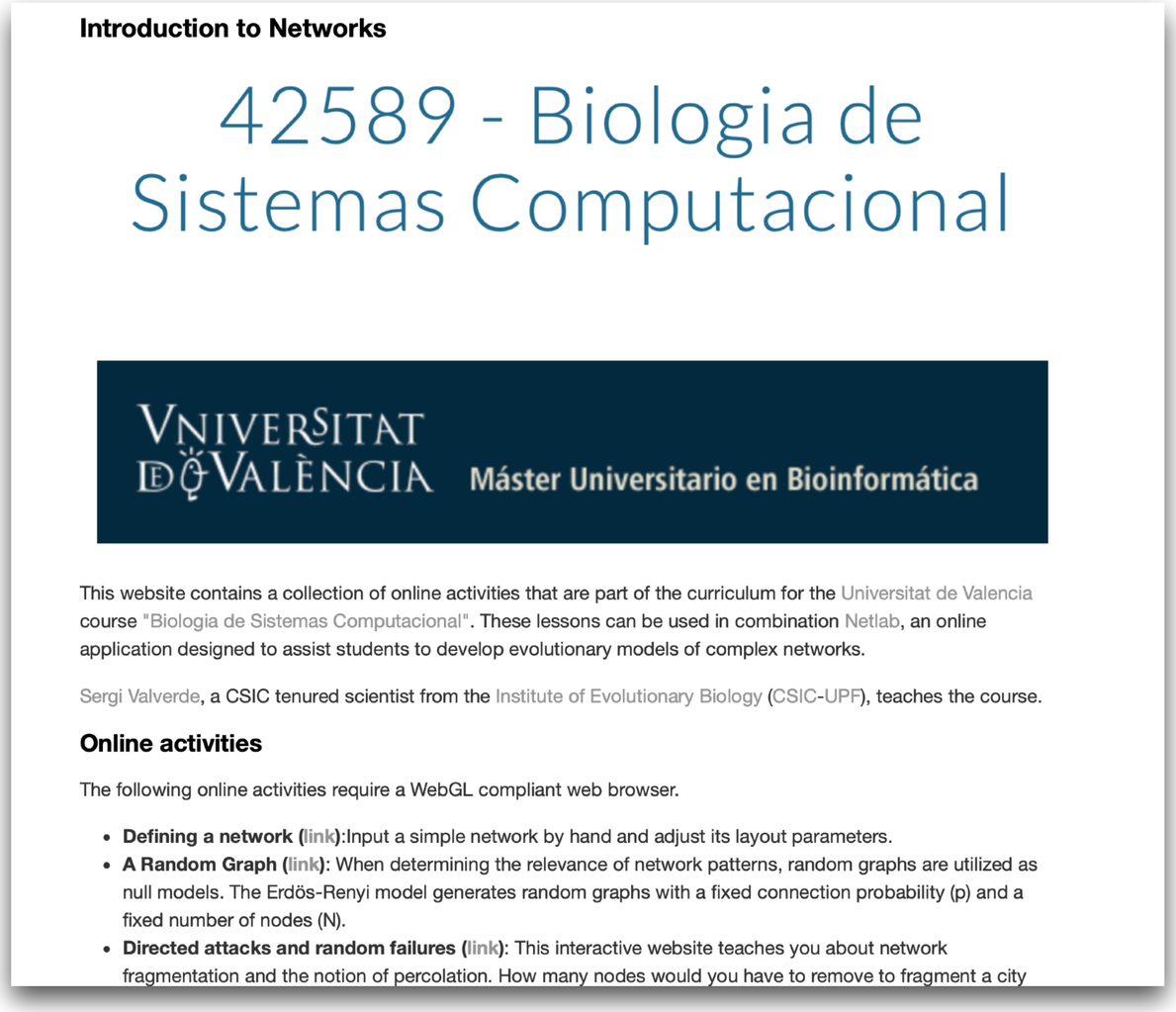
3 4

4 5

5 2

5 1

https://svalver.github.io/course



Introduction to Networks

42589 - Biologia de Sistemas Computacional

VNIVERSITAT DE VALÈNCIA Máster Universitario en Bioinformática

This website contains a collection of online activities that are part of the curriculum for the Universitat de Valencia course "Biologia de Sistemas Computacional". These lessons can be used in combination Netlab, an online application designed to assist students to develop evolutionary models of complex networks.

Sergi Valverde, a CSIC tenured scientist from the Institute of Evolutionary Biology (CSIC-UPF), teaches the course.

Online activities

The following online activities require a WebGL compliant web browser.

- **Defining a network (link)**:Input a simple network by hand and adjust its layout parameters.
- **A Random Graph (link)**: When determining the relevance of network patterns, random graphs are utilized as null models. The Erdös-Renyi model generates random graphs with a fixed connection probability (p) and a fixed number of nodes (N).
- **Directed attacks and random failures (link)**: This interactive website teaches you about network fragmentation and the notion of percolation. How many nodes would you have to remove to fragment a city

NetworkX
Network Analysis in Python

graph-tool

Pajek

Cytoscape

Gephi

Methods in Ecology and Evolution

Methods in Ecology and Evolution 2016, **7**, 127–132 doi: 10.1111/2041-210X.12458

APPLICATION

BiMat: a MATLAB package to facilitate the analysis of bipartite networks

https://arxiv.org/abs/2410.16158

arXiv:2410.16158v1 [cond-mat.dis-nn] 21 Oct 20

Networks: The Visual Language of Complexity

Blai Vidiella, Salva Duran-Nebreda and Sergi Valverde

Abstract Understanding the origins of complexity is a fundamental challenge with implications for biological and technological systems. Network theory emerges as a powerful tool to model complex systems. Networks are an intuitive framework to represent inter-dependencies among many system components, facilitating the study of both local and global properties. However, it is unclear whether we can define a universal theoretical framework for evolving networks. While basic growth mechanisms, like preferential attachment, recapitulate common properties such as the power-law degree distribution, they fall short in capturing other system-specific properties. Tinkering, on the other hand, has shown to be very successful in generating modular or nested structures 'for-free', highlighting the role of internal, non-adaptive mechanisms in the evolution of complexity. Different network extensions, like hypergraphs, have been recently developed to integrate exogenous factors in evolutionary models, as pairwise interactions are insufficient to capture environmentally-mediated species associations. As we confront global societal and climatic challenges, the study of network and hypergraphs provides valuable insights, emphasizing the importance of scientific exploration in understanding and managing complexity.

Key words: *Networks; Evolution; Hypergraphs; Complex Systems; Tinkering*

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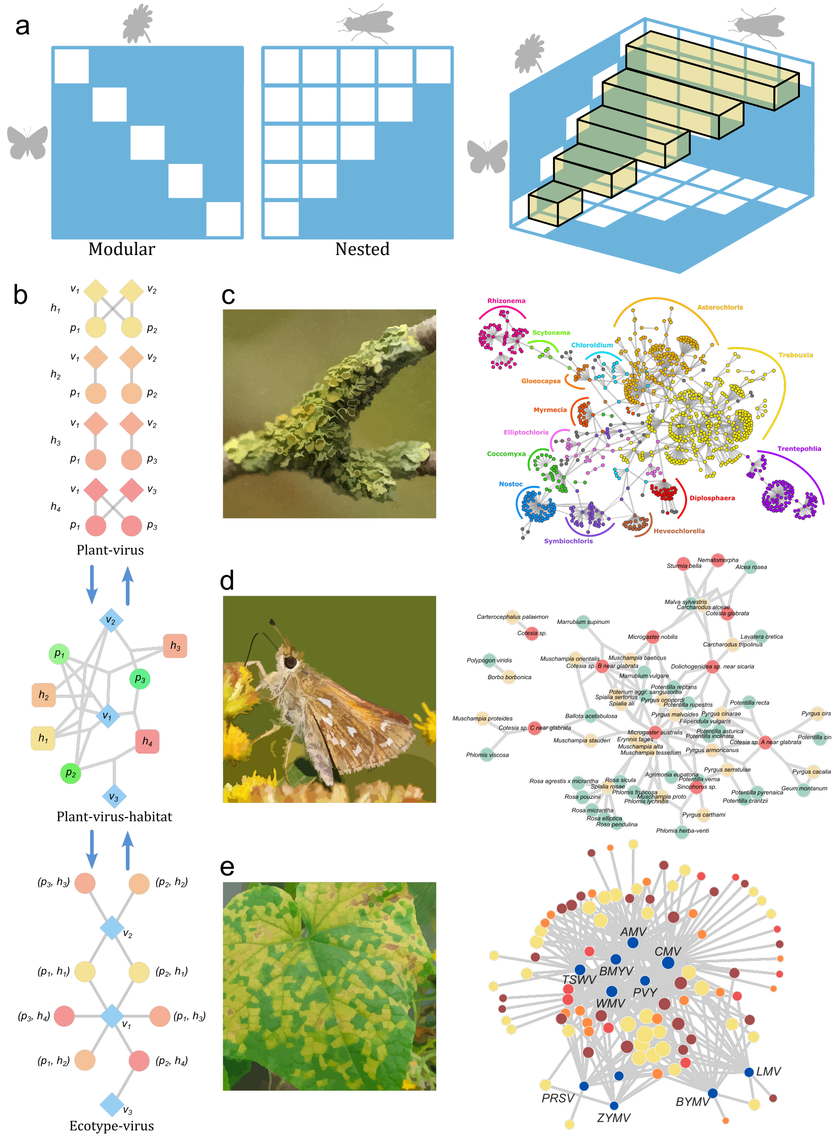
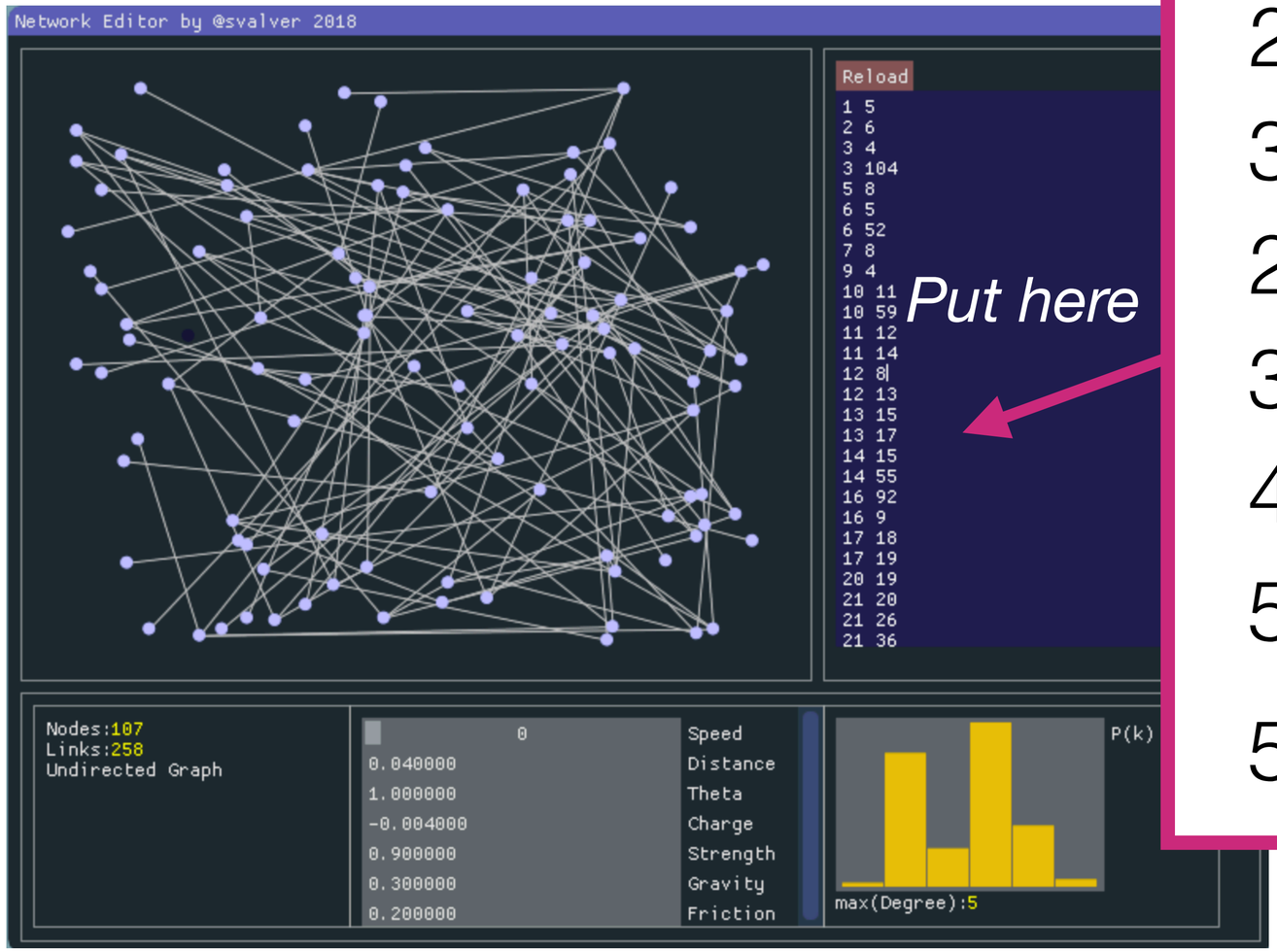
Contributed chapter to "Nonlinear Dynamics for Biological Systems", M. Stich, J. Carballido-Landeira (Eds), Springer, Switzerland, 2024



Activity: Defining Networks

https://tinyurl.com/24e3n5tf



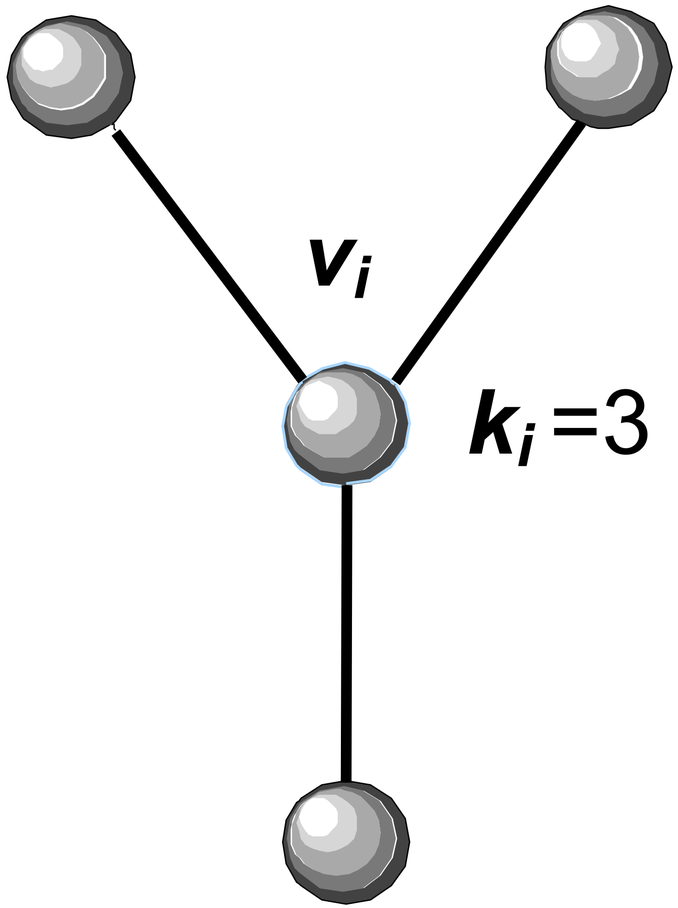
Put here

2 3
3 2
2 4
3 4
4 5
5 2
5 1

1. Explain how many bytes are needed to store this network using the adjacency list and the matrix representations.

2. Consider an alternative method for representing networks. Explain.

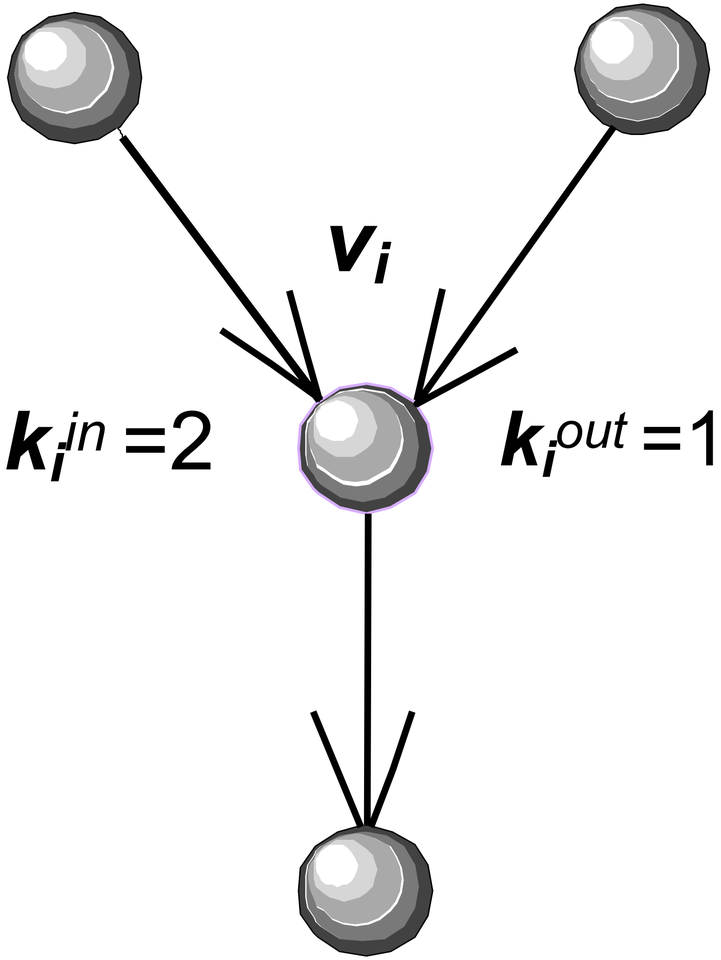
Degree



$$k_i = \sum_j A_{ij}$$

In-degree and Out-degree

Dominance hierarchies



$$k_i^{in} = \sum_{j=1}^{N} A_{ji}$$

$$k_i^{out} = \sum_{j=1}^{N} A_{ij}$$

Number of Links (Edges)

Directed:

$$L = \sum_{i=1}^{N} k_i^{in} = \sum_{i=1}^{N} k_i^{out}$$

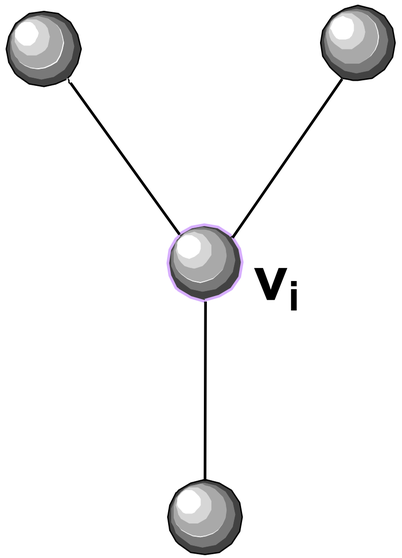
Undirected:

$$2L = \sum_{i=1}^{N} k_i$$

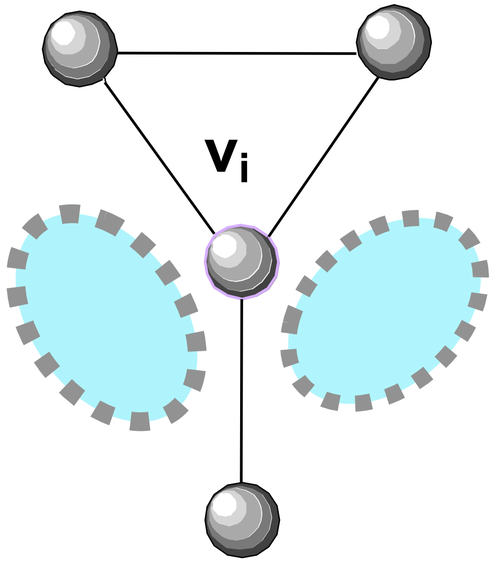
Local Clustering

$$c_i = \frac{e_i}{\binom{k_i}{2}}$$

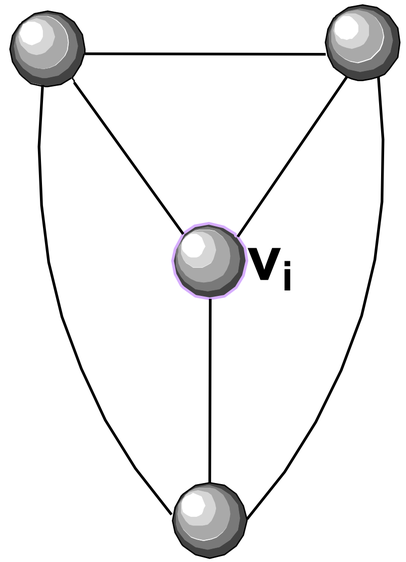
$$= \frac{2e_i}{k_i(k_i - 1)}$$



$C_i = 0$



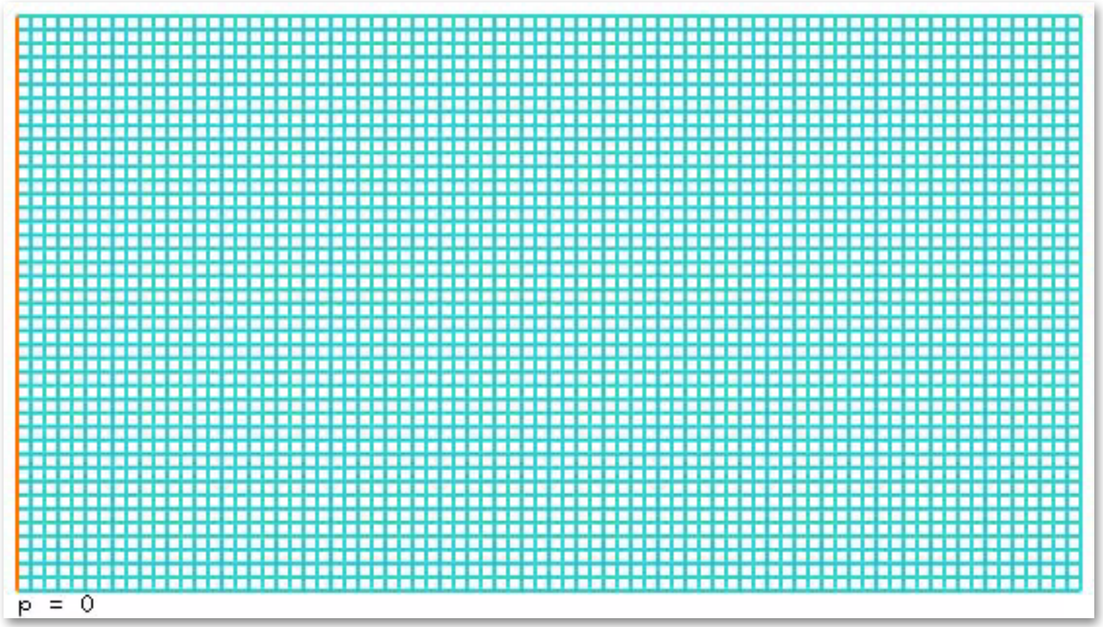
$C_i = 1/3$

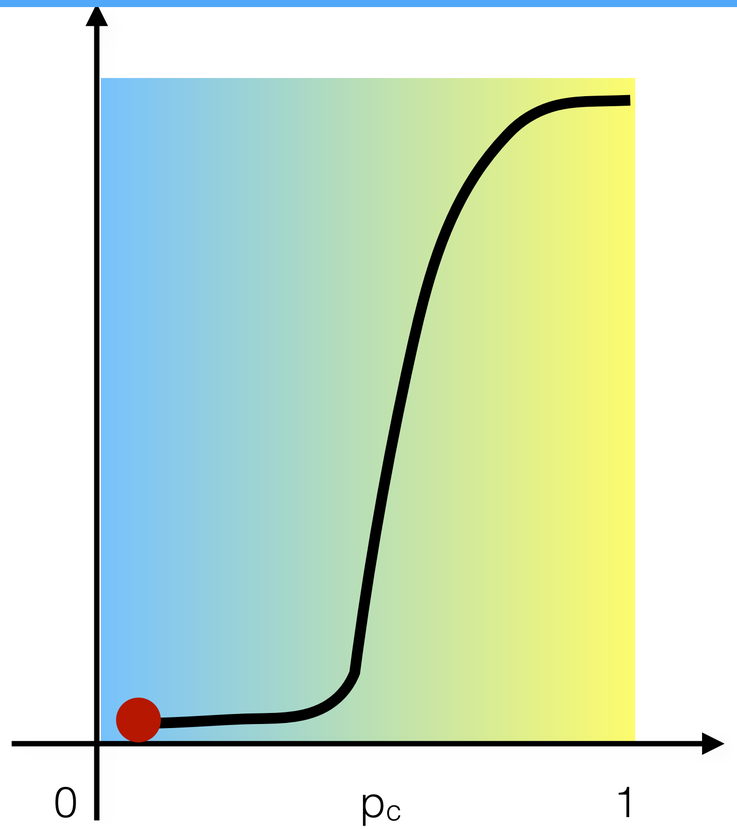


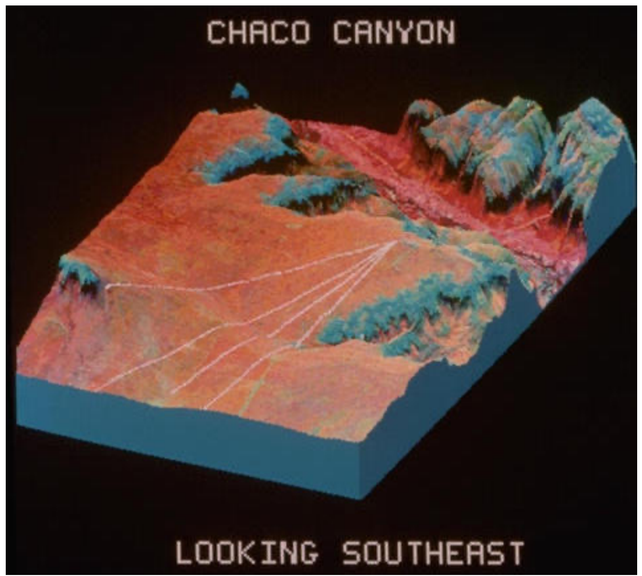
$C_i = 1$

Random Networks :
Robustness & Fragility

Percolation



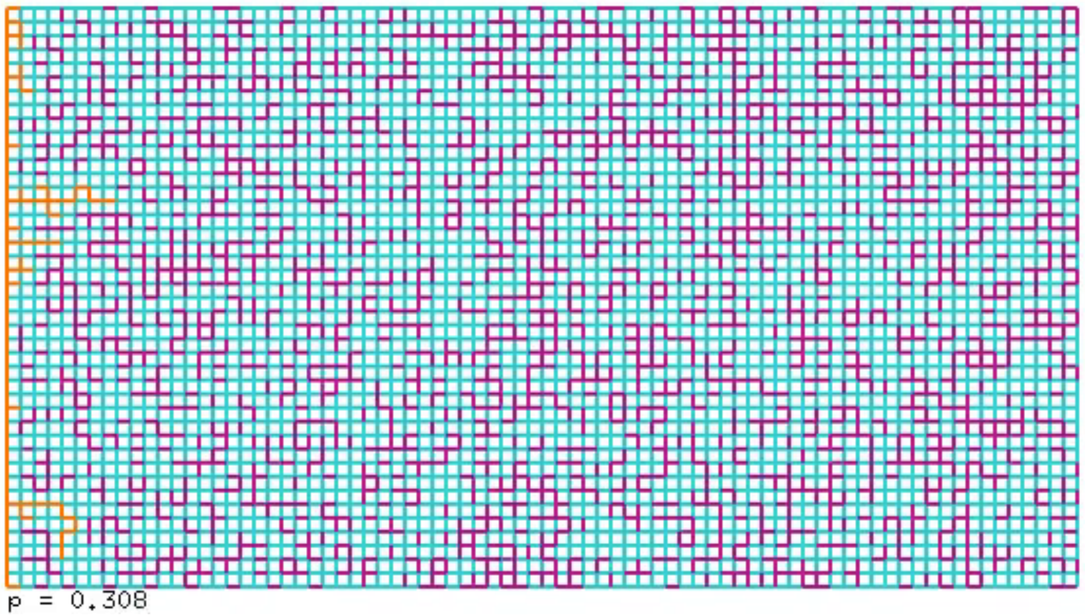


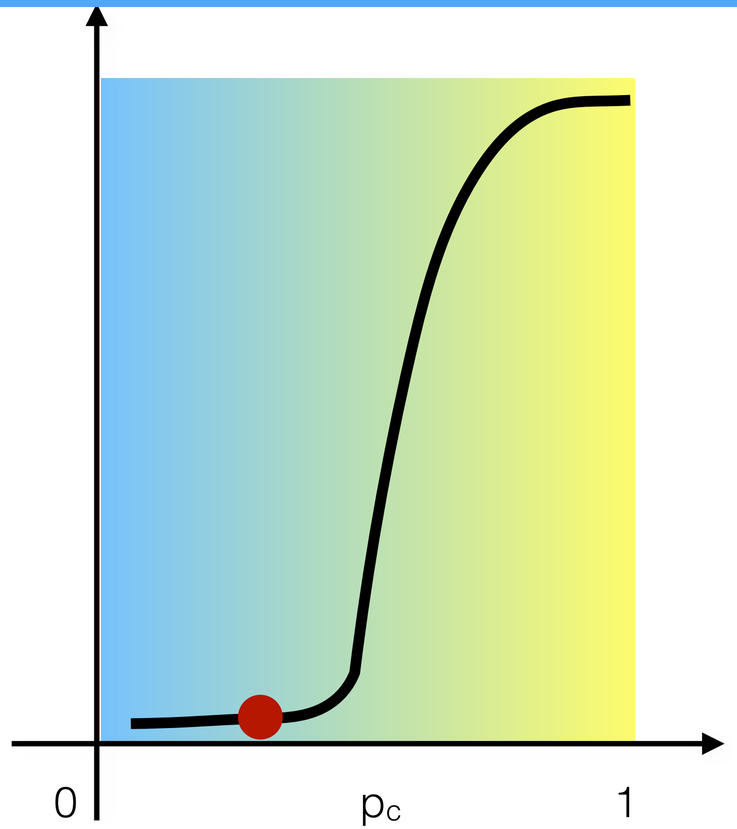


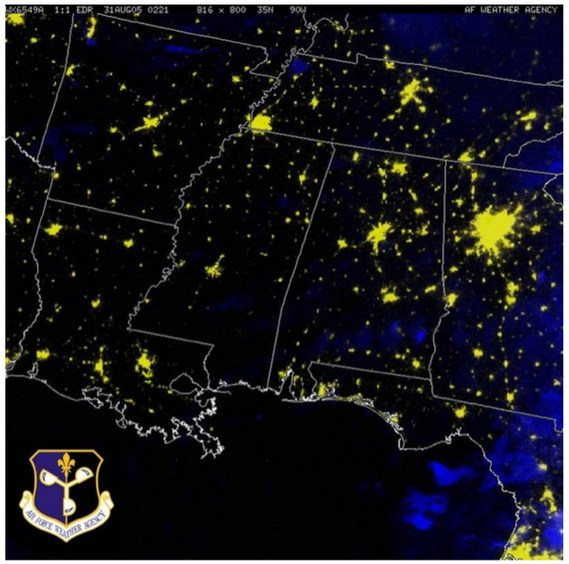
How does connectivity affects behaviour?

Kesten, Harry (1982), Percolation theory for mathematicians, Birkhauser

Disconnected Phase

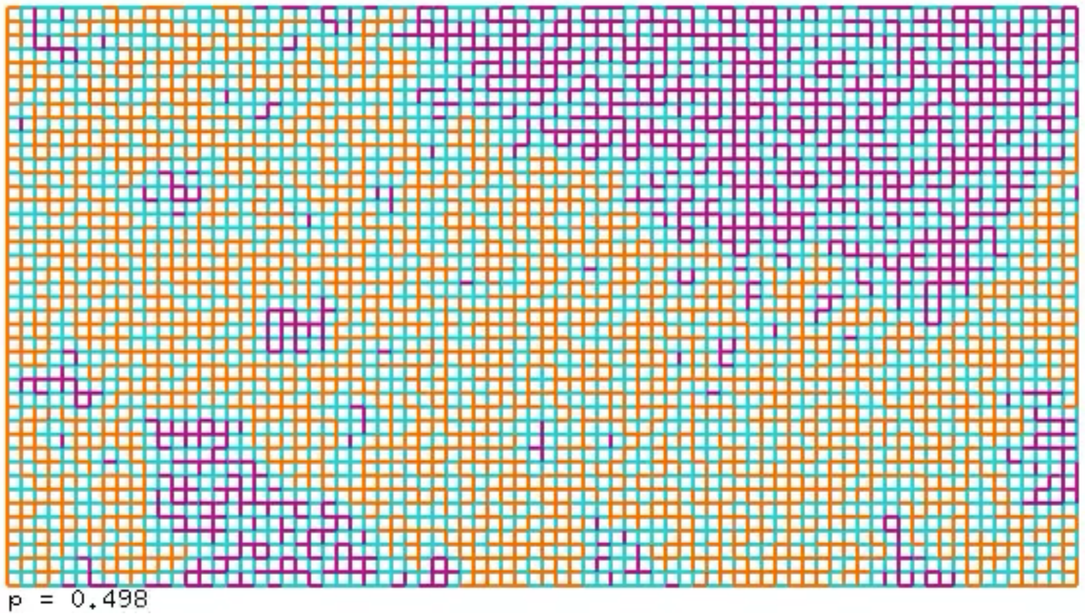


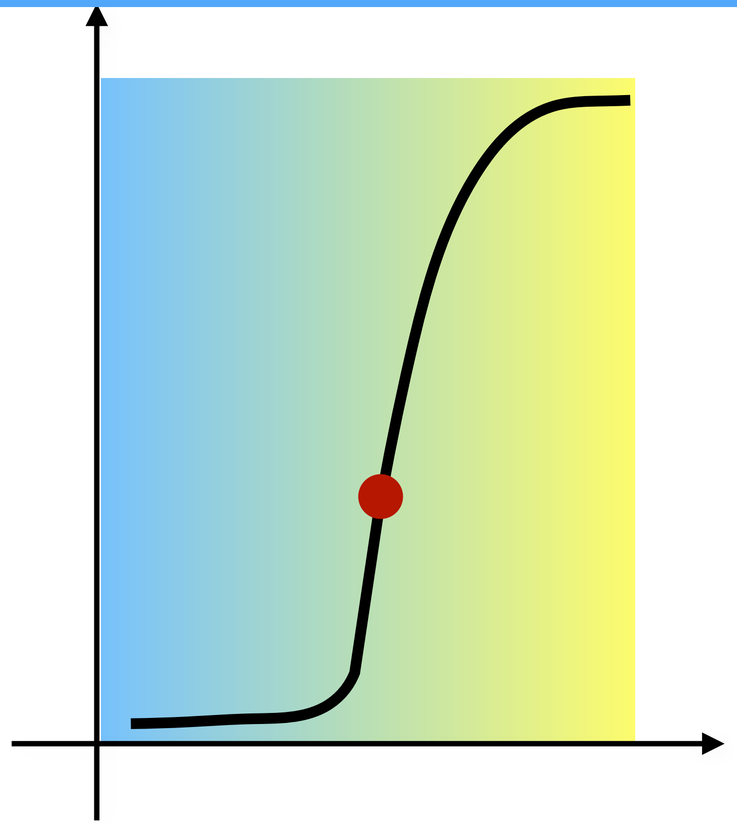


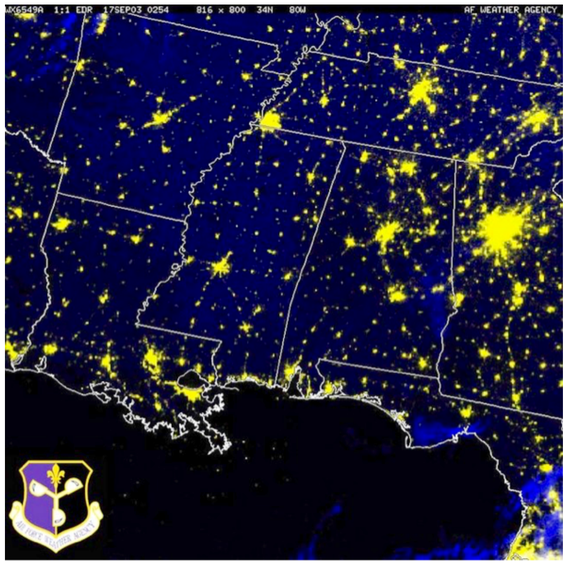


Power outage after Hurricane Katrina hit the Gulf Coast
This image was take Aug 30 and shows the widespread power outages across the Gulf Coast after Hurricane Katrina ravaged the area. U.S. Air Force Image.

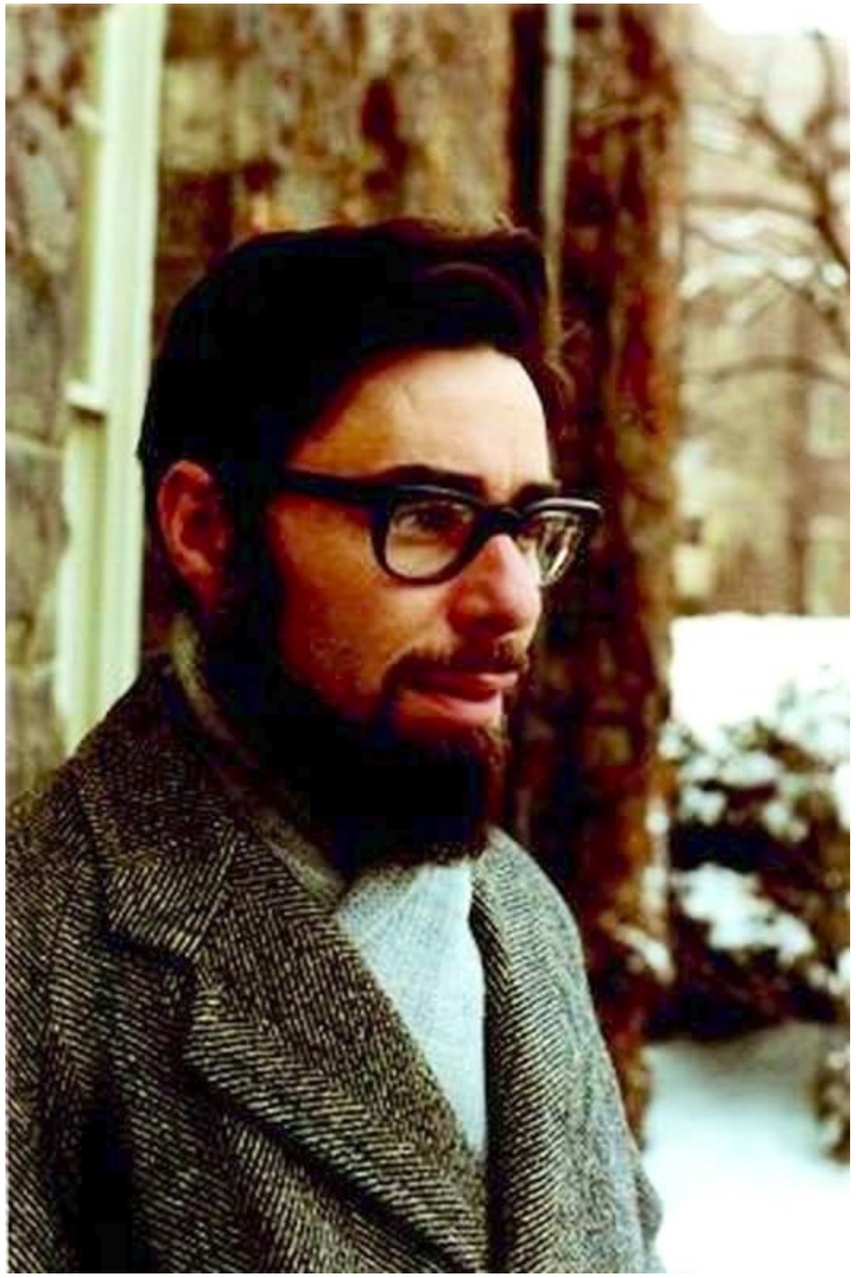
Connected Phase







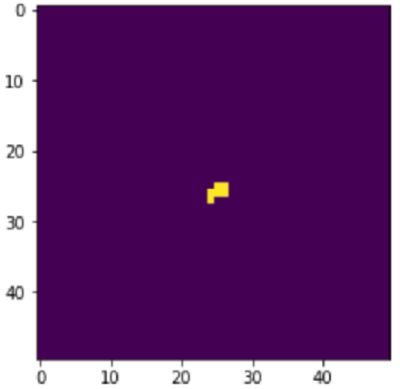
Power grid before the Hurricane Katrina hit the Gulf Coast
This image was taken Sept. 17,2003 and shows the city lights in the Gulf Coast clearly visible. U.S. Air Force Image.



Theorem (Kesten, 1980)

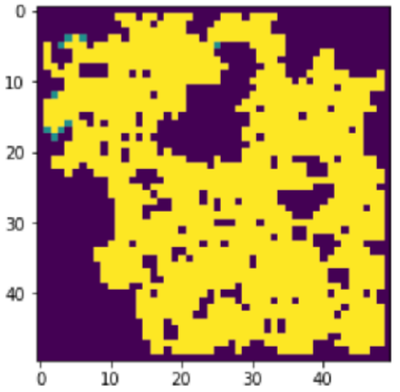
In Bernoulli percolation with parameter *p* on the infinite square grid,

if *p* <= 1/2, the
P(infinite cluster) = 0,



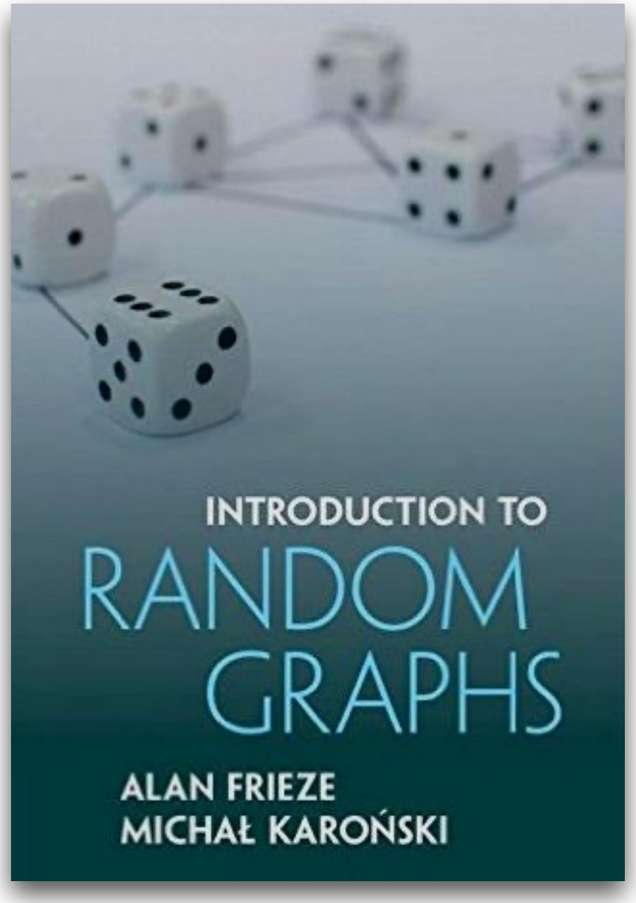
and

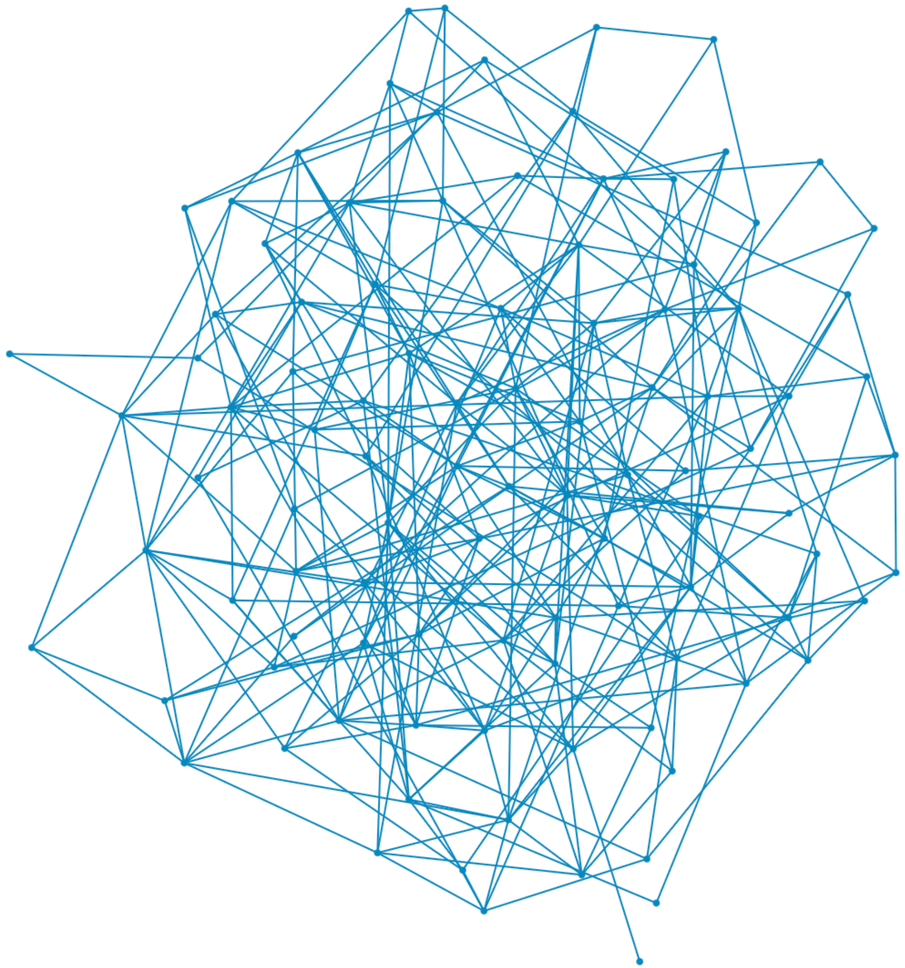
if *p* > 1/2 then
P(infinite cluster) = 1

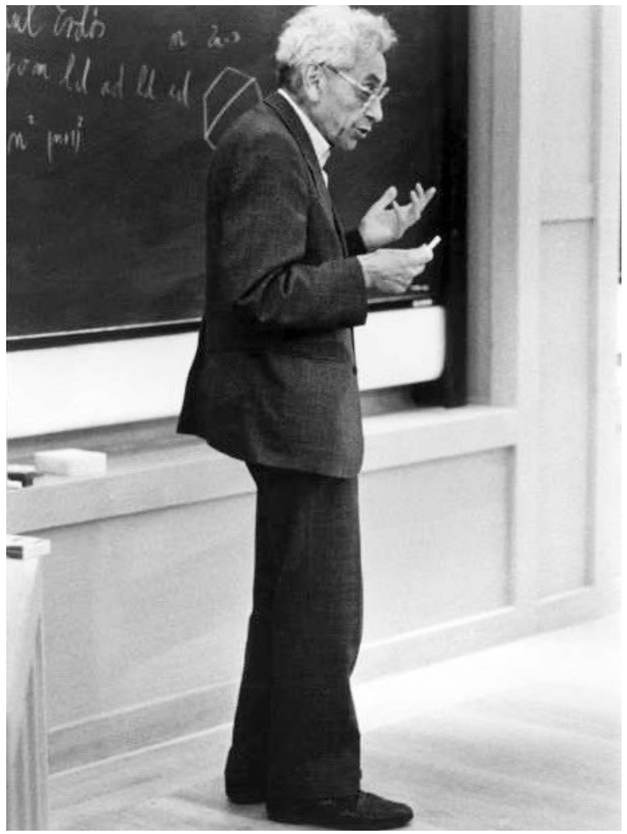


Randomness

*The simplest model of a network : everything is **boring***







Paul Erdös (1913-1996)

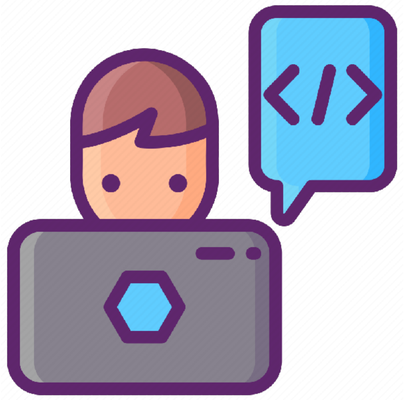
Simulating Random Graphs

A static world without geography

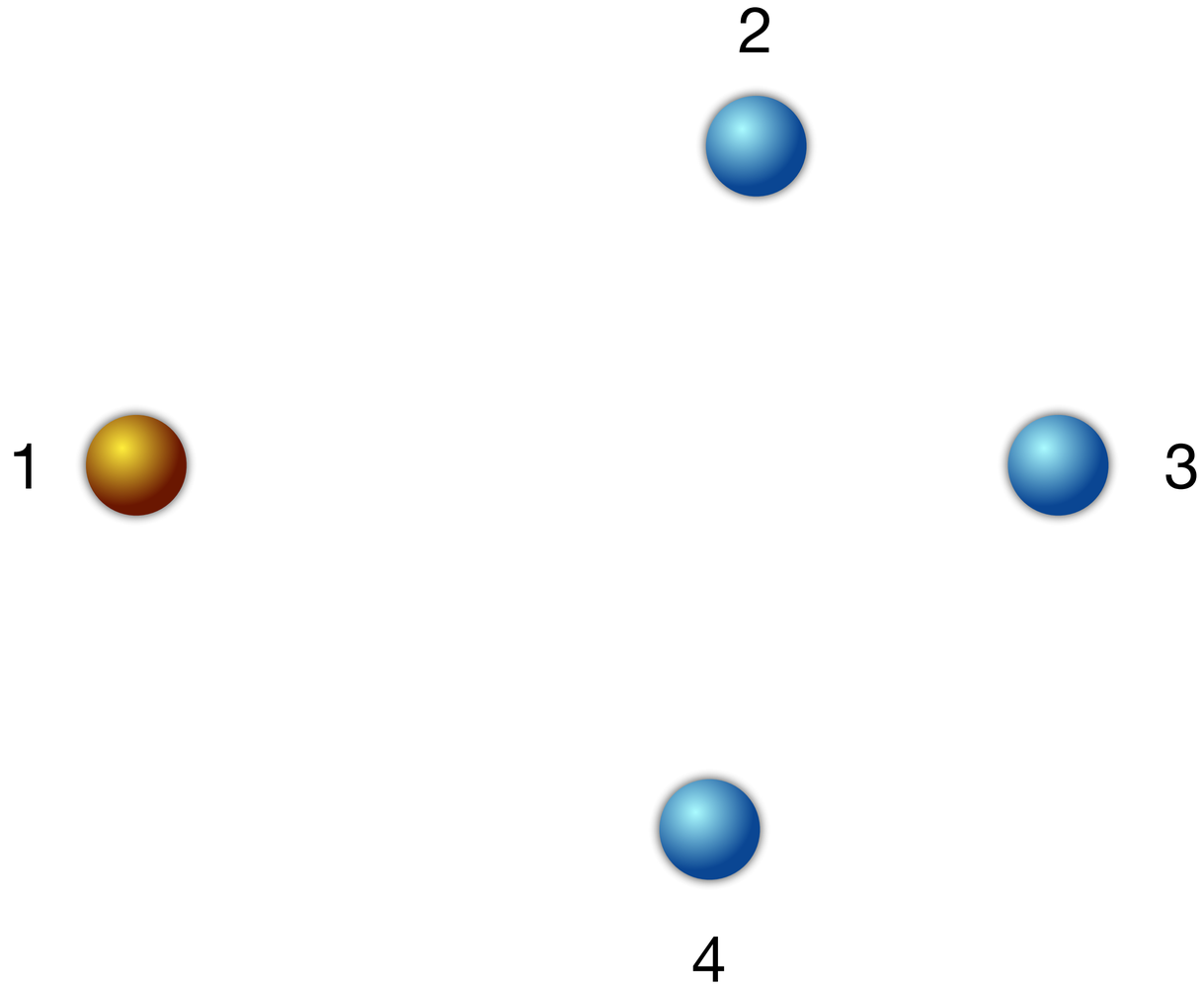
N = number of nodes

p = probability of connecting a pair of nodes

$$A_{ij} = A_{ji} \sim \text{Bernoulli}(p) \text{ for } i < j$$

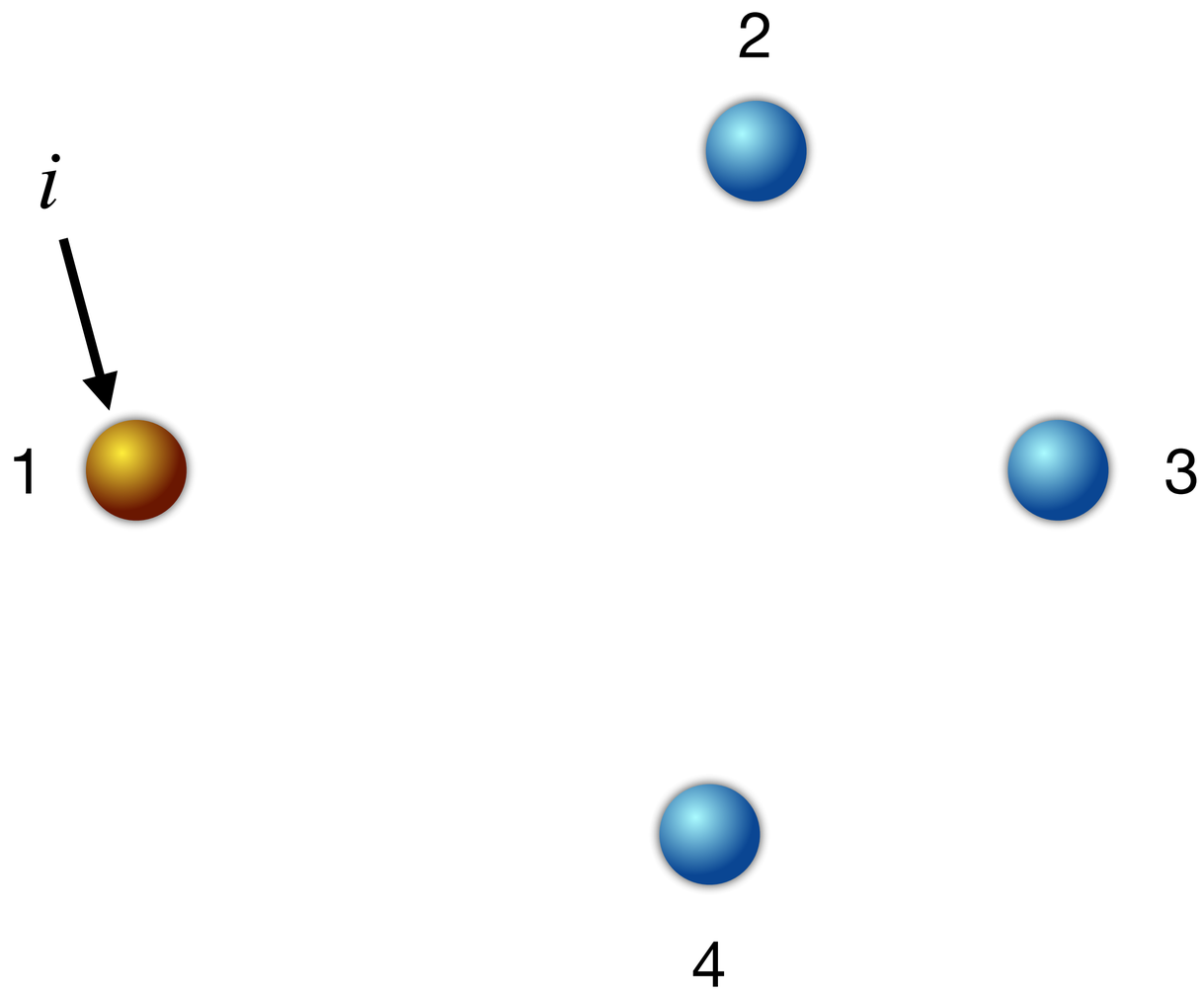


`create(4)`

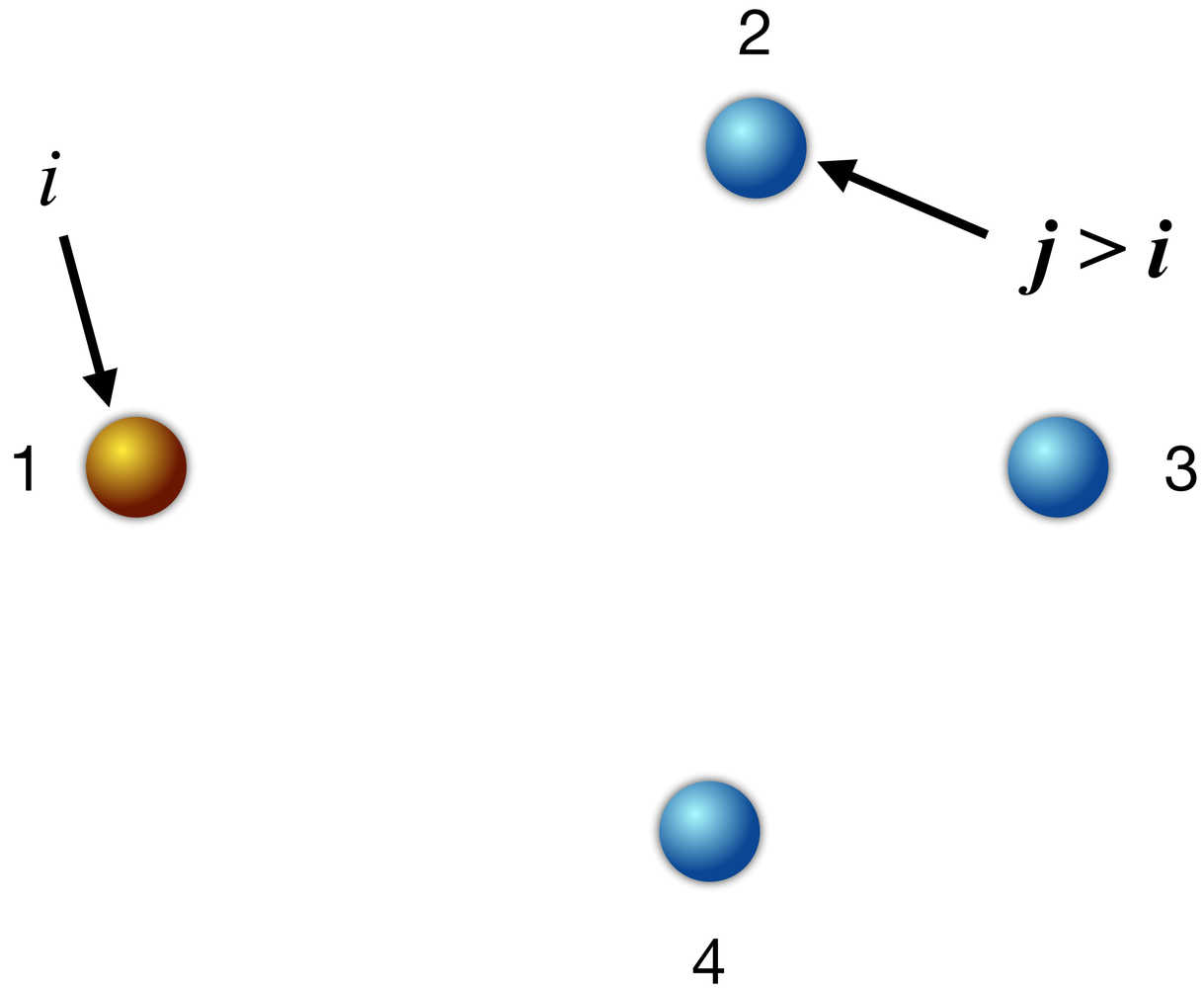


$$A = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

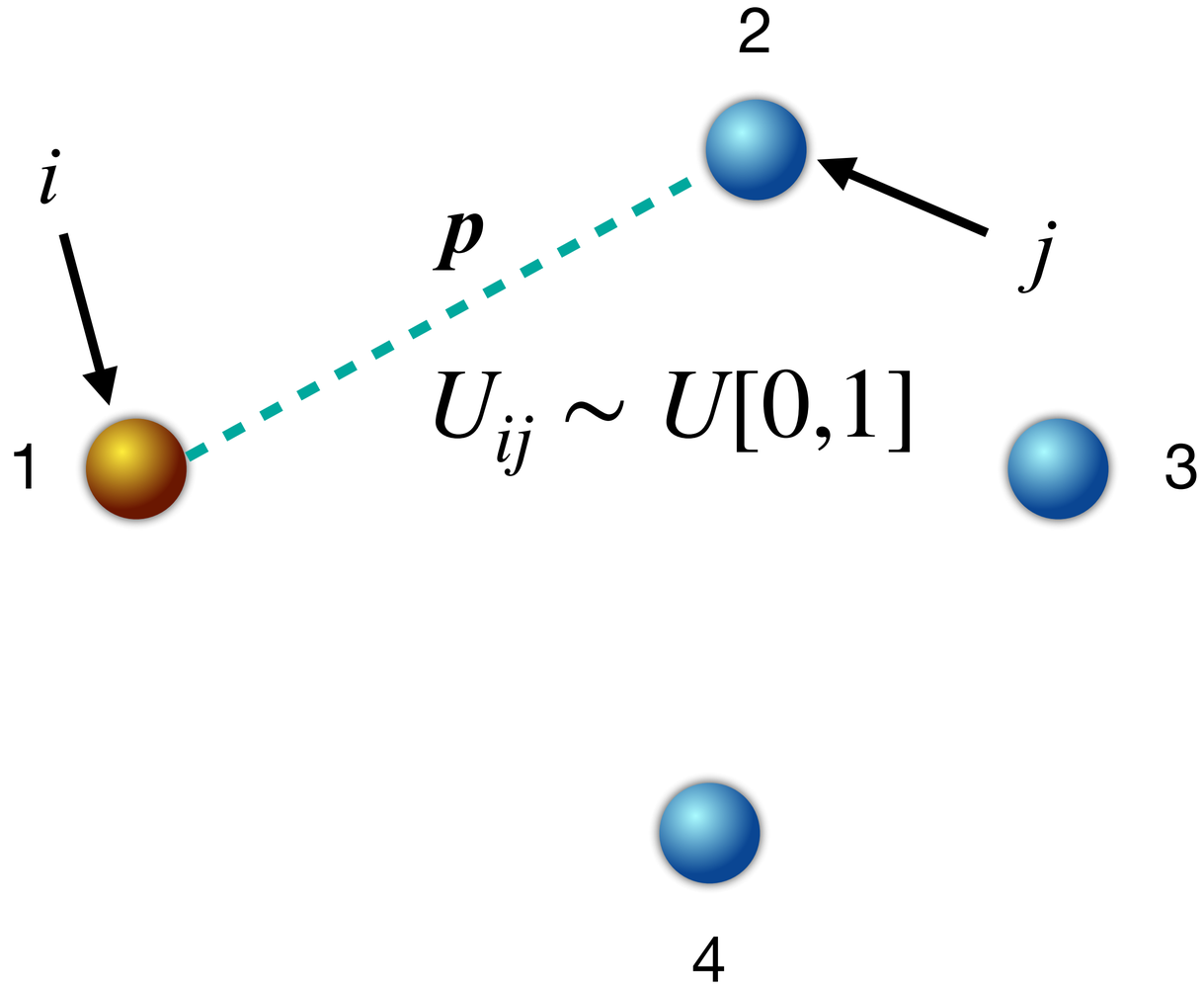
for i in (1, N-1)



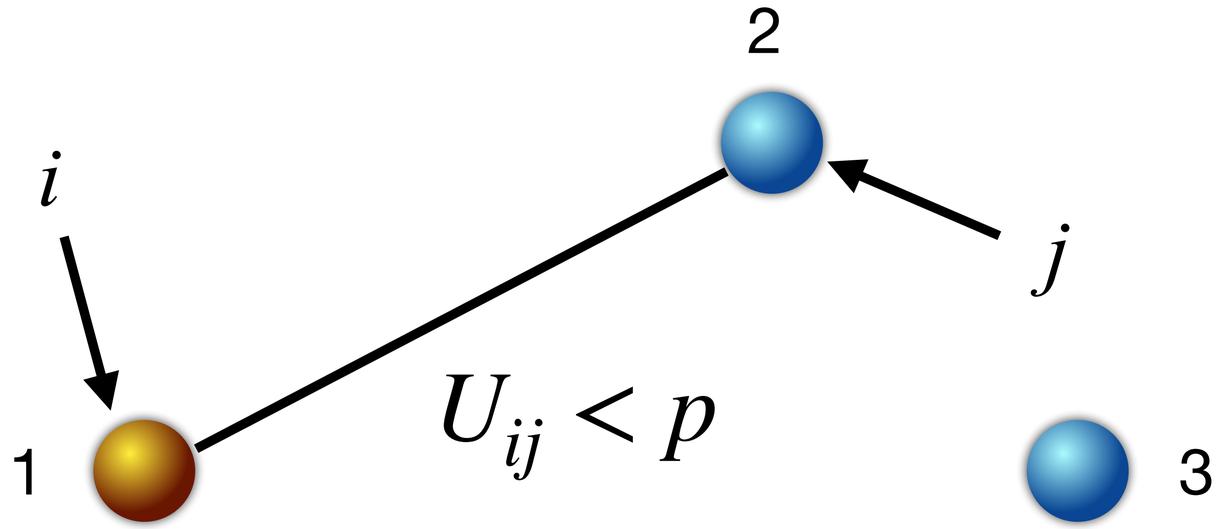
```
for i in (1, N-1)
    for j in (i+1, N)
```



`random-float (1) < p`

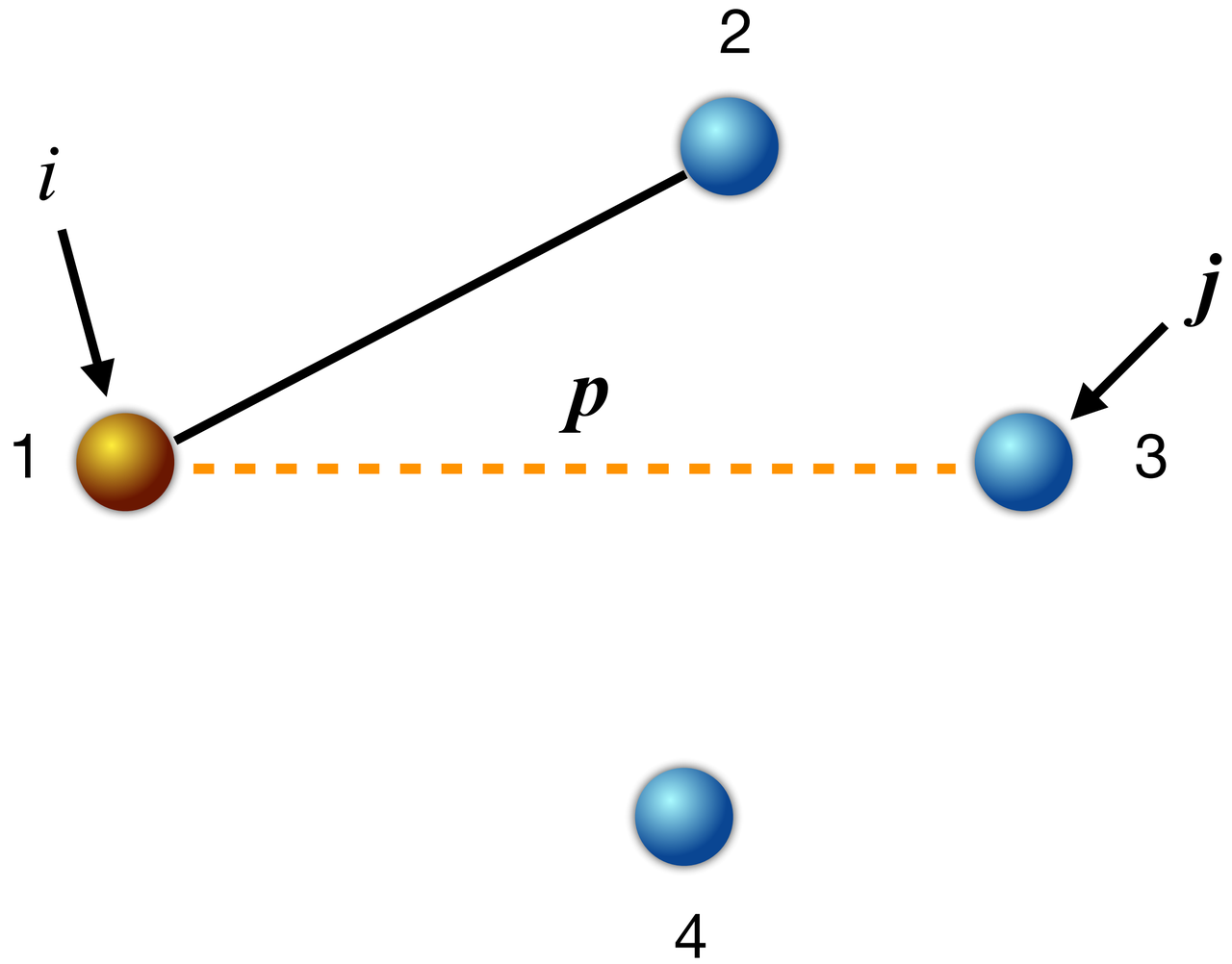


`add_edge(i, j)`

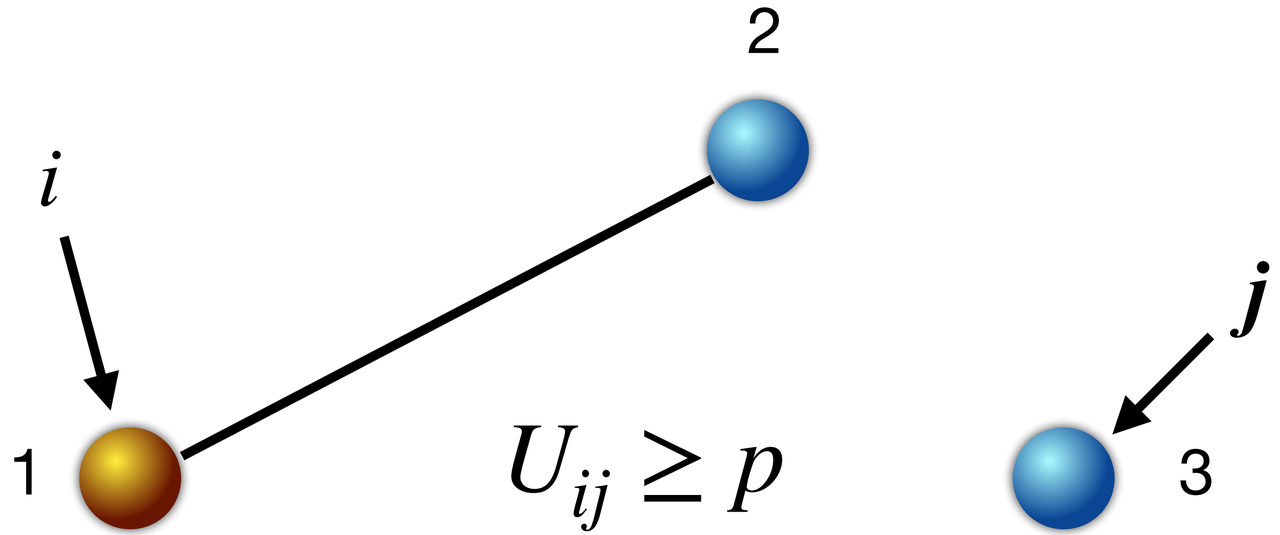


$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

```
for i in (1, N-1)
    for j in (i+1, N)
```



```
for i in (1, N-1)
    for j in (i+1, N)
```



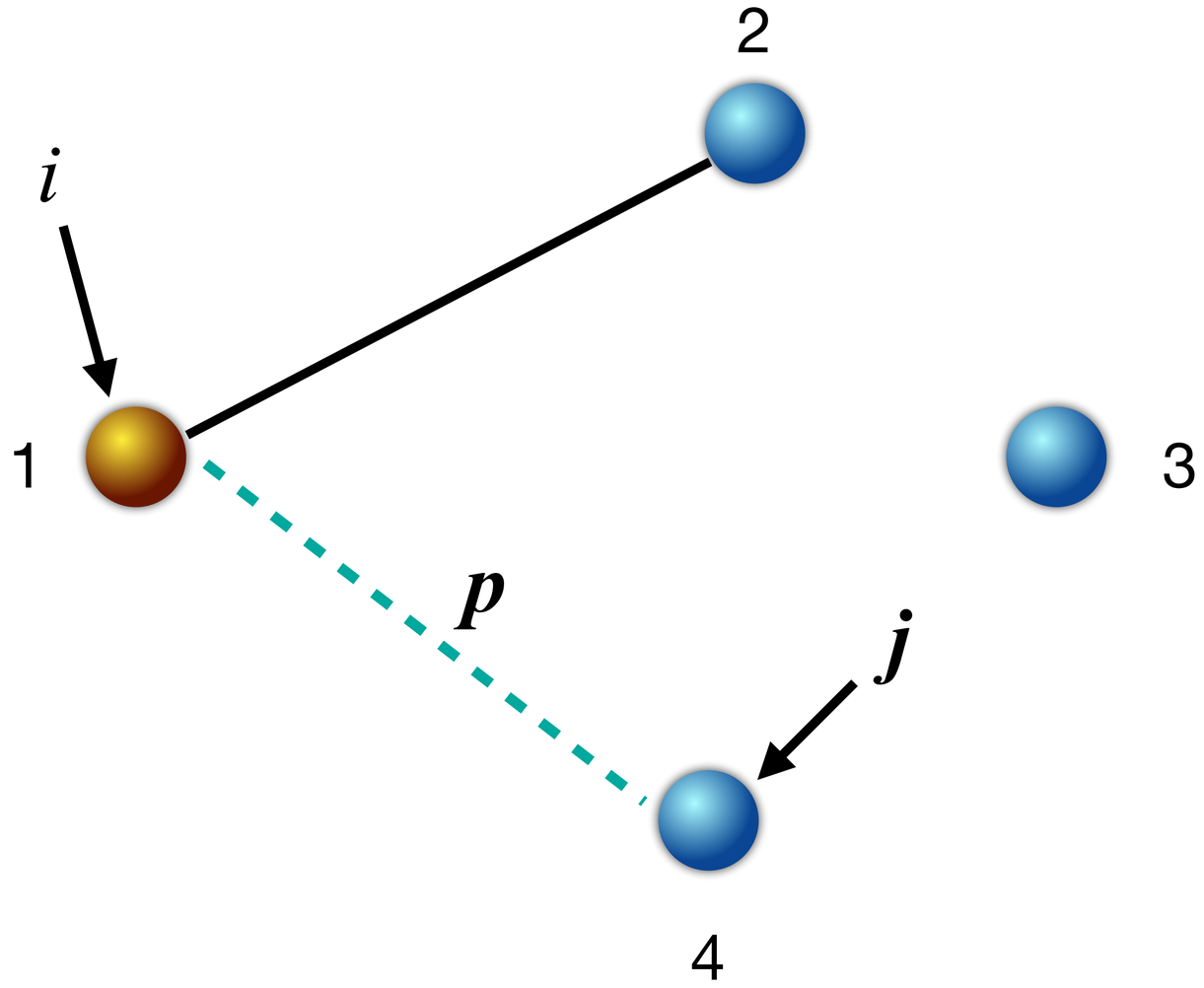
$$U_{ij} \geq p$$

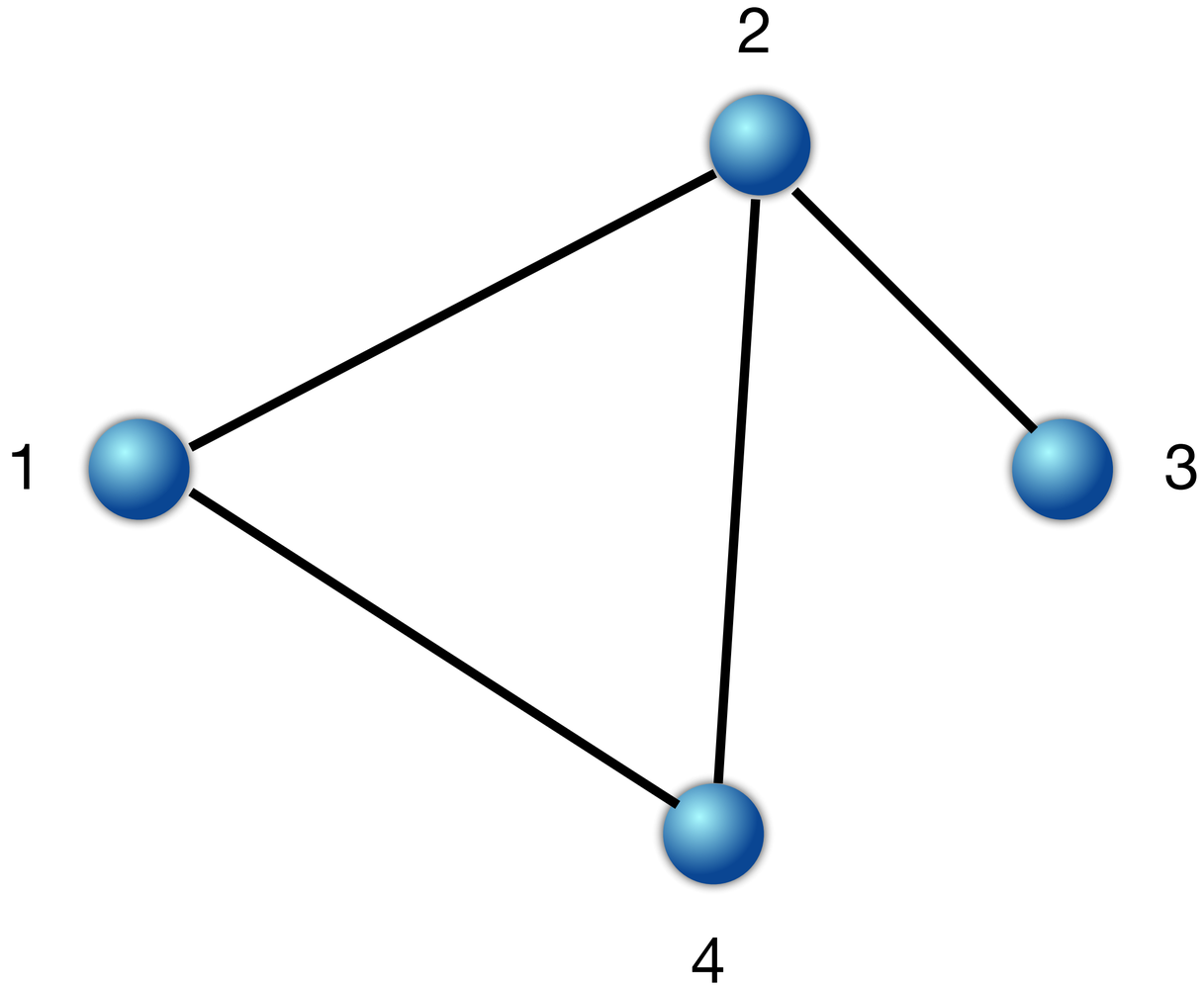
$$A_{ij} = A_{ji} = 0$$

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\binom{N}{2}$$

```
for i in (1, N-1)
    for j in (i+1, N)
```





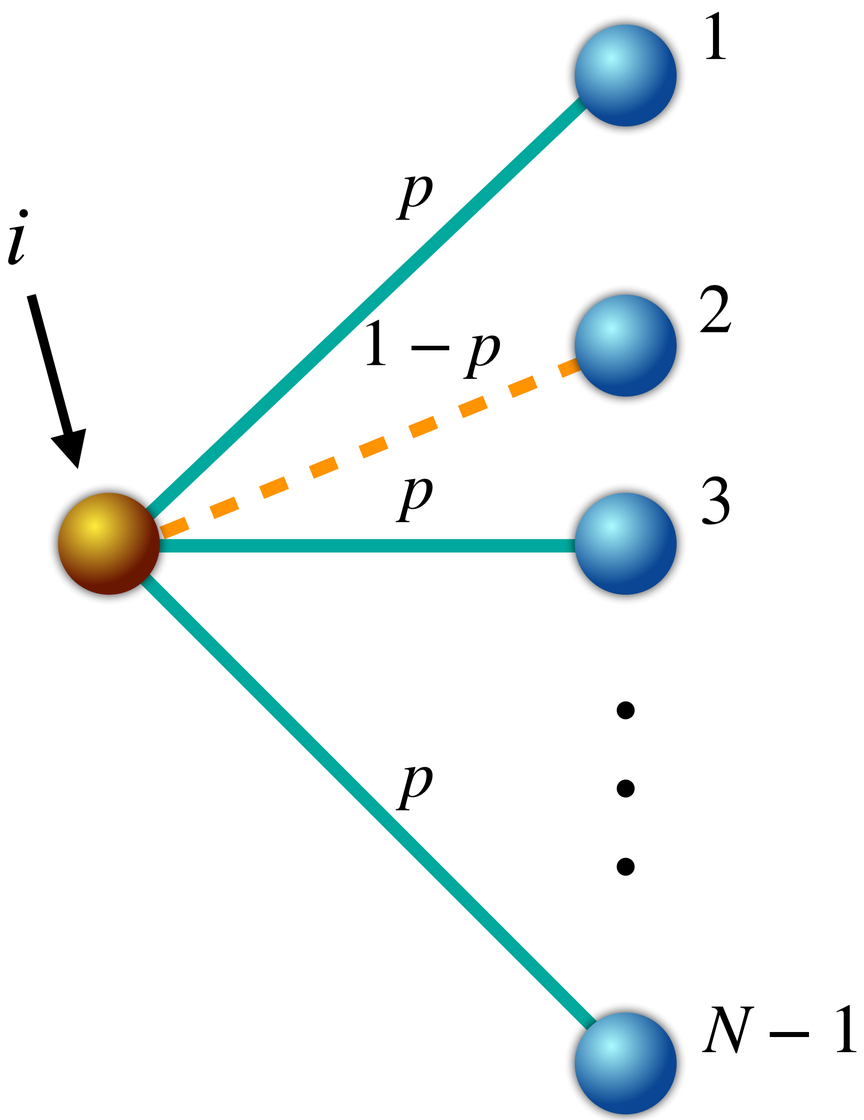
$$N = 4$$

$$L = 4$$

$$\langle k \rangle = 2$$

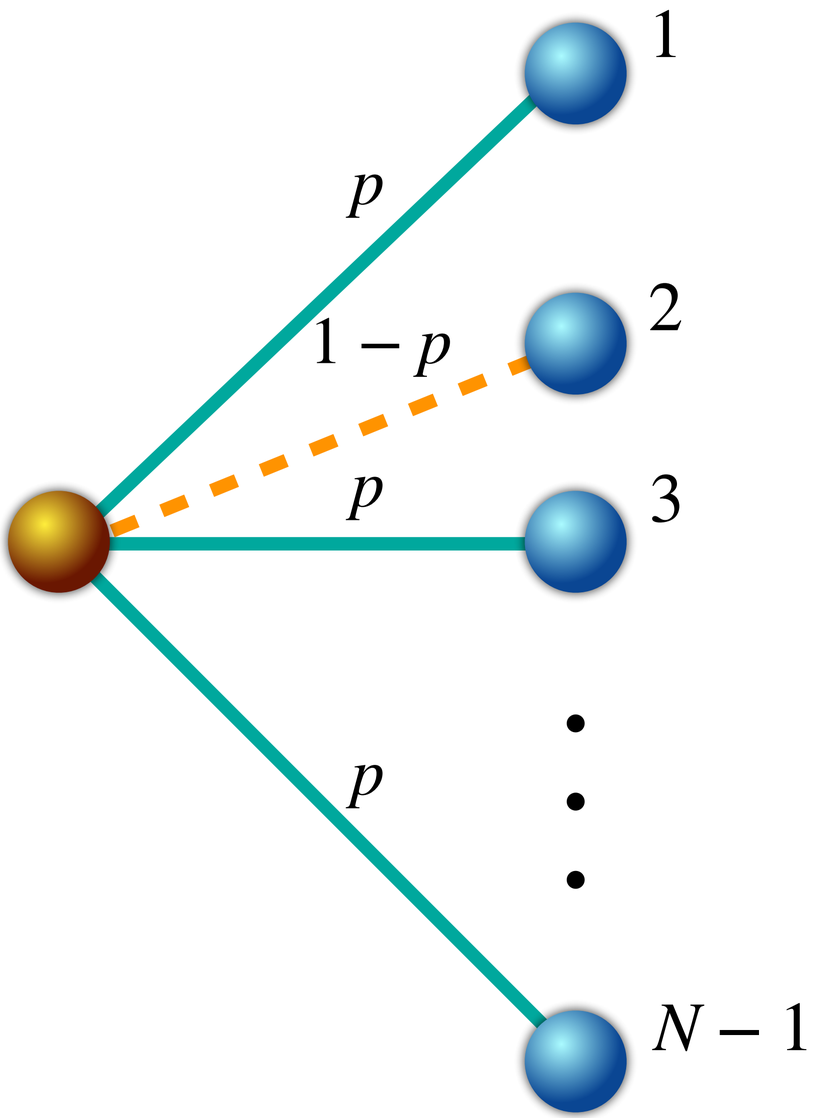
$$A = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

Average degree $\langle k \rangle$



$$k_i = \sum_{j \neq i} A_{ij}$$

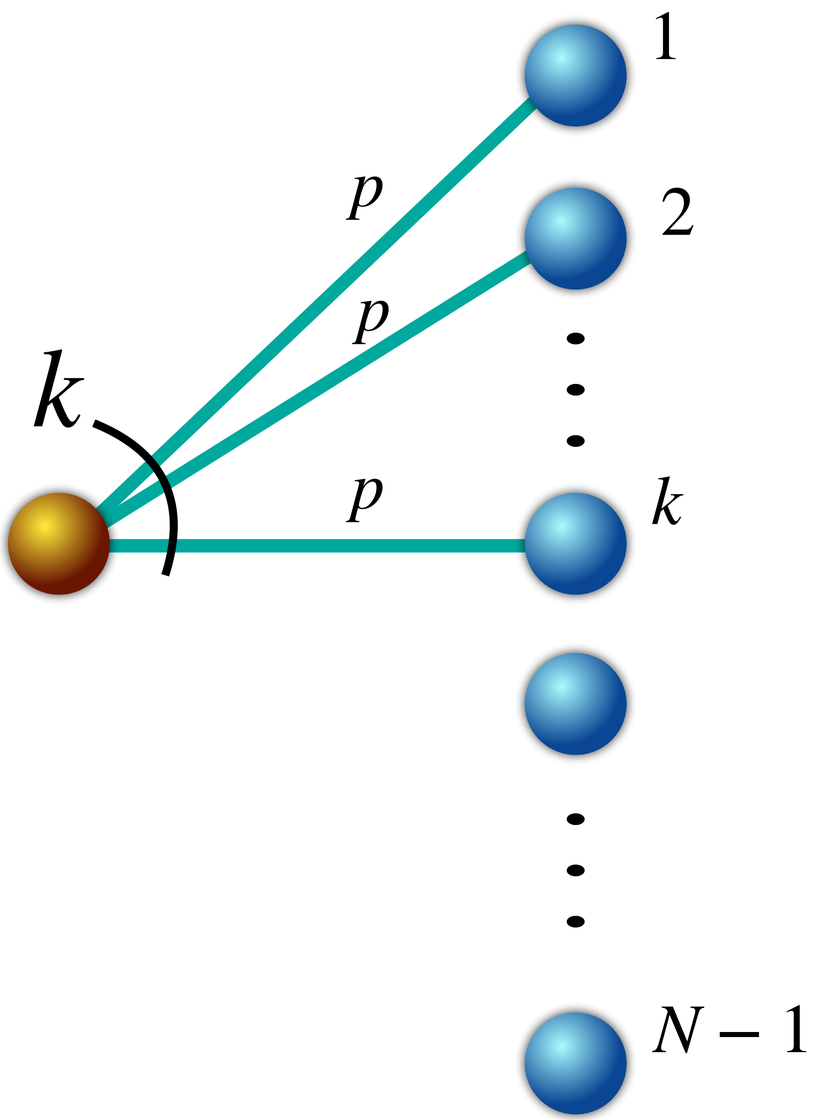
Average degree $\langle k \rangle$



$$\mathbb{E}[L] = p\binom{N}{2} = p\left(\frac{N(N-1)}{2}\right)$$

$$\mathbb{E}[k] = \langle k \rangle = \frac{2\mathbb{E}[L]}{N} = p(N-1)$$

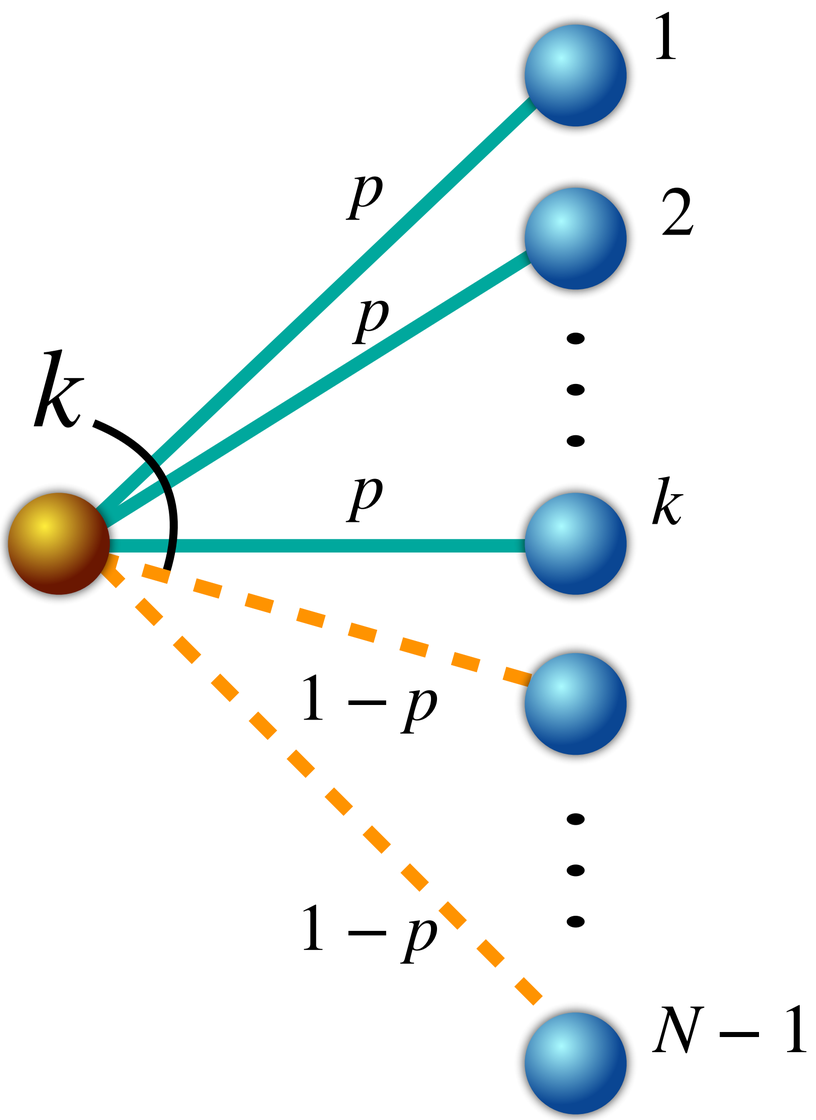
Degree Distribution $P(k)$



What is the probability that these specified k edges are present ?

$$p^k$$

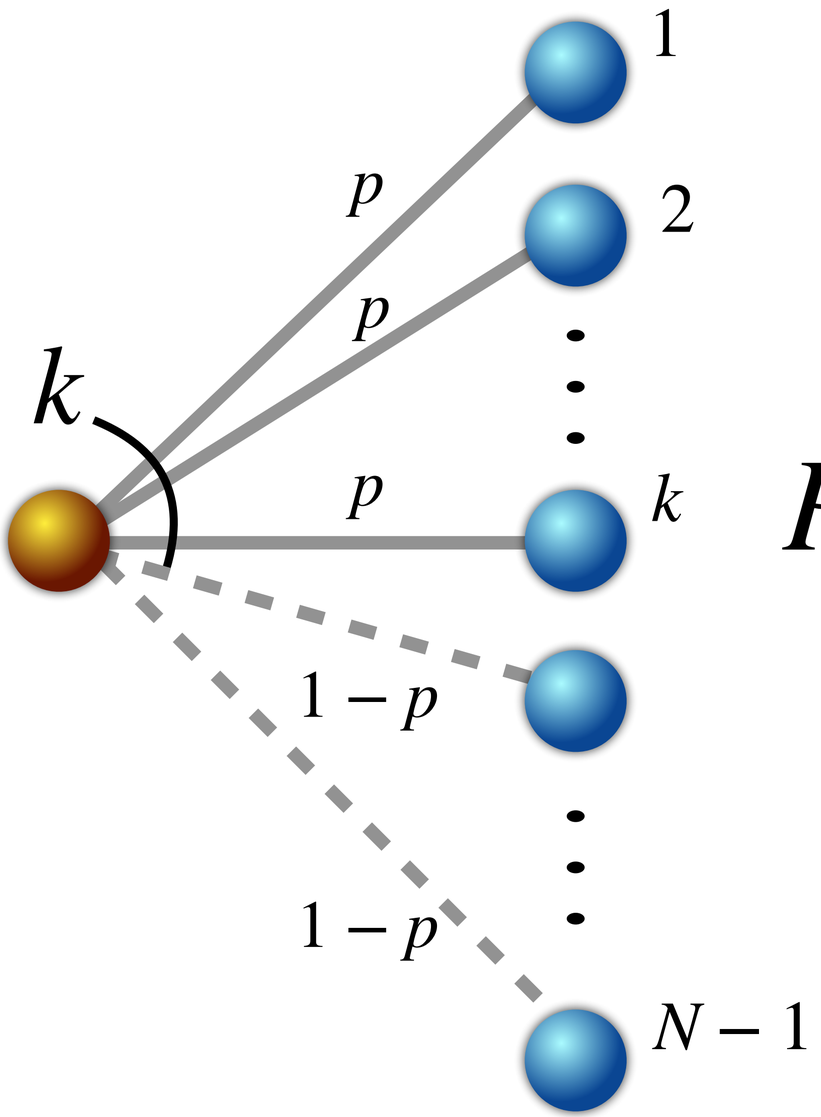
Degree Distribution $P(k)$



*Probability of **one** specified neighbor set*

$$p^k(1-p)^{N-1-k}$$

Degree Distribution $P(k)$



*Let's count the **multiple** neighbor sequences of size k.*

$$P(k) = \binom{N-1}{k} p^k (1-p)^{N-1-k}$$

Binomial coefficient

$$n = N - 1$$

$$\downarrow$$

$$\langle k \rangle = pn = z$$

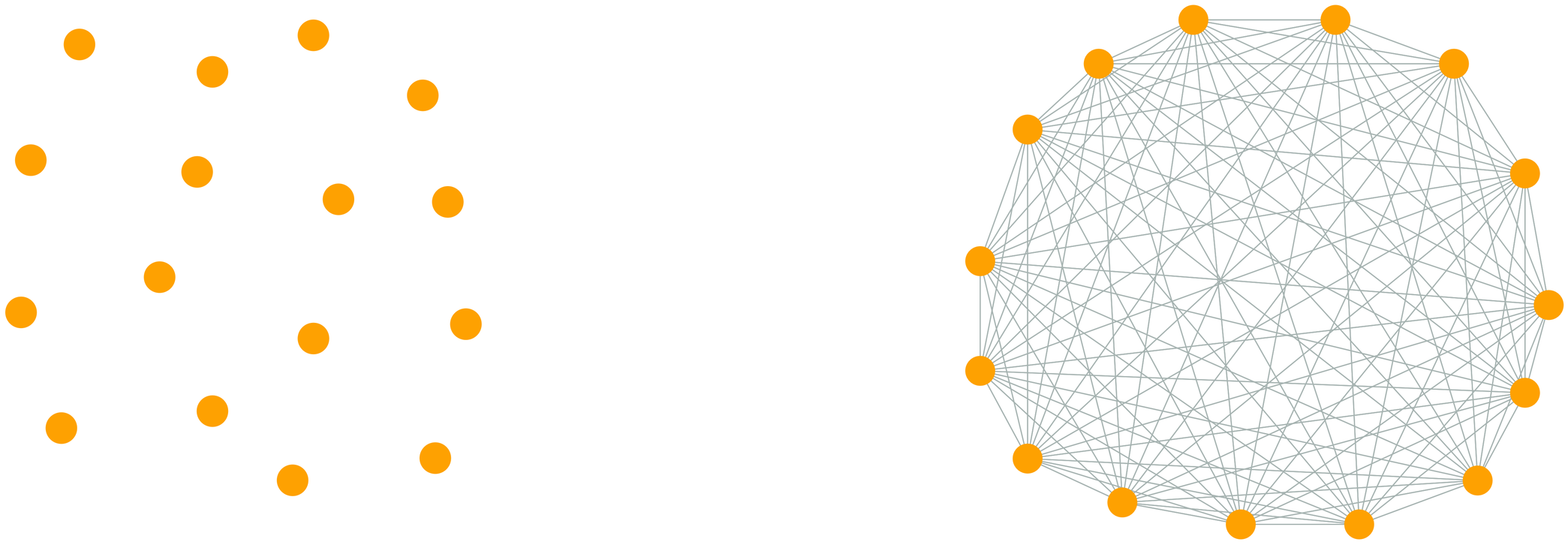
$$\binom{n}{k} p^k (1-p)^{n-k}$$

$$\binom{n}{k}(z/n)^k(1-z/n)^{n-k} \xrightarrow[p\to 0]{n\to\infty} e^{-z}\frac{z^k}{k!}$$

Binomial distribution

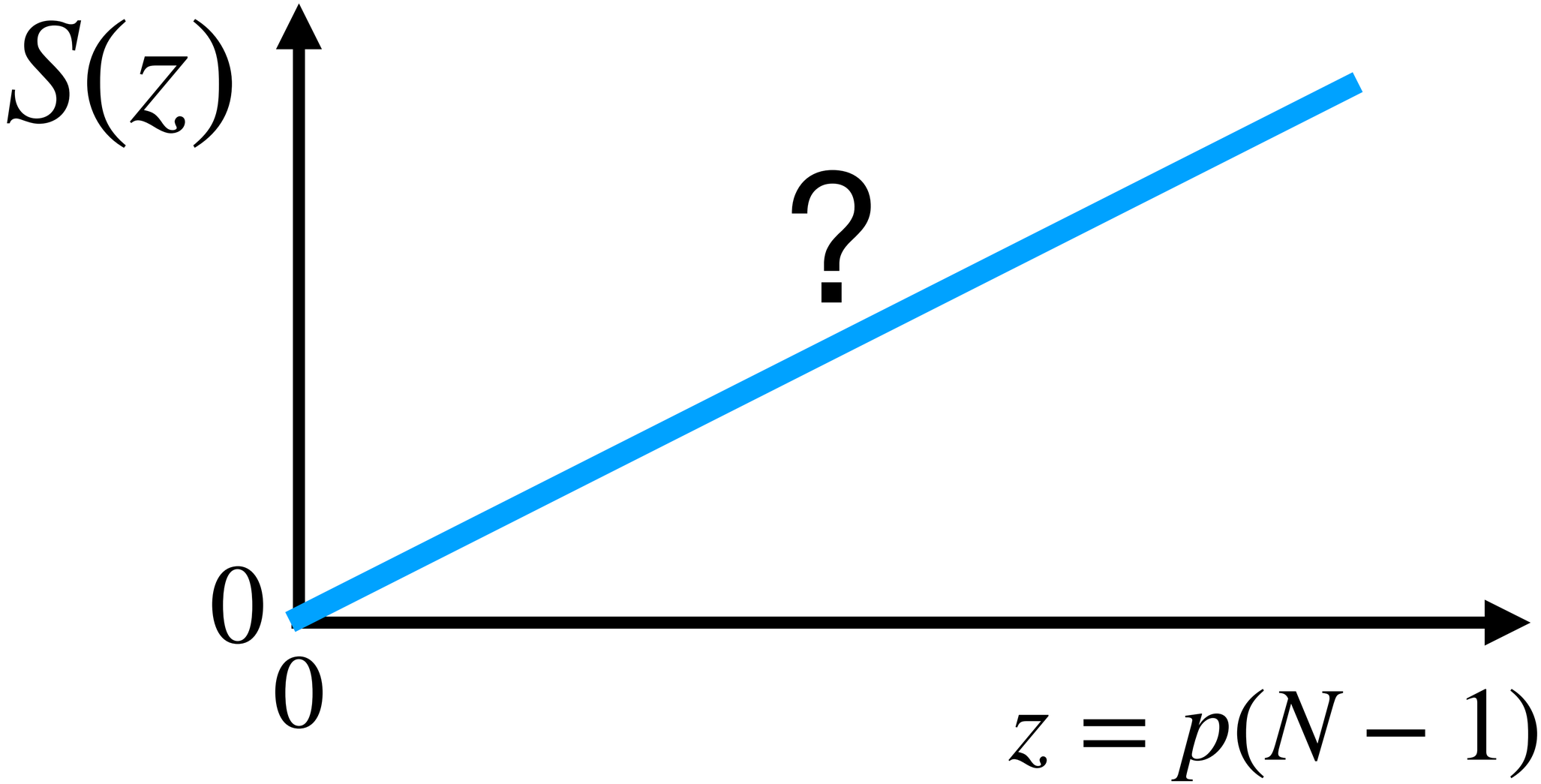
Poisson distribution

Percolation Transition

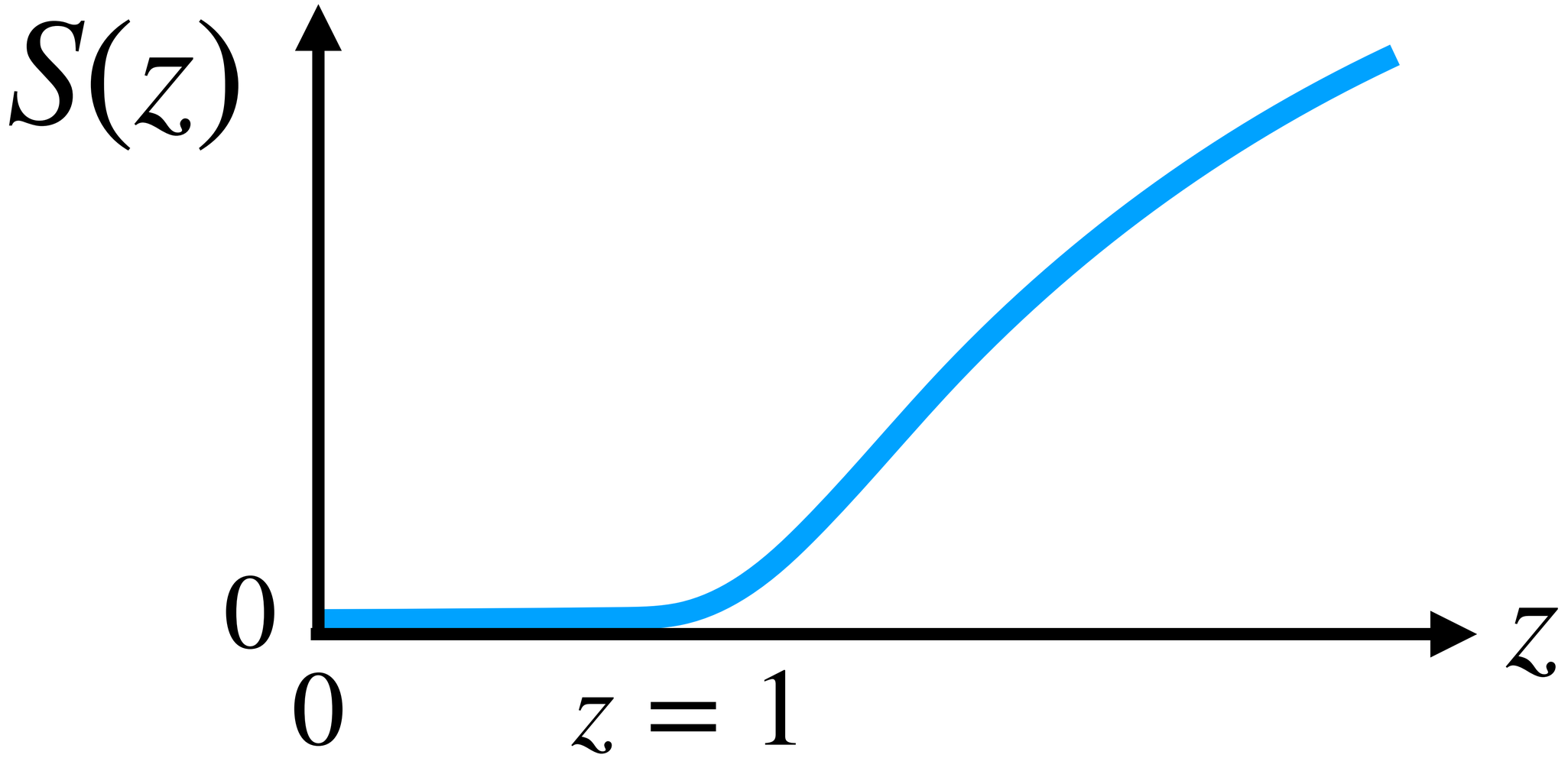


$p = 0$ $p = 1$

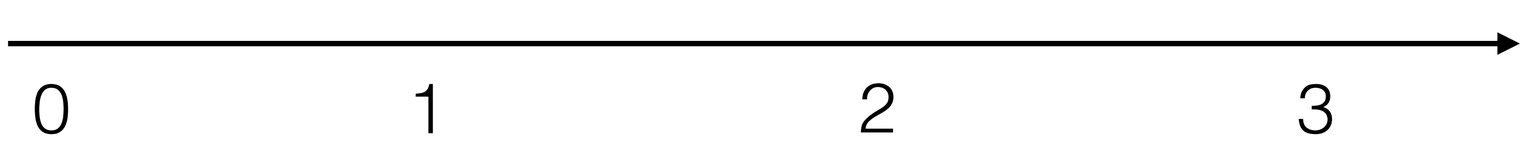
Percolation Transition



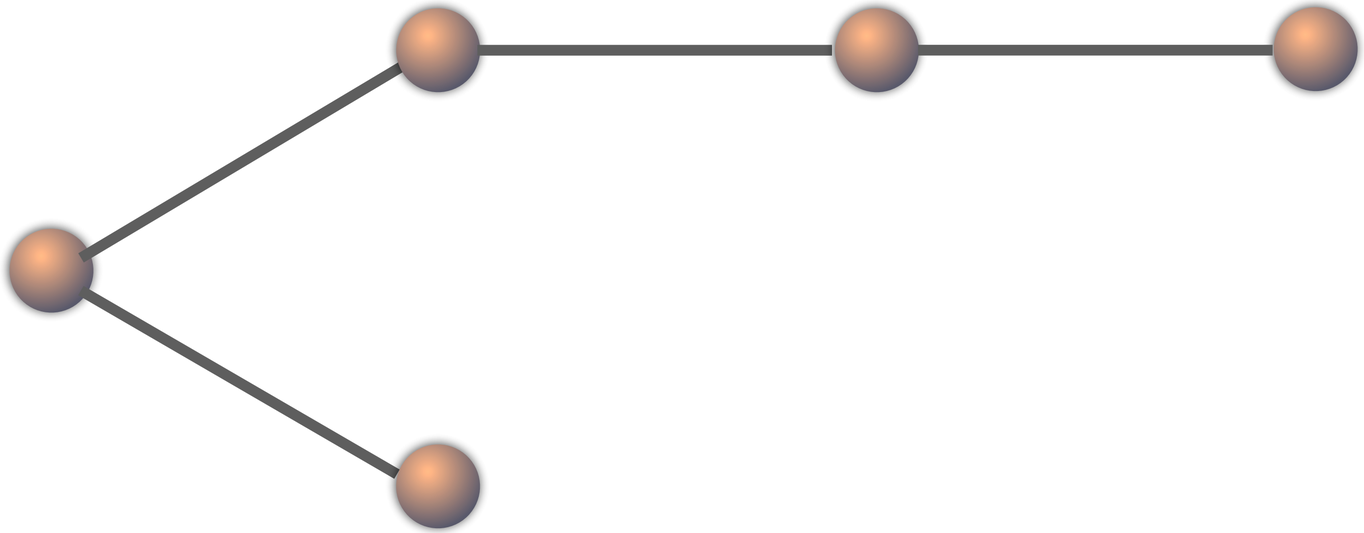
Percolation Transition



Bounded exploration, z < 1



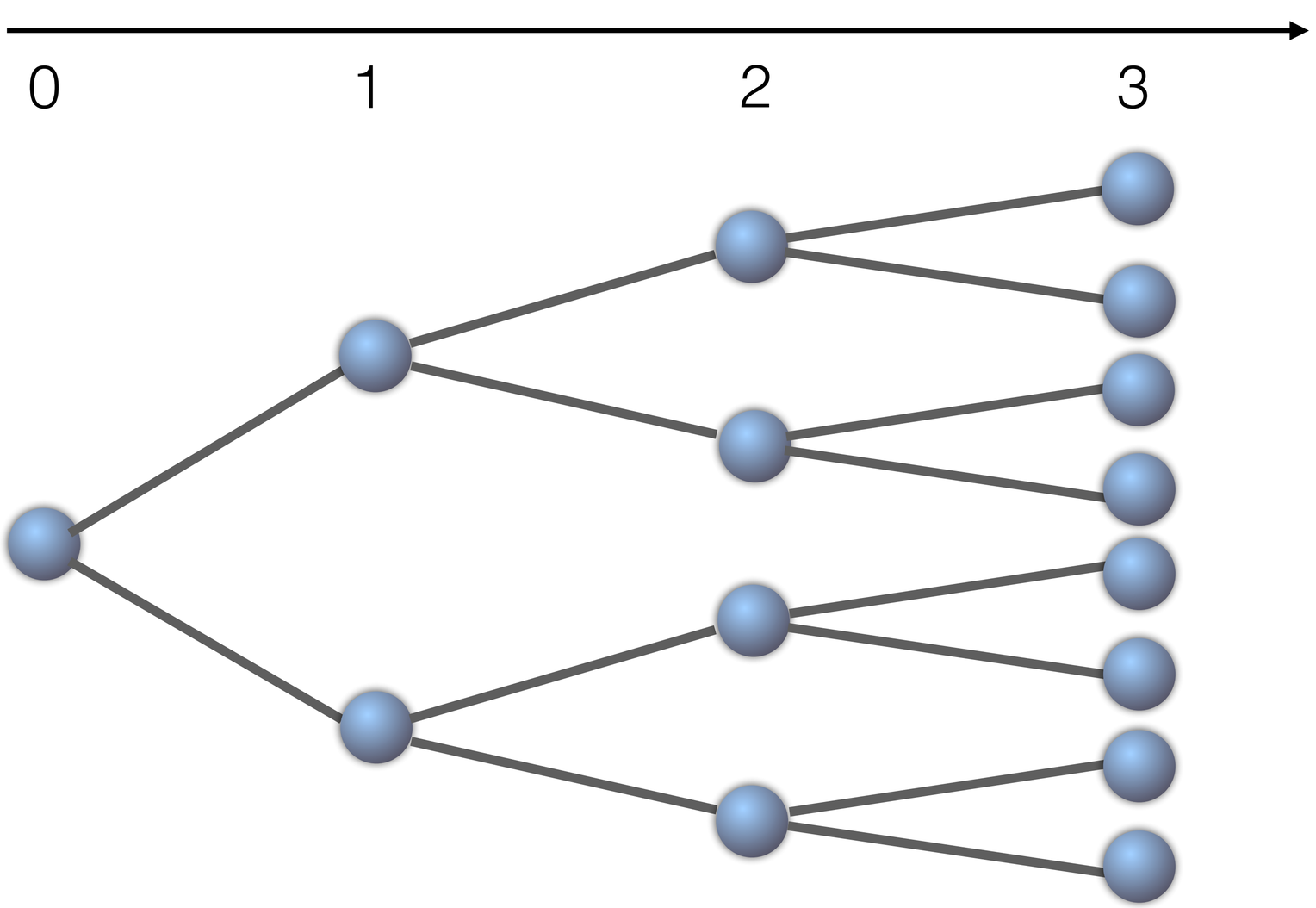
Generation l



$$N_{l+1} = zN_l$$

Extinction

Unbounded exploration, z > 1

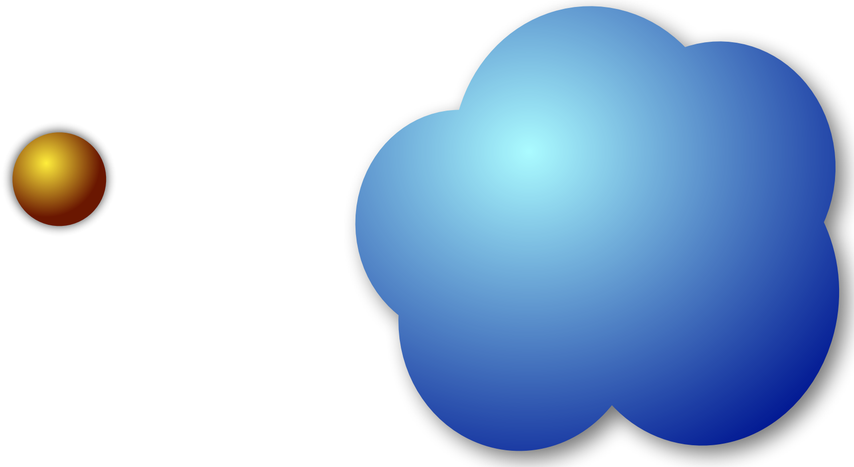


Generation l

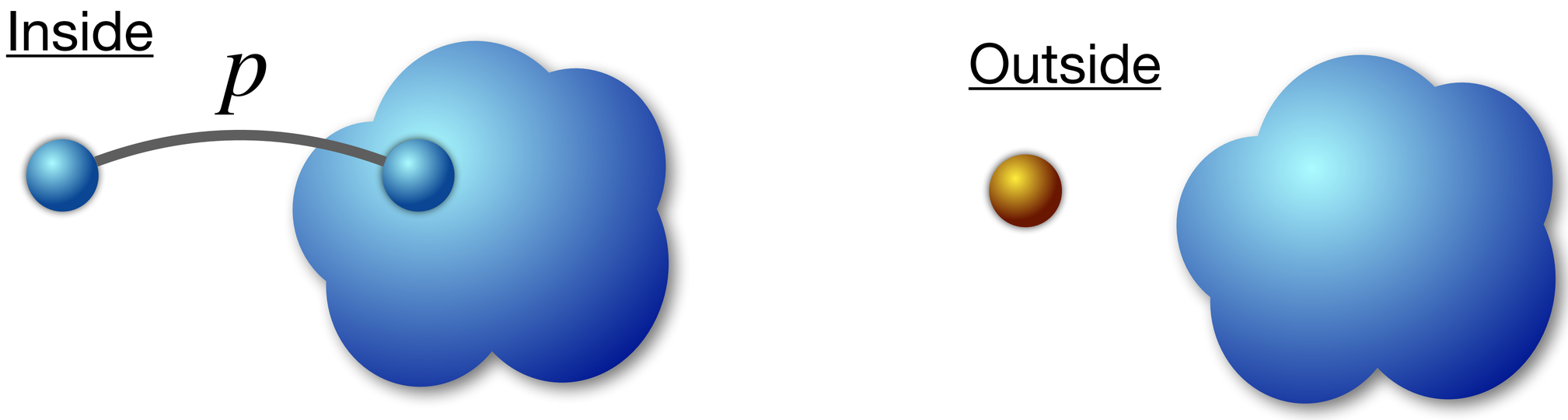
$$N_{l+1} = N_l^z$$

Reaching a macroscopic component

$$1 - S = \Pr(\text{a random vertex is outside the giant})$$



$$1 - S \approx (1 - p)^{S(N-1)}$$



$$\log(1 - S) \approx S(N - 1)\log(1 - p)$$

$$\log(1 - S) \approx S(N - 1)(-p)$$

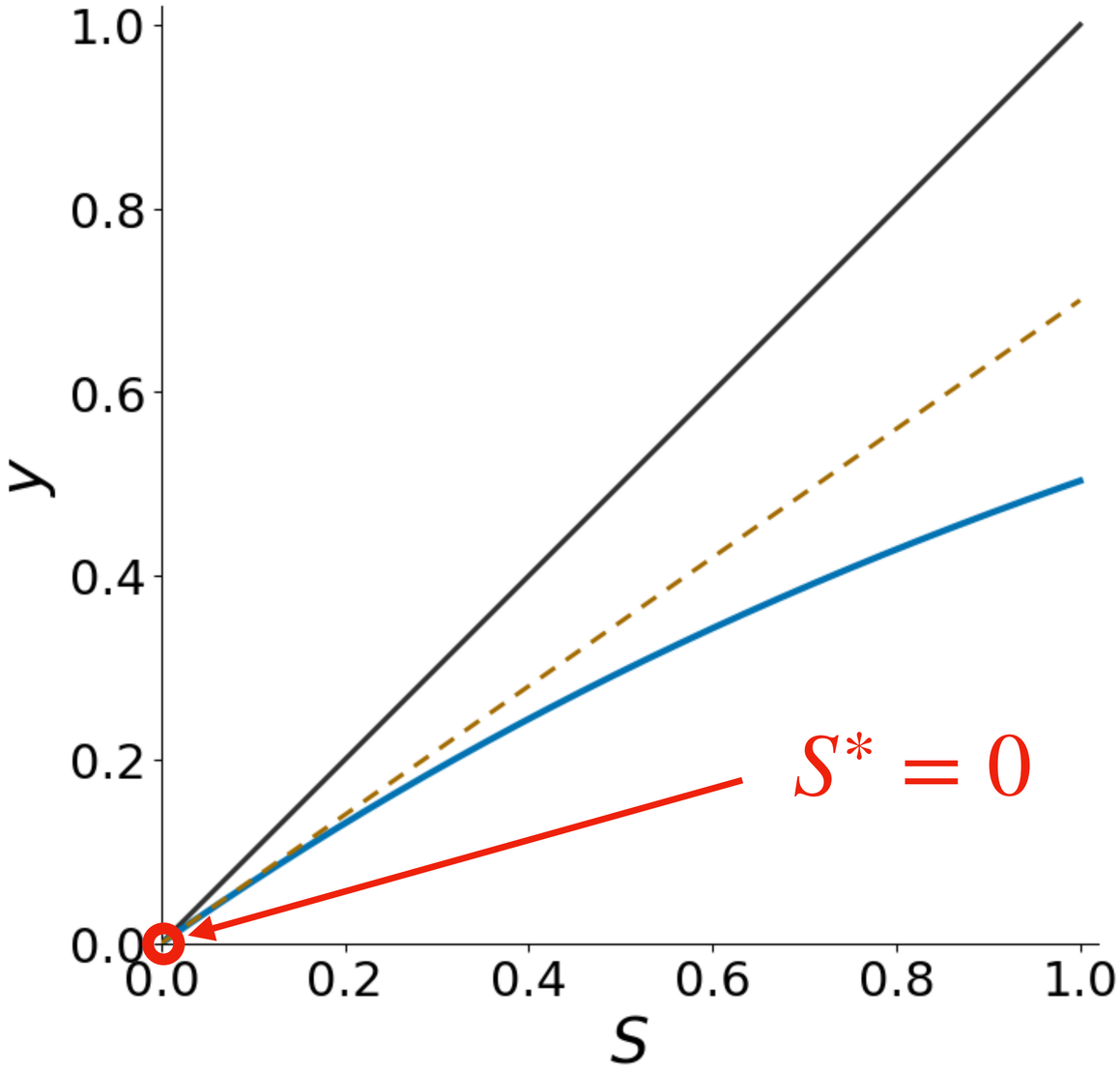
$$\log(1 - S) \approx -zS$$

$$1 - S = e^{-zS}$$

$$S = 1 - e^{-zS}$$

$$S^* = 0$$

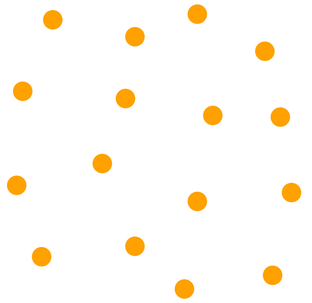
Below threshold: $z = 0.7 < 1$



$$y(S) = S$$

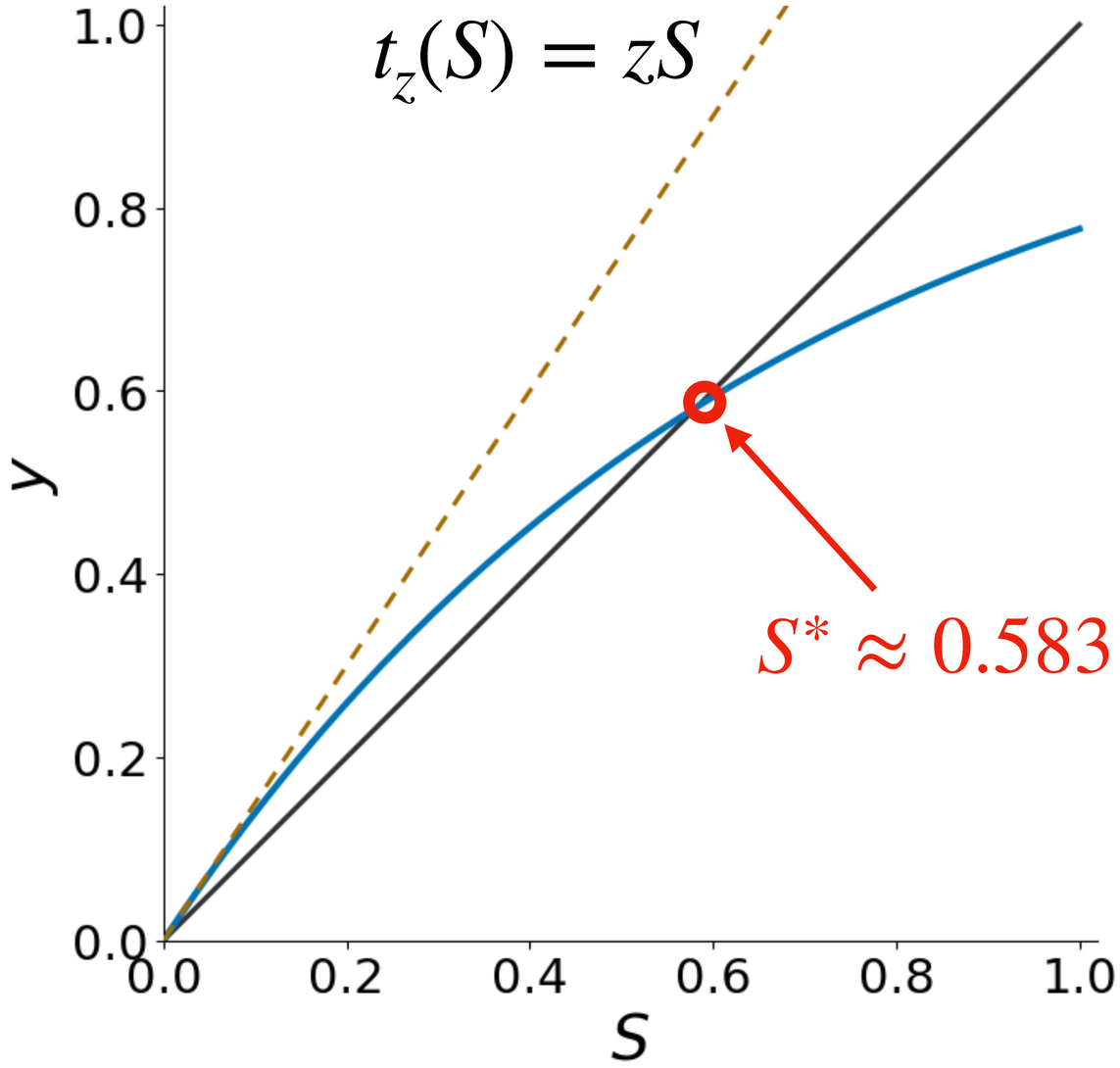
$$t_z(S) = g_z(0) + g_z'(0)(S - 0) = zS$$

$$y(S) = g_z(S) = 1 - e^{-zS}$$



There is no giant component in the large-network limit.

Above threshold: $z = 1.5 > 1$



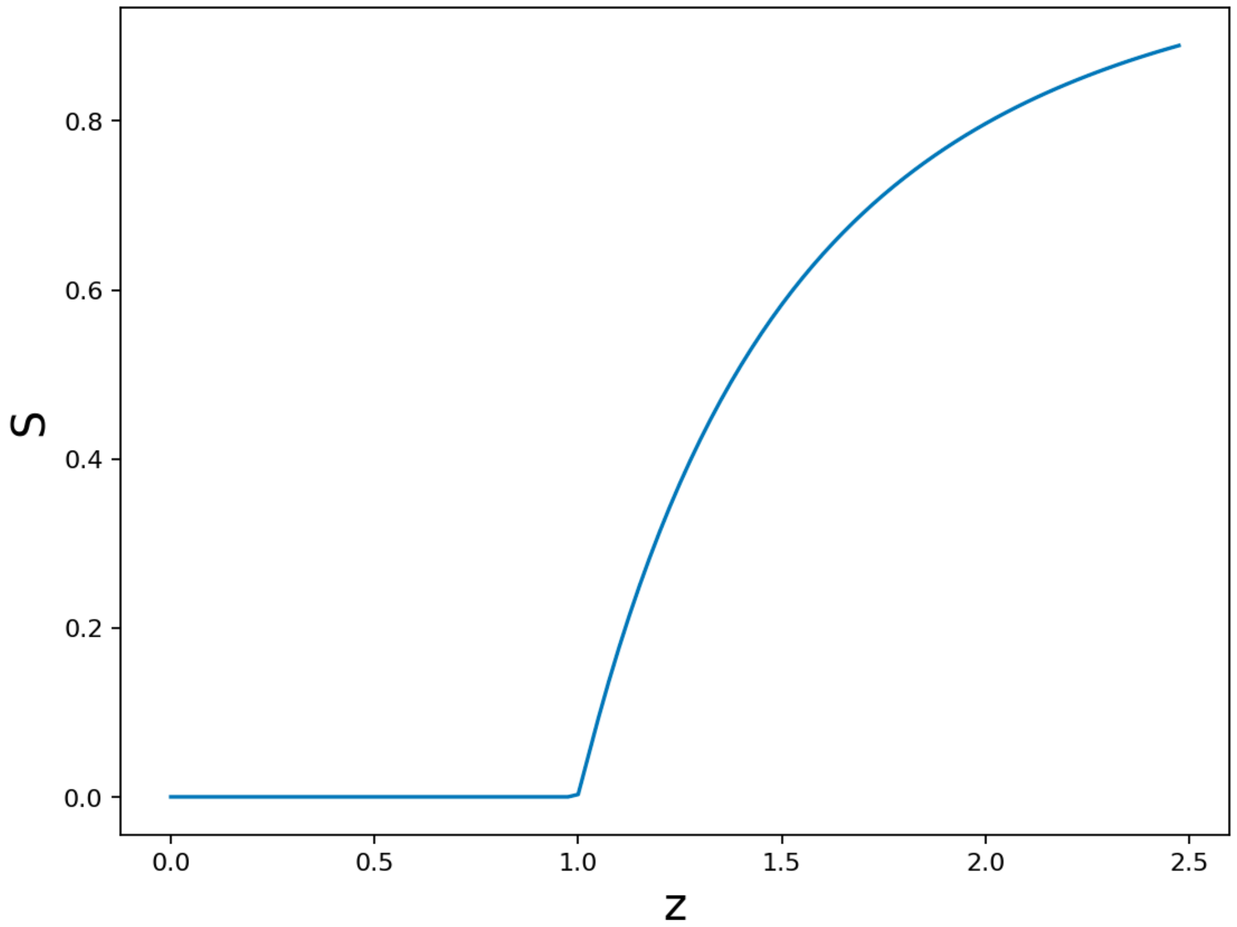
$$y(S) = S$$

$$y(S) = 1 - e^{-zS} < 1$$

A positive solution exists for z>1.

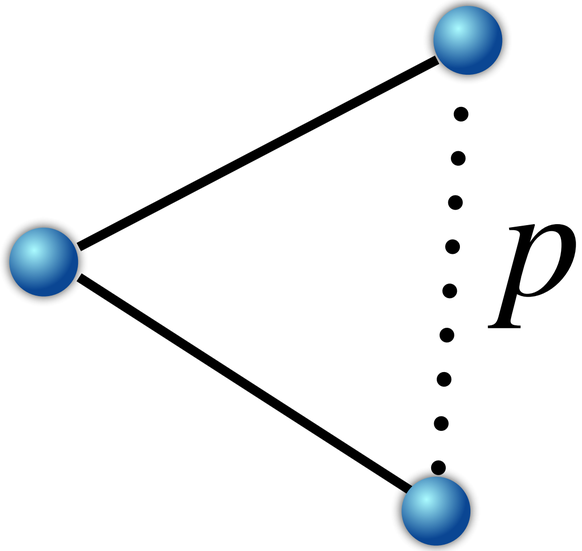
Numerical Solution

$$S = 1 - e^{-zS}$$




```
import matplotlib.pyplot as plt
import numpy as np
plt.figure(figsize=(8,6), dpi = 160)
S_values = []
z_values = [float(i)/40.0 for i in range(100)]
for z in z_values:
    S = 0.01
    for j in range(500):
        S = 1 - np.exp( -z * S )
    S_values.append (S)
plt.xlabel ("z", fontsize= 18)
plt.ylabel ("S", fontsize = 18)
plt.plot (z_values, S_values)
plt.show()
```

Random graphs display clustering of order p



$$\langle C \rangle_{rand} = p$$

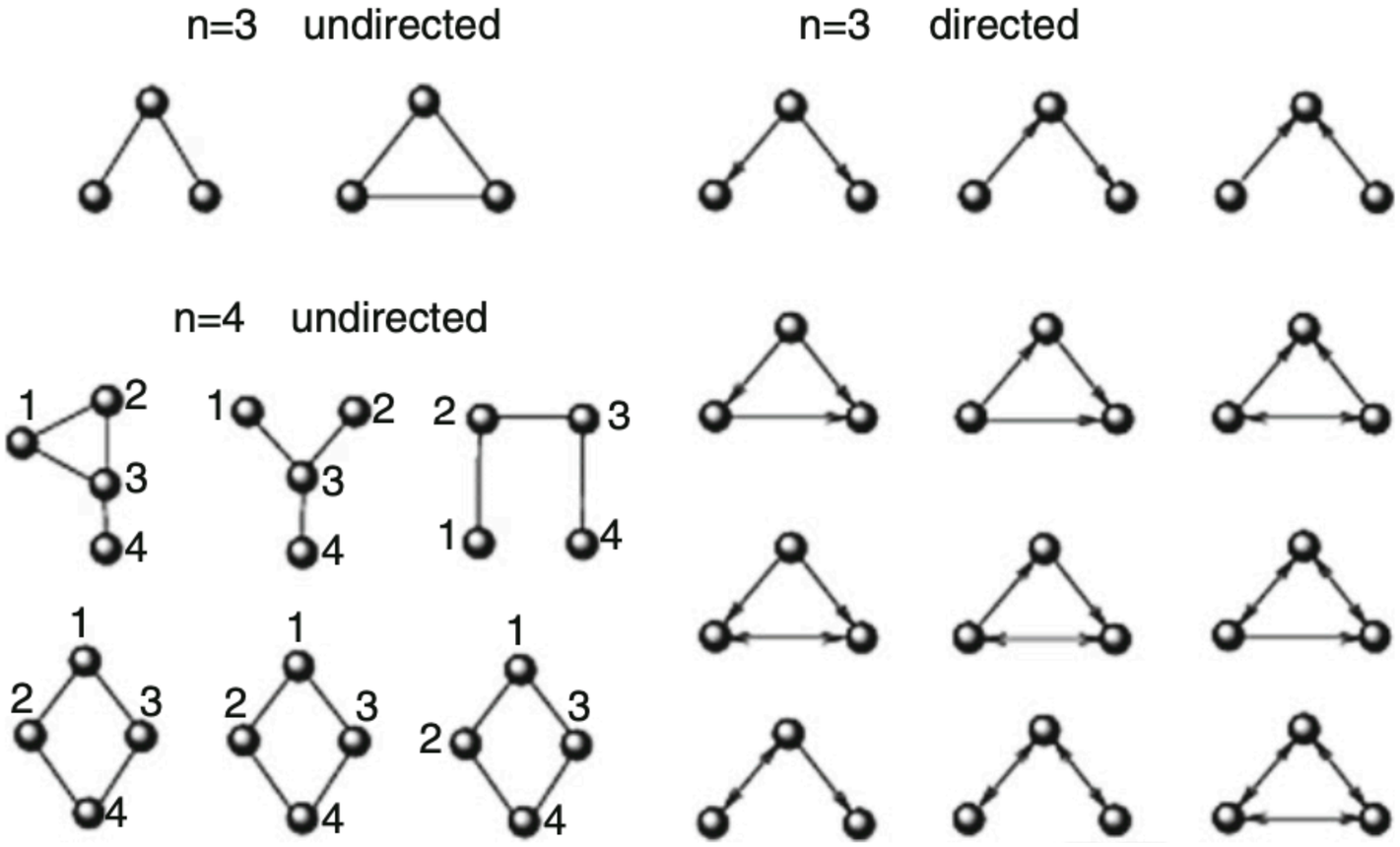
$$\langle C \rangle_{rand} \approx p = \frac{z}{N-1} \rightarrow 0$$

Real-world links are not independent

$$0.01 \leq \langle C \rangle_{Facebook} \leq 0.5$$

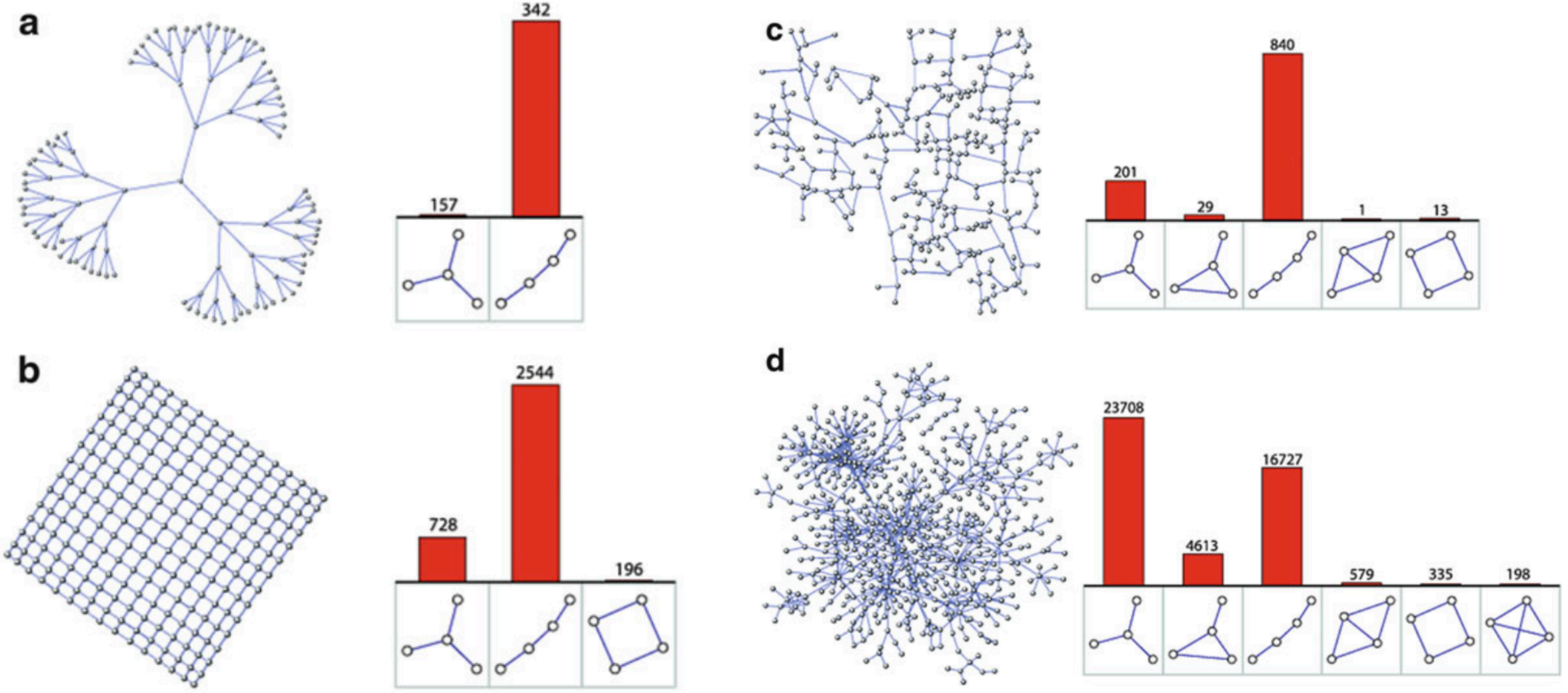
$$\langle C \rangle_{rand} = \frac{\langle k \rangle}{N-1} = \frac{10^3}{10^9} \approx 10^{-6}$$

Use null models to correctly interpret connectivity patterns



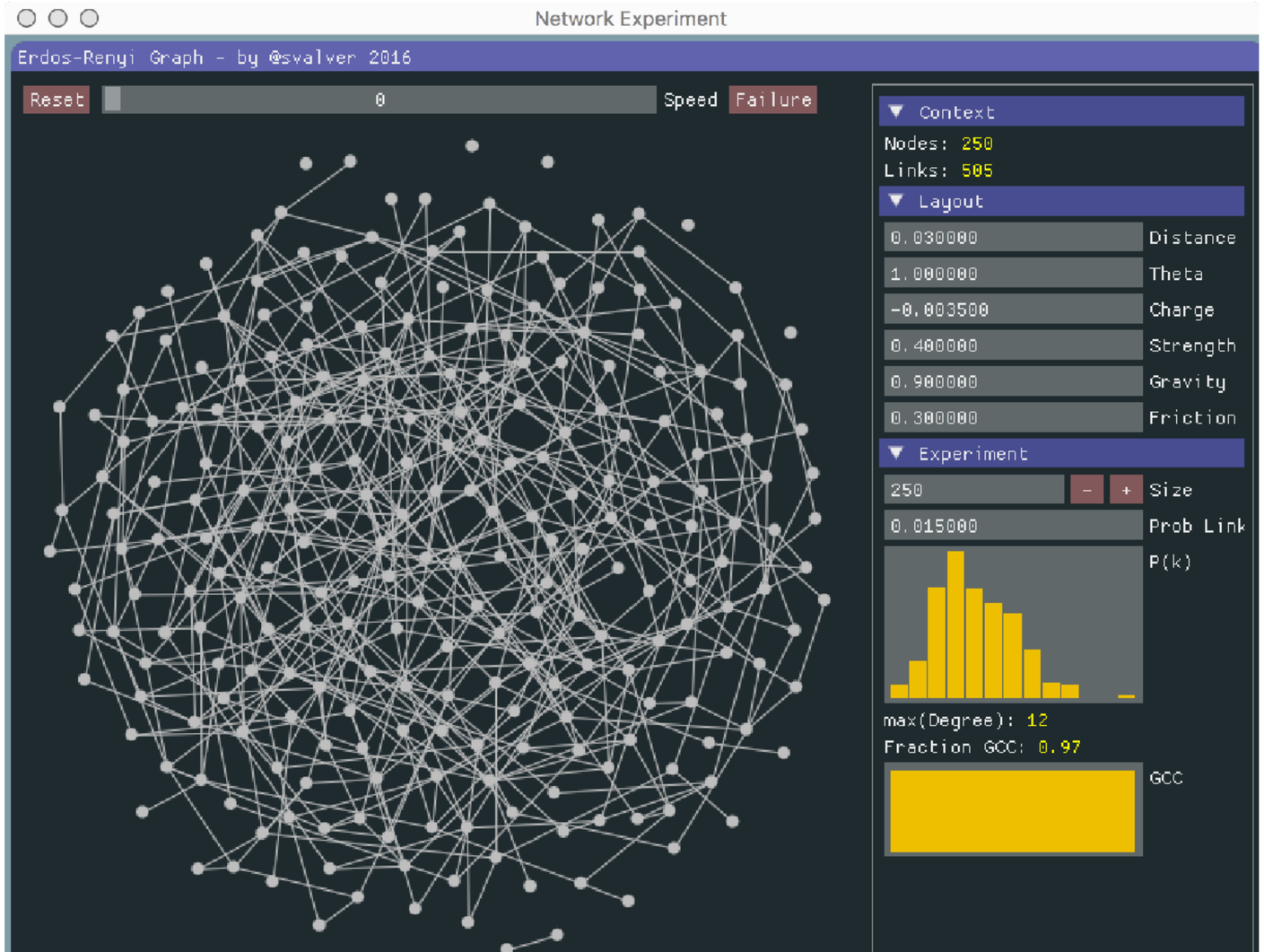
$$Z_\omega = \frac{n_\omega^{\text{obs}} - \mathbb{E}_{\text{null}}[n_\omega]}{\text{sd}_{\text{null}}(n_\omega)}$$

Abundance does not always imply function / selection



Activity: Random Networks

https://tinyurl.com/3p9fxnsc

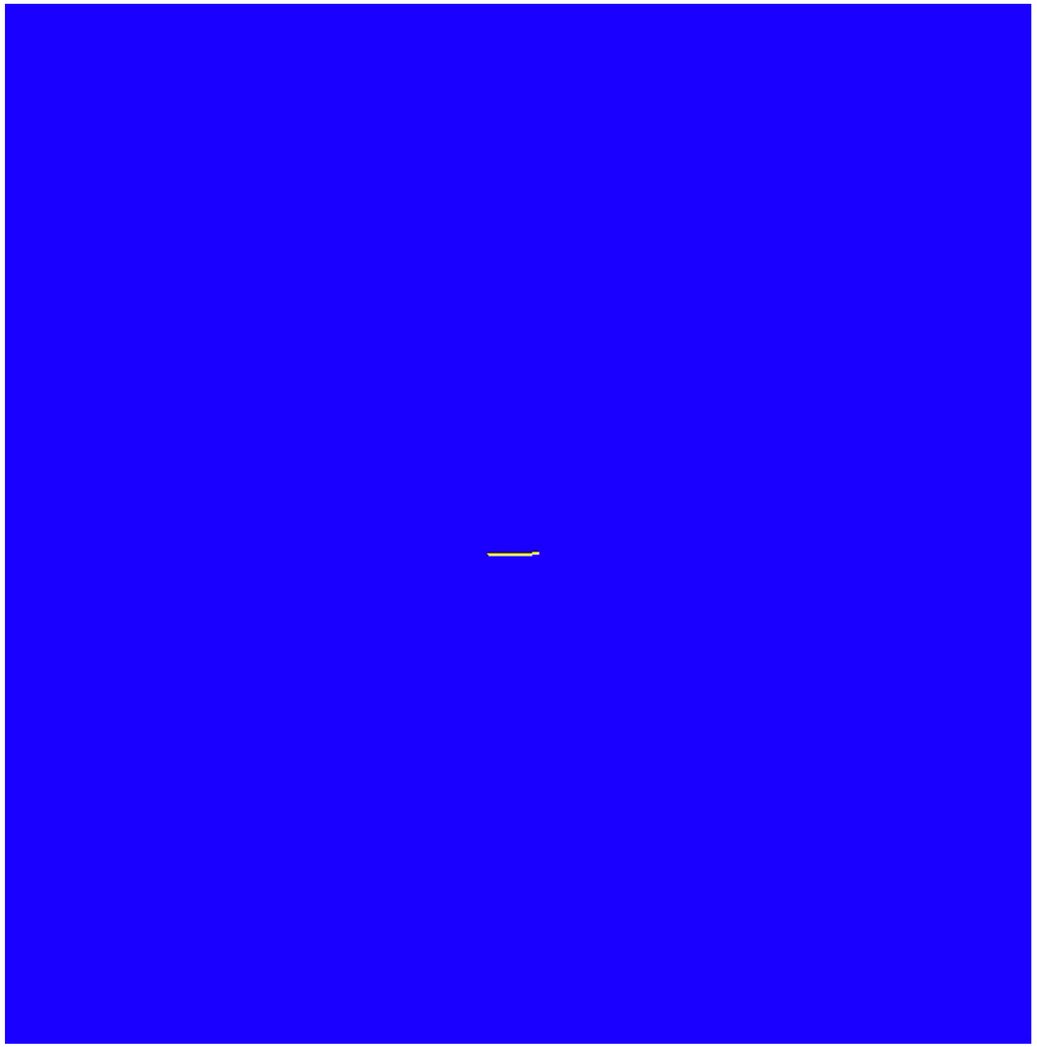


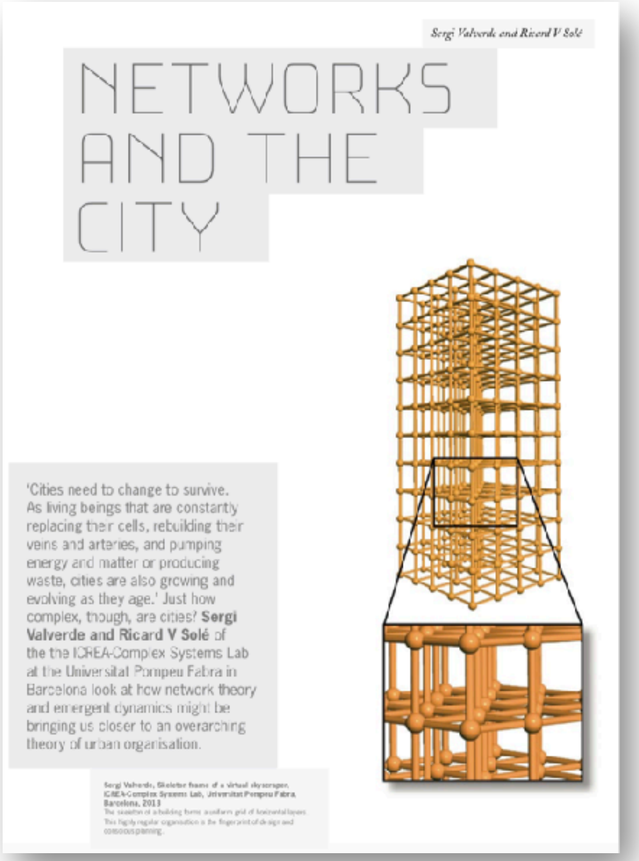
3. Can you predict the average degree before running the simulation?

4. How probable is it to obtain a node with a very large number of links?

Growing Networks

Man-made objects can be geometrically complex and do not resemble ideal forms such as points, lines, planes, cubes, circles of spheres.





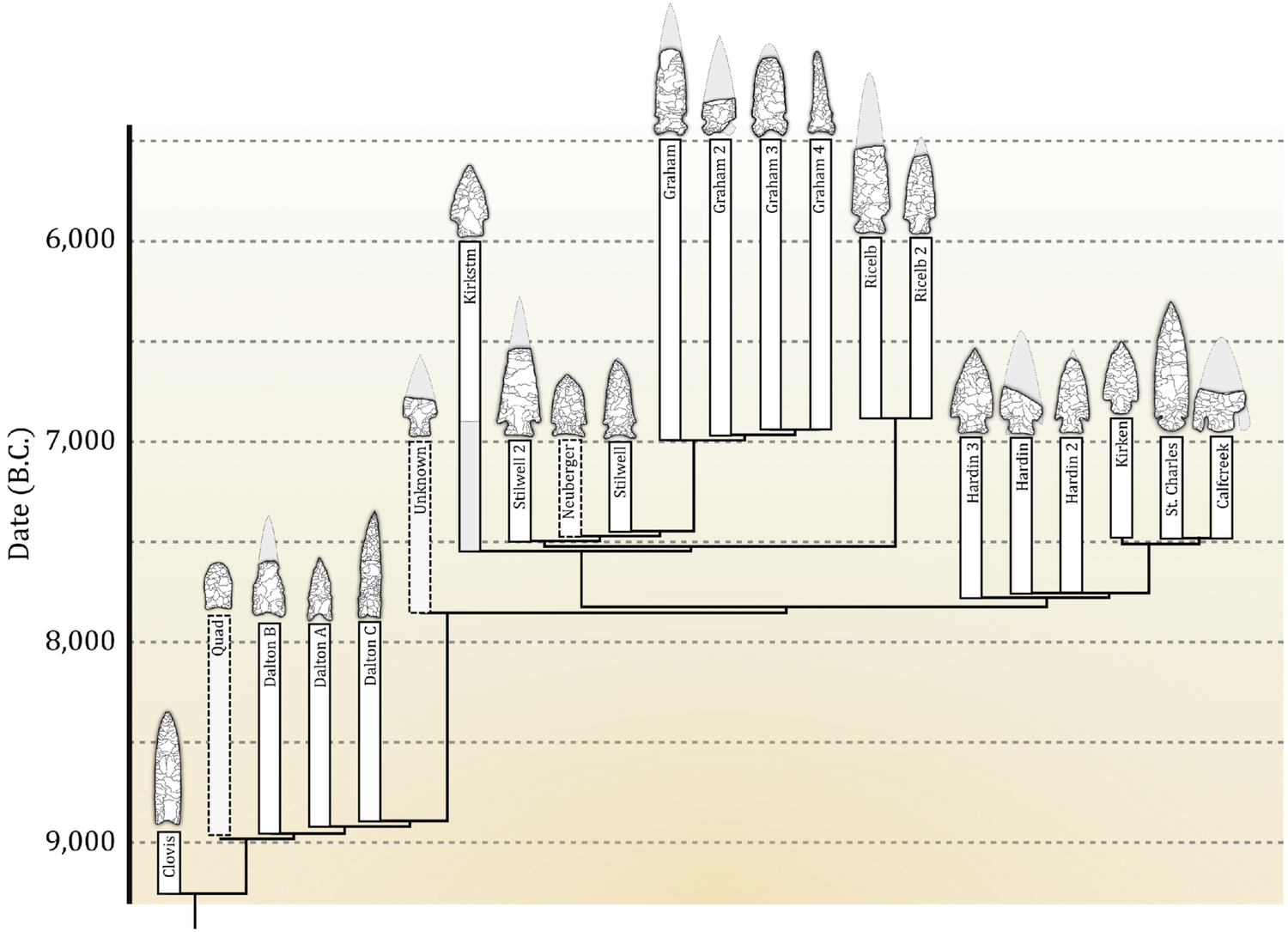
Sergi Valverde and Ricard V Solé

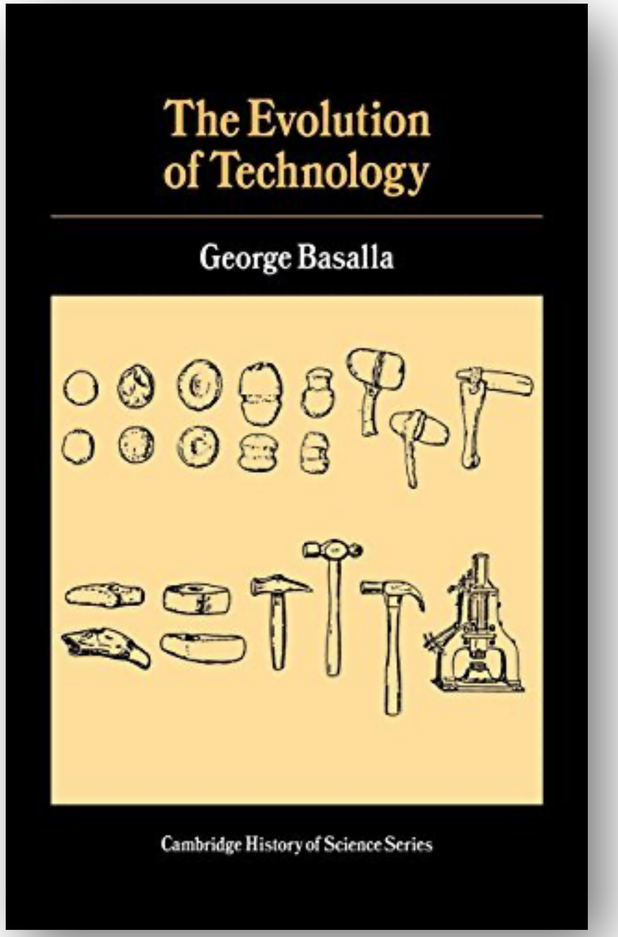
NETWORKS AND THE CITY

'Cities need to change to survive. As living beings that are constantly replacing their cells, rebuilding their veins and arteries, and pumping energy and matter or producing waste, cities are also growing and evolving as they age.' Just how complex, though, are cities? **Sergi Valverde and Ricard V Solé** of the the ICREA-Complex Systems Lab at the Universitat Pompeu Fabra in Barcelona look at how network theory and emergent dynamics might be bringing us closer to an overarching theory of urban organisation.

Sergi Valverde, Skeleton frame of a virtual skyscraper, ICREA-Complex Systems Lab, Universitat Pompeu Fabra, Barcelona, 2013

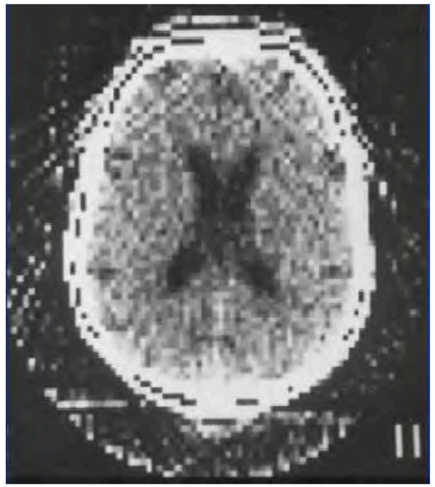
Evolution of Technology



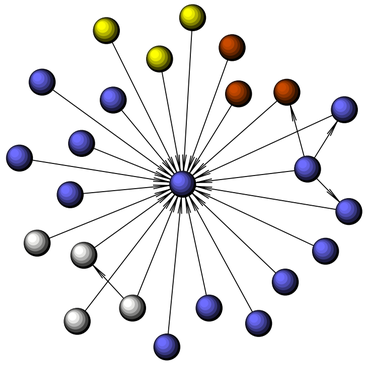


Growth: Patent Networks

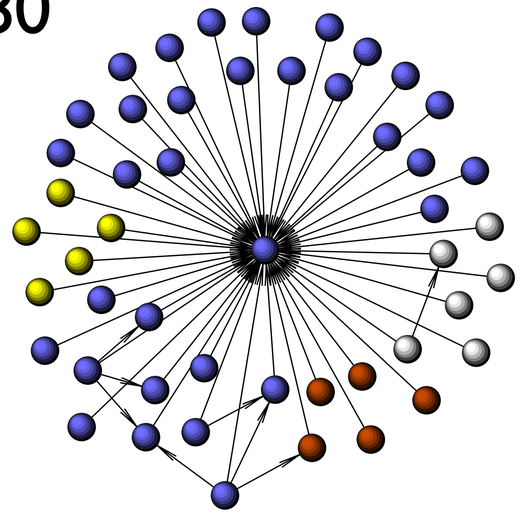
1974



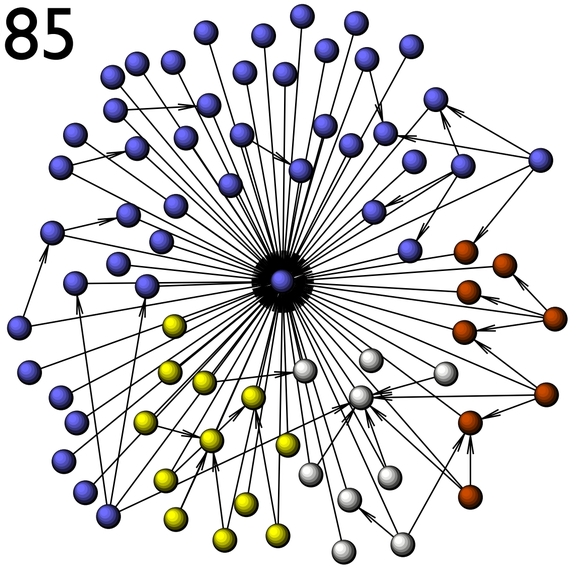
1973



1980

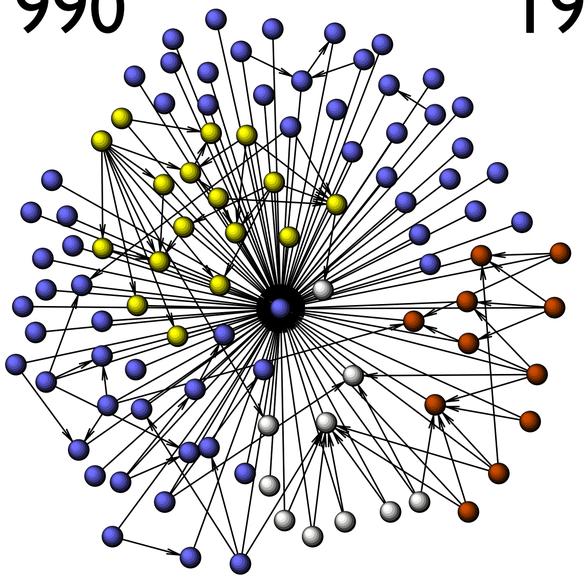


1985

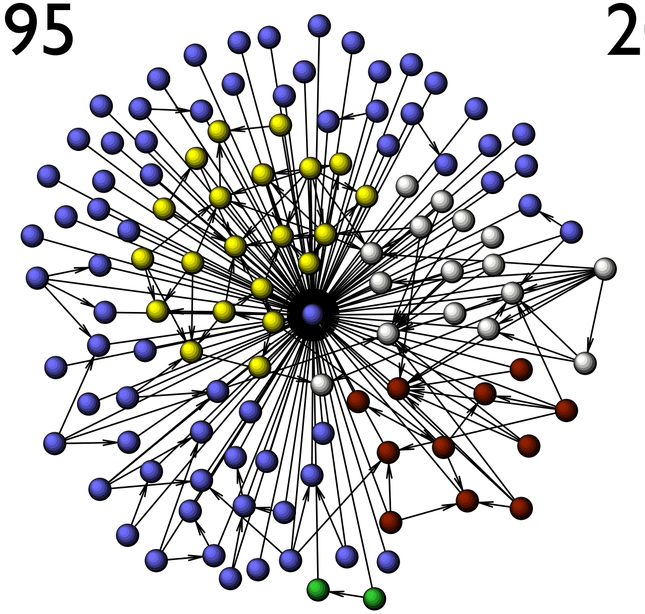


1994

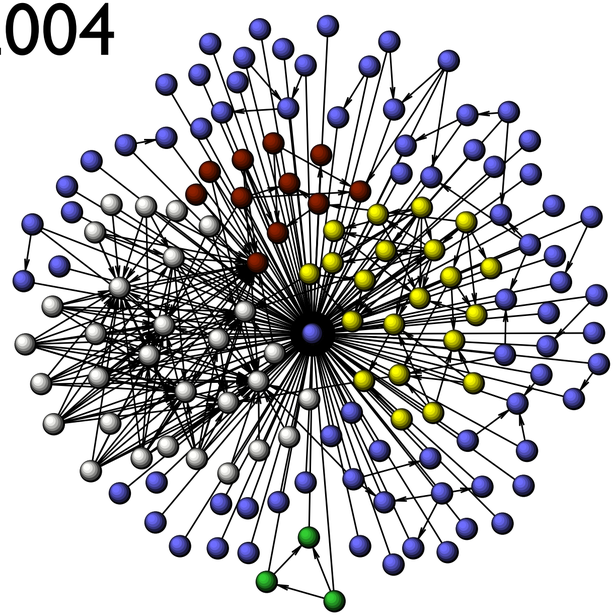
1990



1995

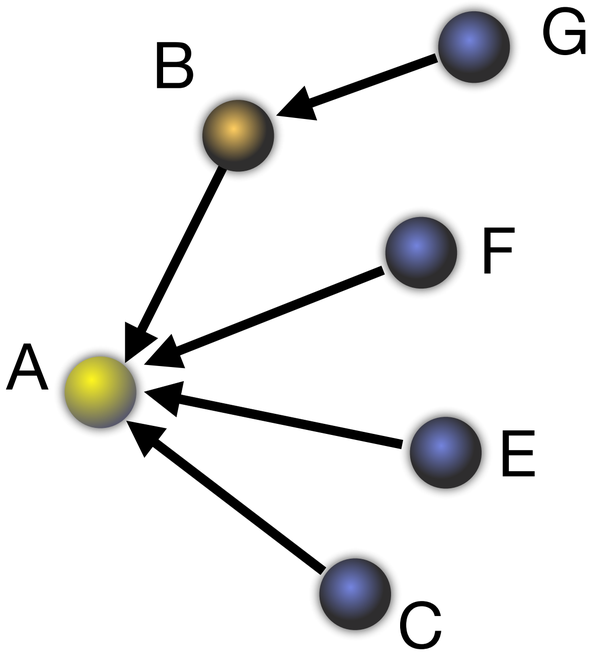


2004

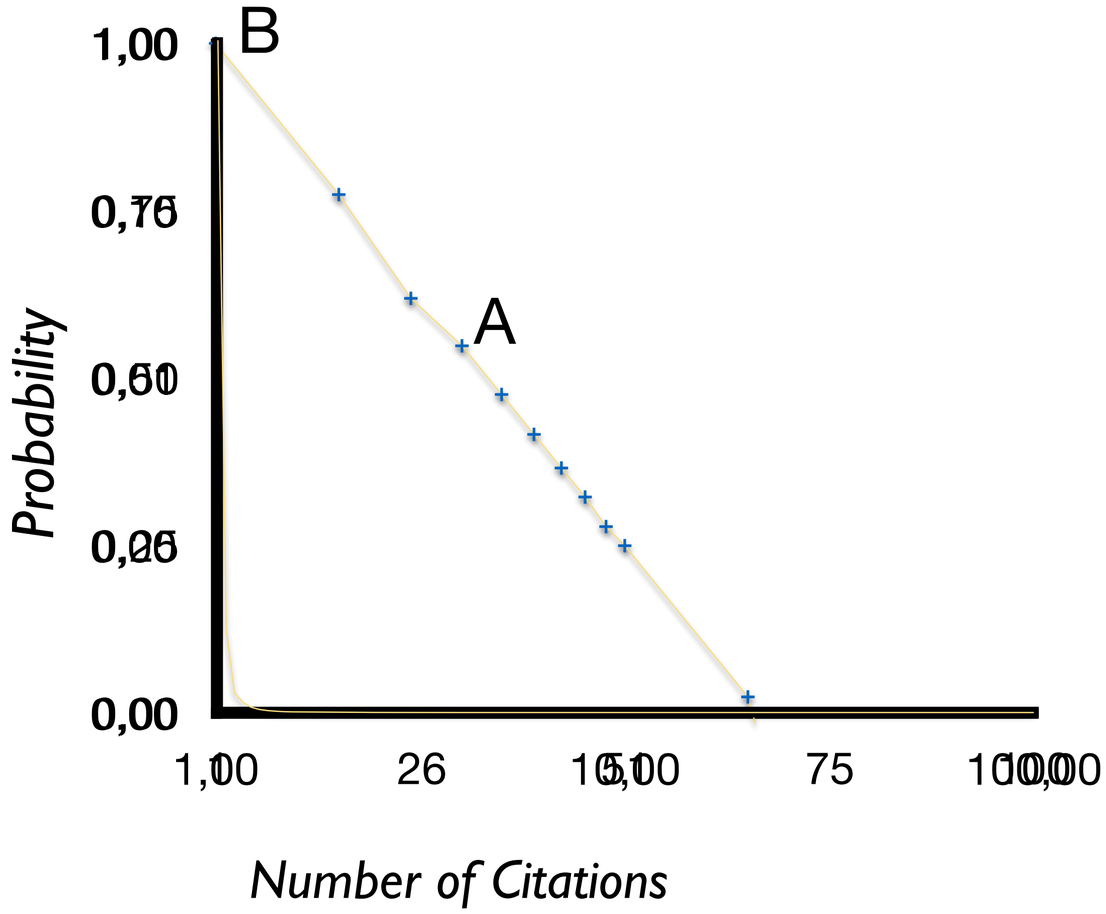


Growth: Preferential Attachment

$$\Pi(k) \sim k^{\beta} \longrightarrow P(k) = Uk^{-\gamma}$$



(Price, 1965) & (Price, 1976)



Derek de Solla Price (1922-1983)

Cumulative degree distribution

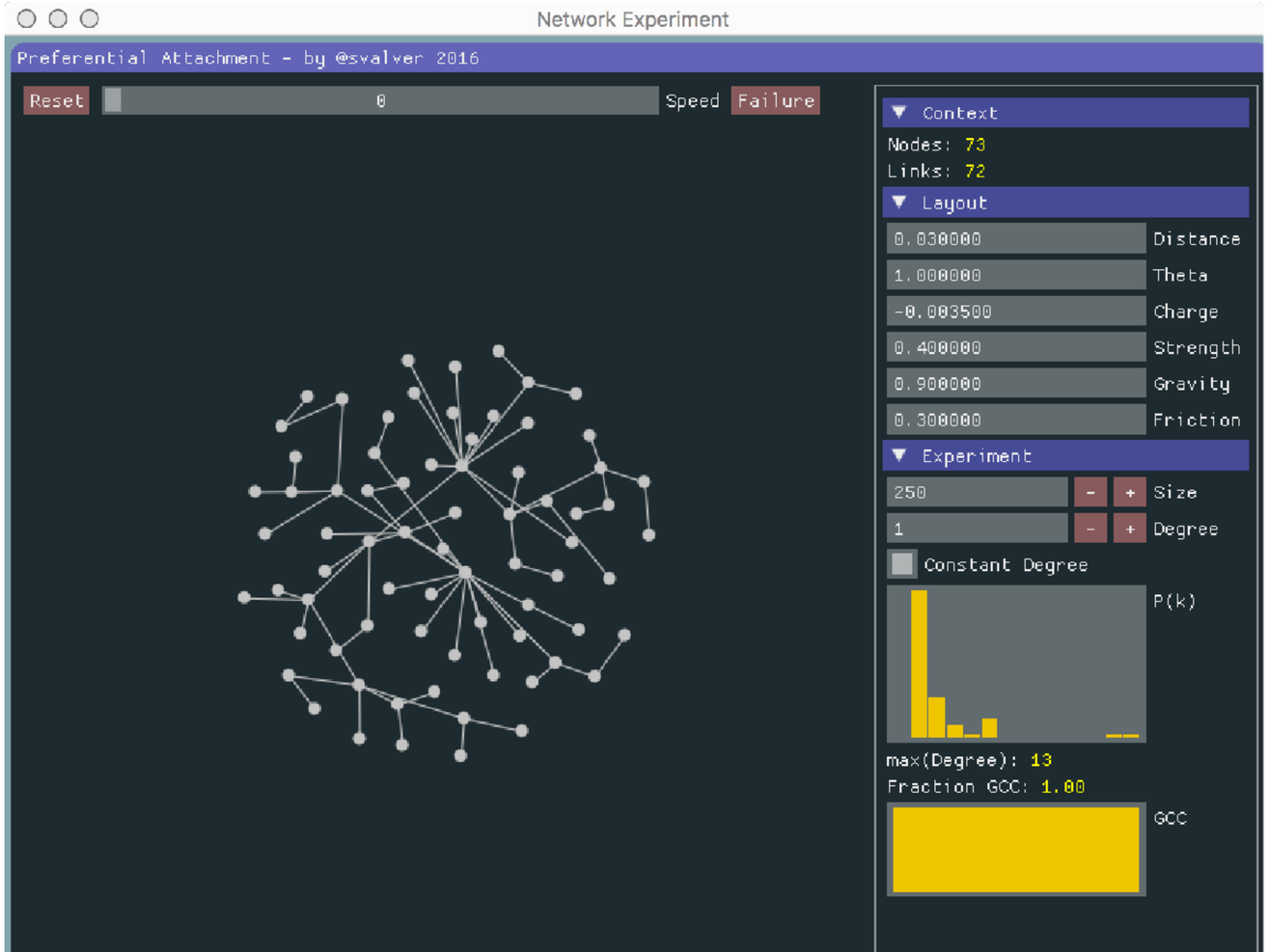
$$P_{\geq k} = \Pr(K \geq k) = \sum_{k'=k}^{\infty} P(k')$$

$$P_{\geq k} = U \sum_{j=k}^{\infty} j^{-\gamma} \approx U \int_{k}^{\infty} j^{-\gamma} dj = \frac{U}{\gamma - 1} k^{-(\gamma-1)}$$

Activity: Preferential Attachment

How history and reinforcement influence network architecture?

https://tinyurl.com/3ttchcep

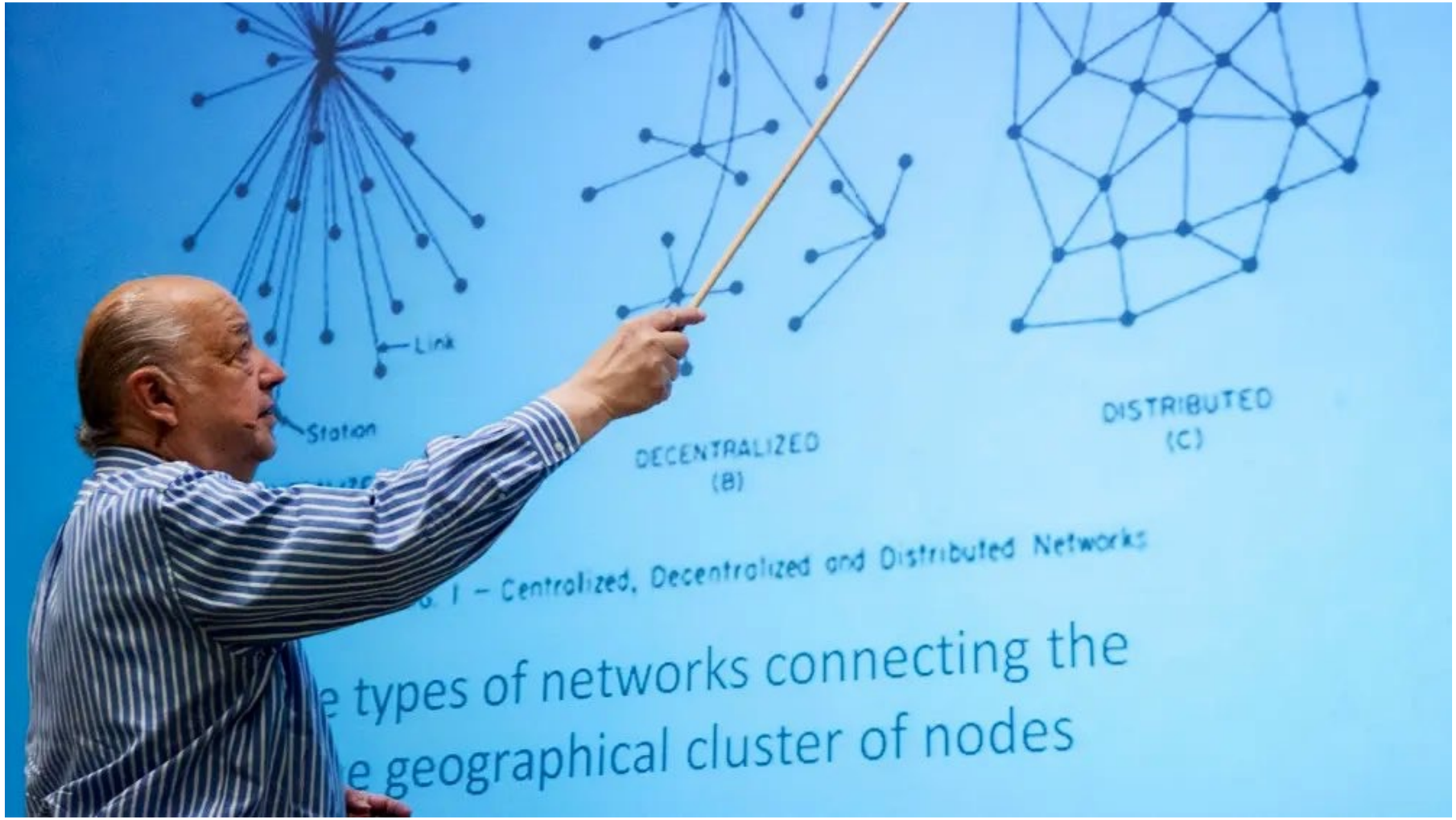


5. How many nodes are “hubs”?

6. How many nodes have only a few links?

7. Does some low k node ever become a hub? How often?

Network Robustness

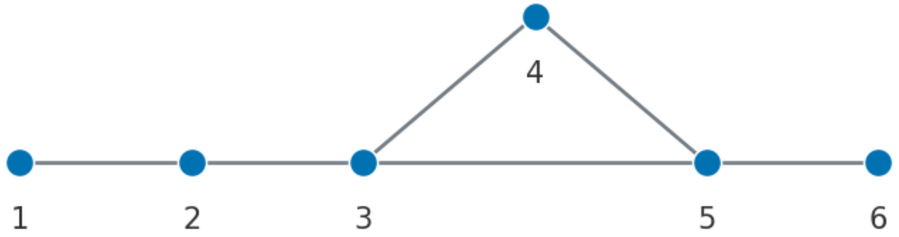


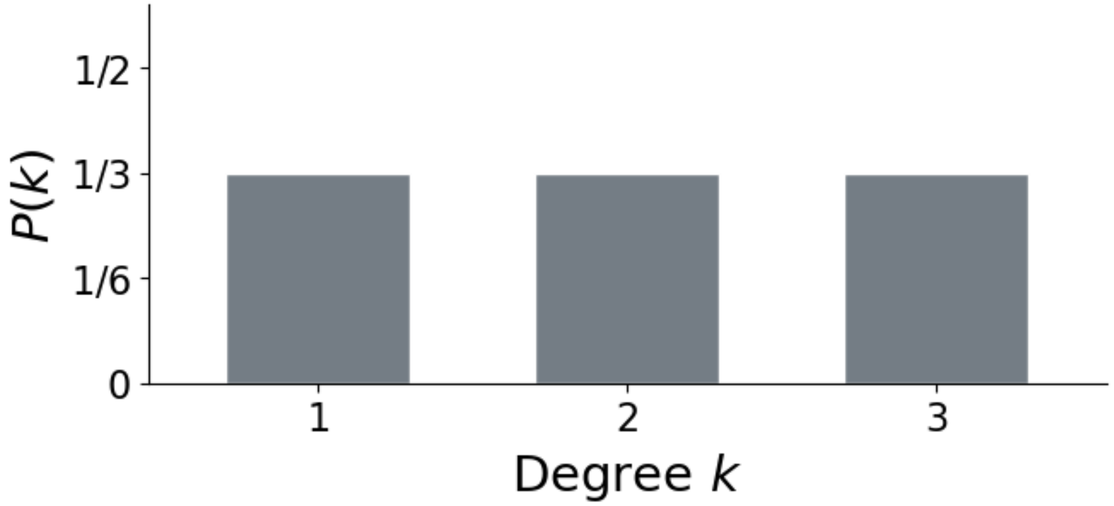
Paul Baran presents his work at a RAND Alumni Association event on July 25, 2009

Why do hubs matter for robustness?

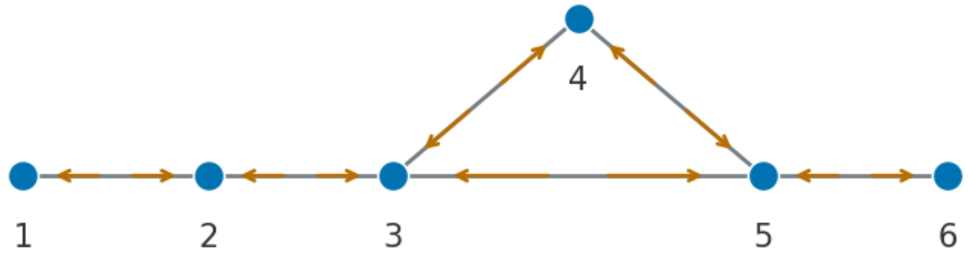
Following an edge reaches degree-k vertices with probability kP(k)/⟨k⟩

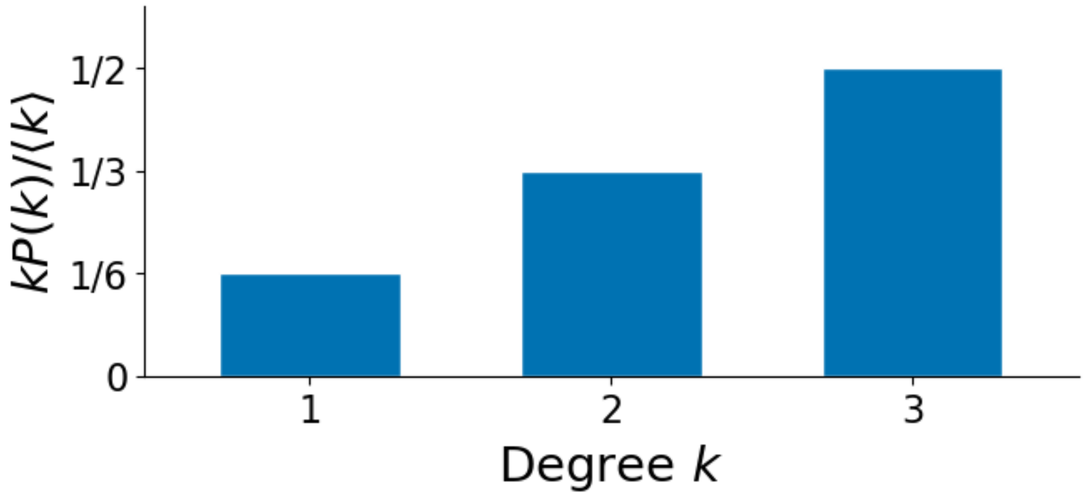
1) Choose a node uniformly





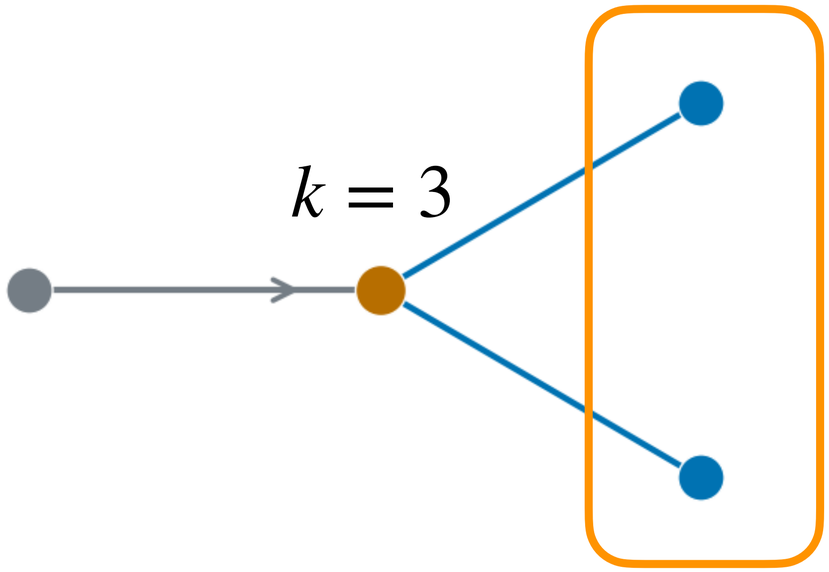
2) Reach a vertex along an edge





From degree to branching

High-degree vertices are more accessible ***and facilitate further exploration***



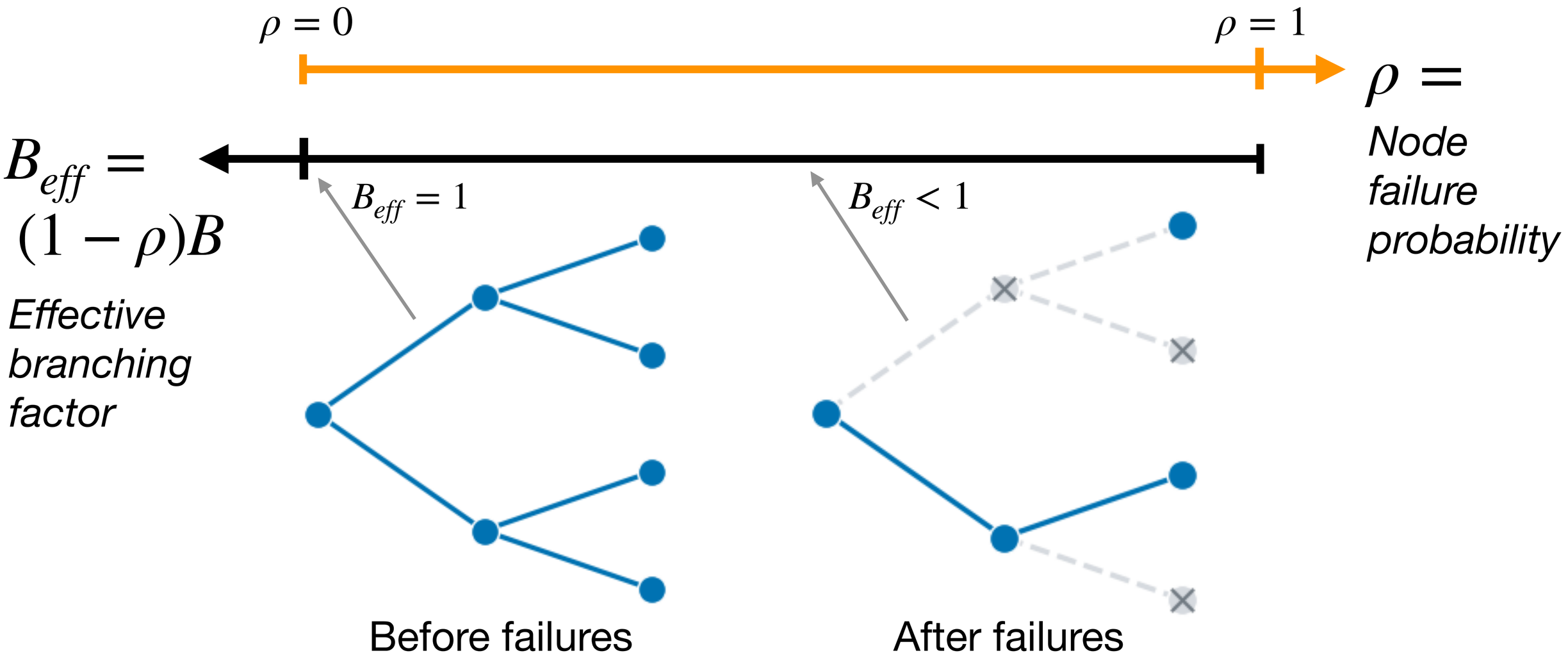
k-1 other neighbors with probability kP(k)/<k>

Branching = (other neighbors) x (edge-end probability)

$$B = \sum_k (k-1)\frac{kP(k)}{\langle k \rangle}$$

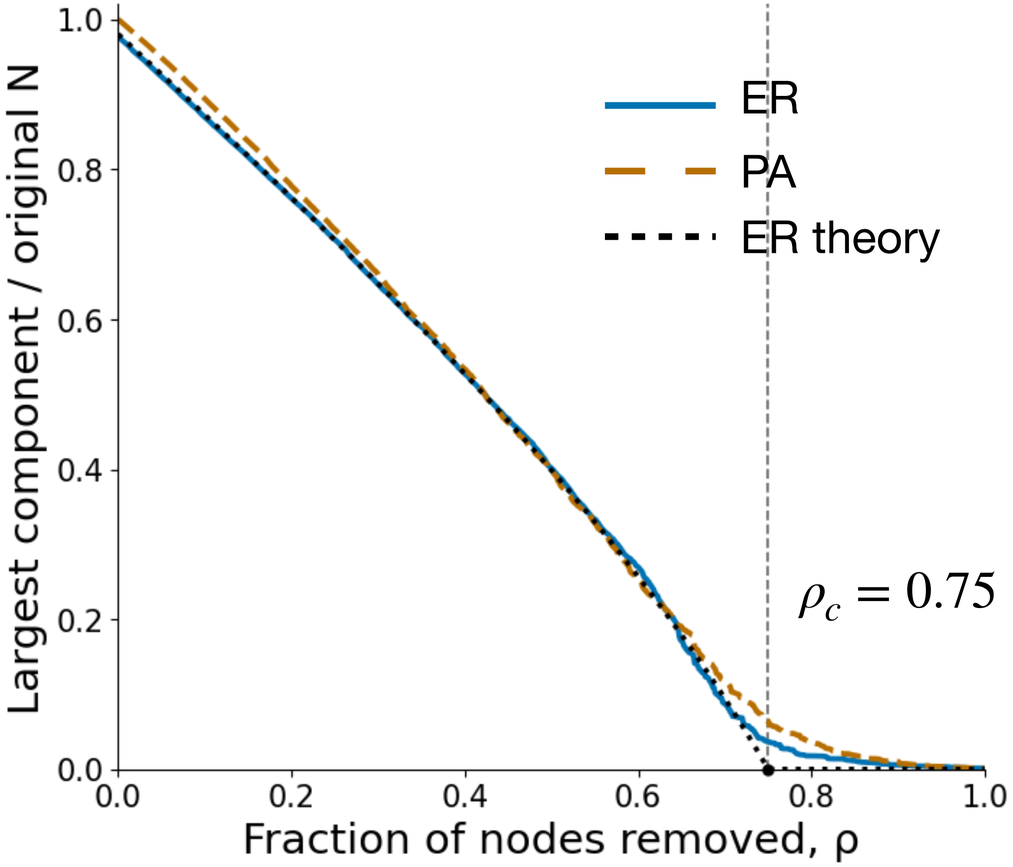
$$B = \frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle}$$

How do random failures reduce robustness?

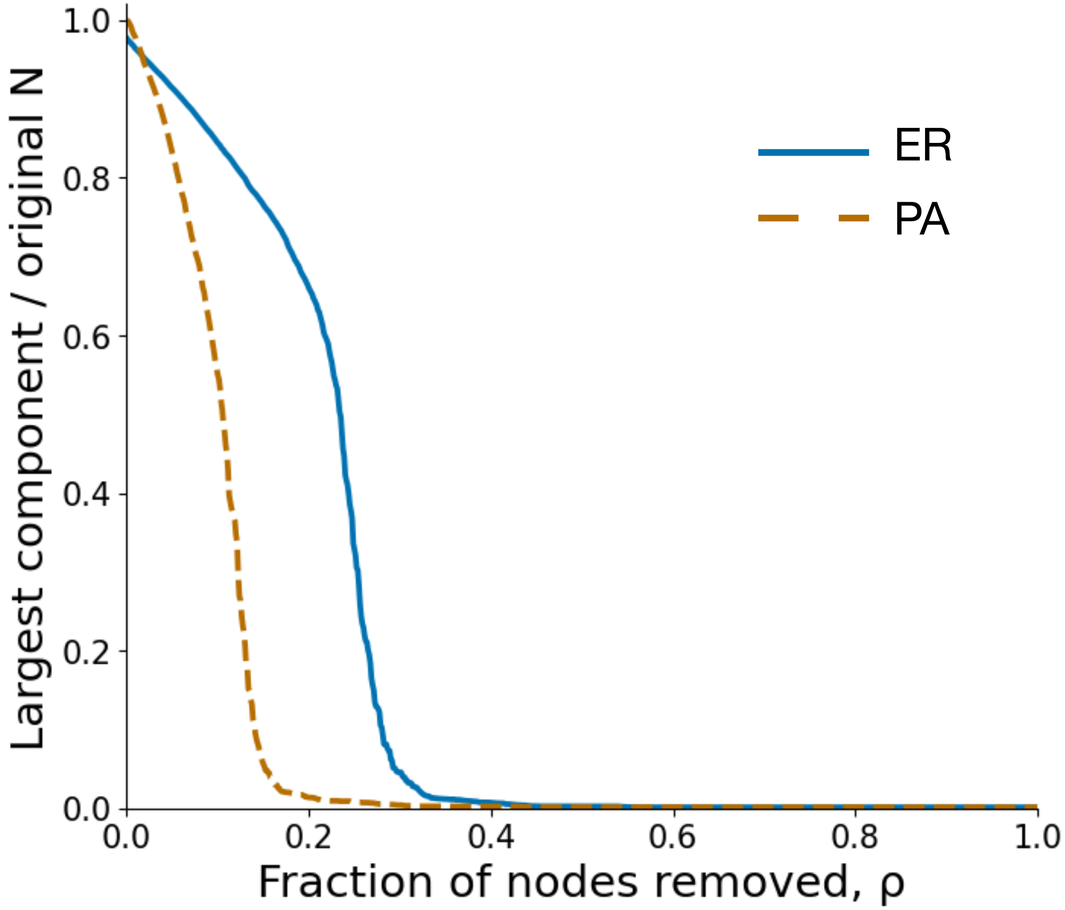


Random failures vs targetted attacks

Uniform random removal



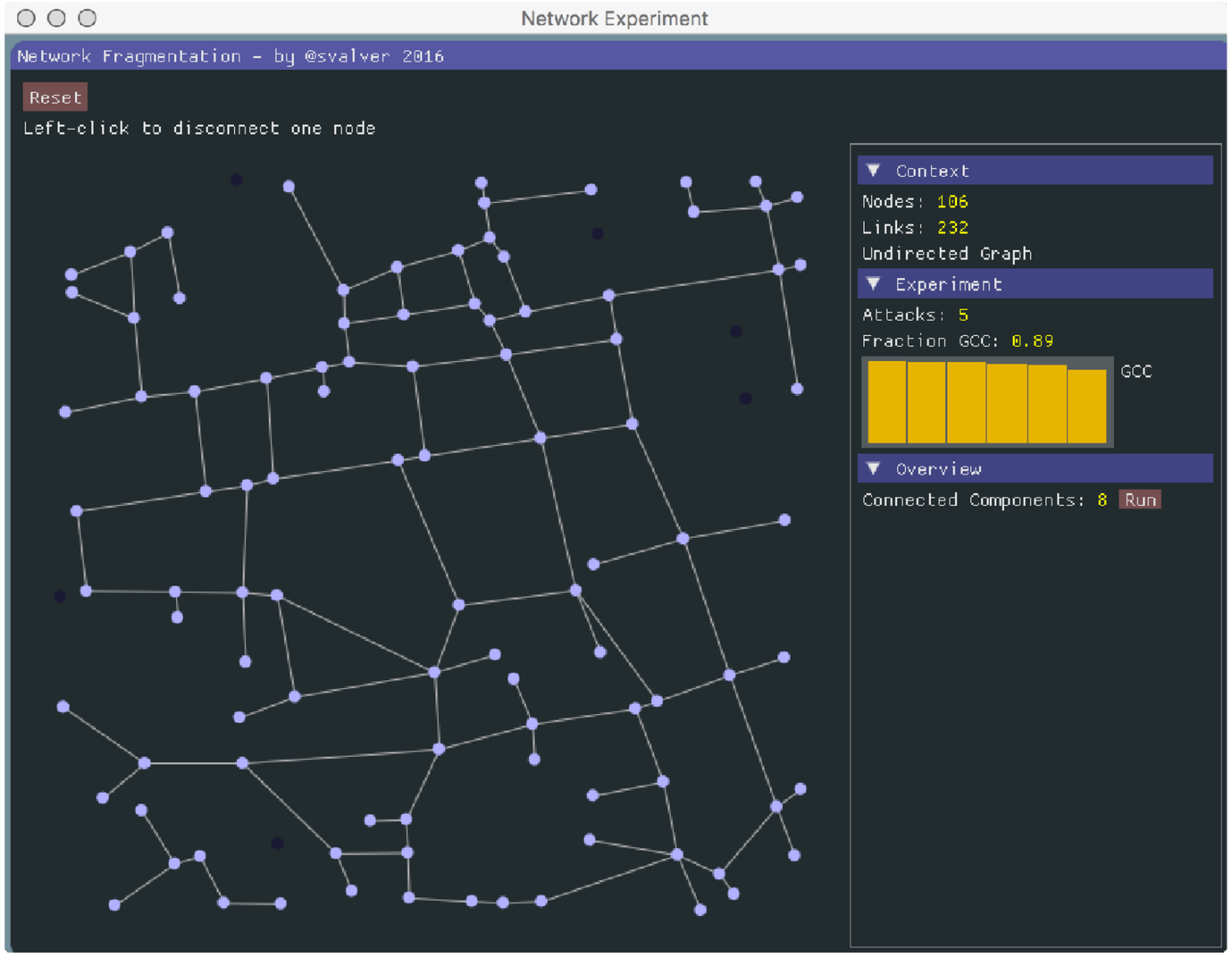
Targeted removal



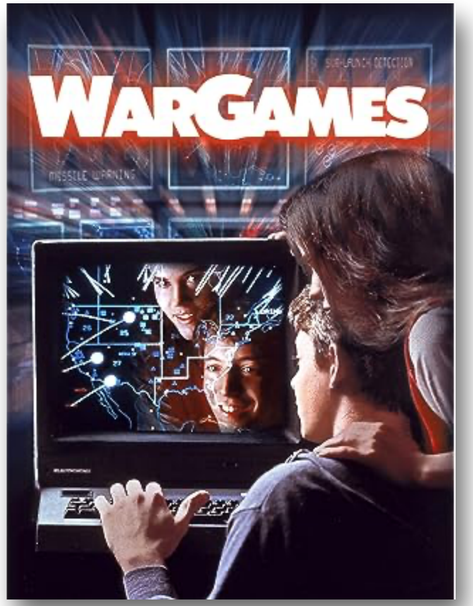
R. Albert, H. Jeong & L-A Barabási (2000) *Nature* **406:** 378-382

Activity: Random failures vs targeted attacks

https://tinyurl.com/3jkubj8j



8. If you wanted to shut down the network, how many nodes would you have to take out?



9. Are collapses quick or gradual?

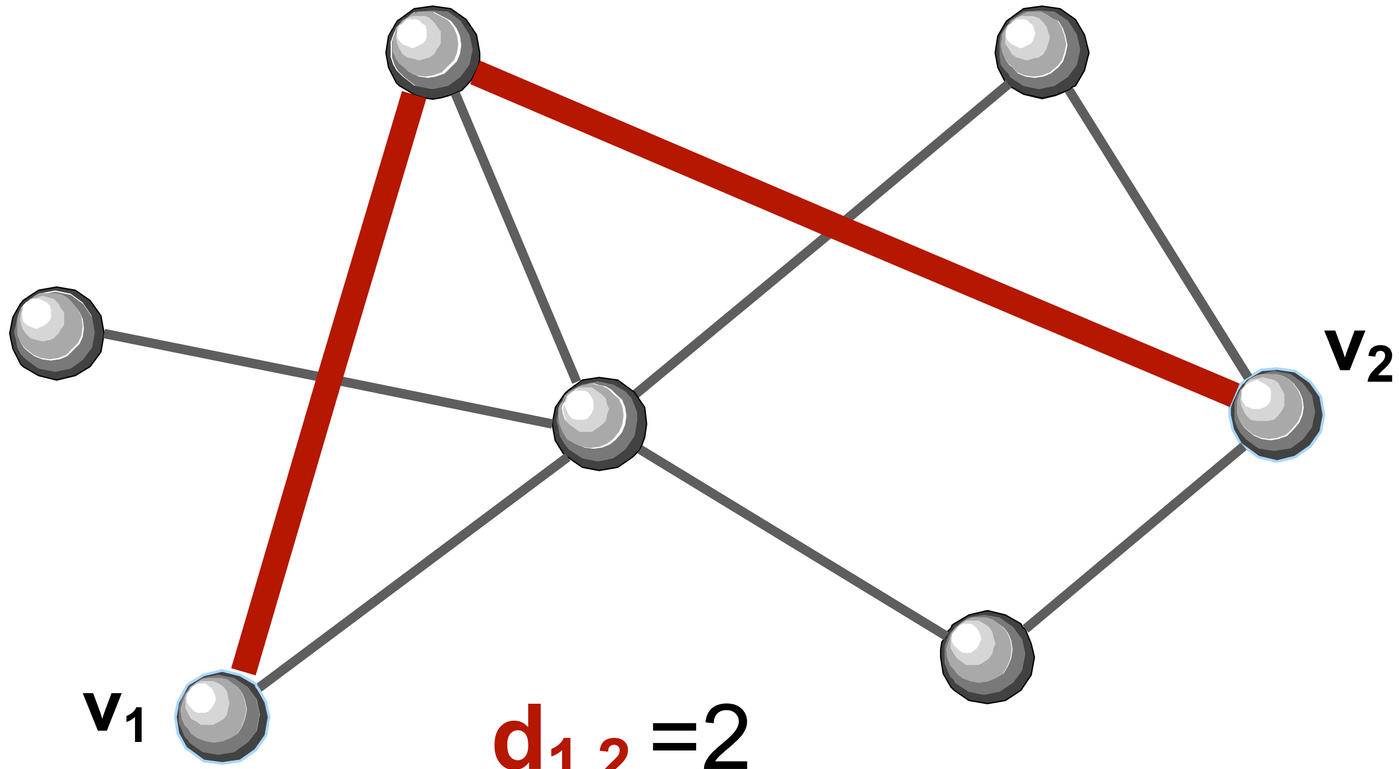
10. Can you predict the breaking point? Is this network fragile or robust? Why?

Network Efficiency:
Hubs, Connectors & Paths

Definitions

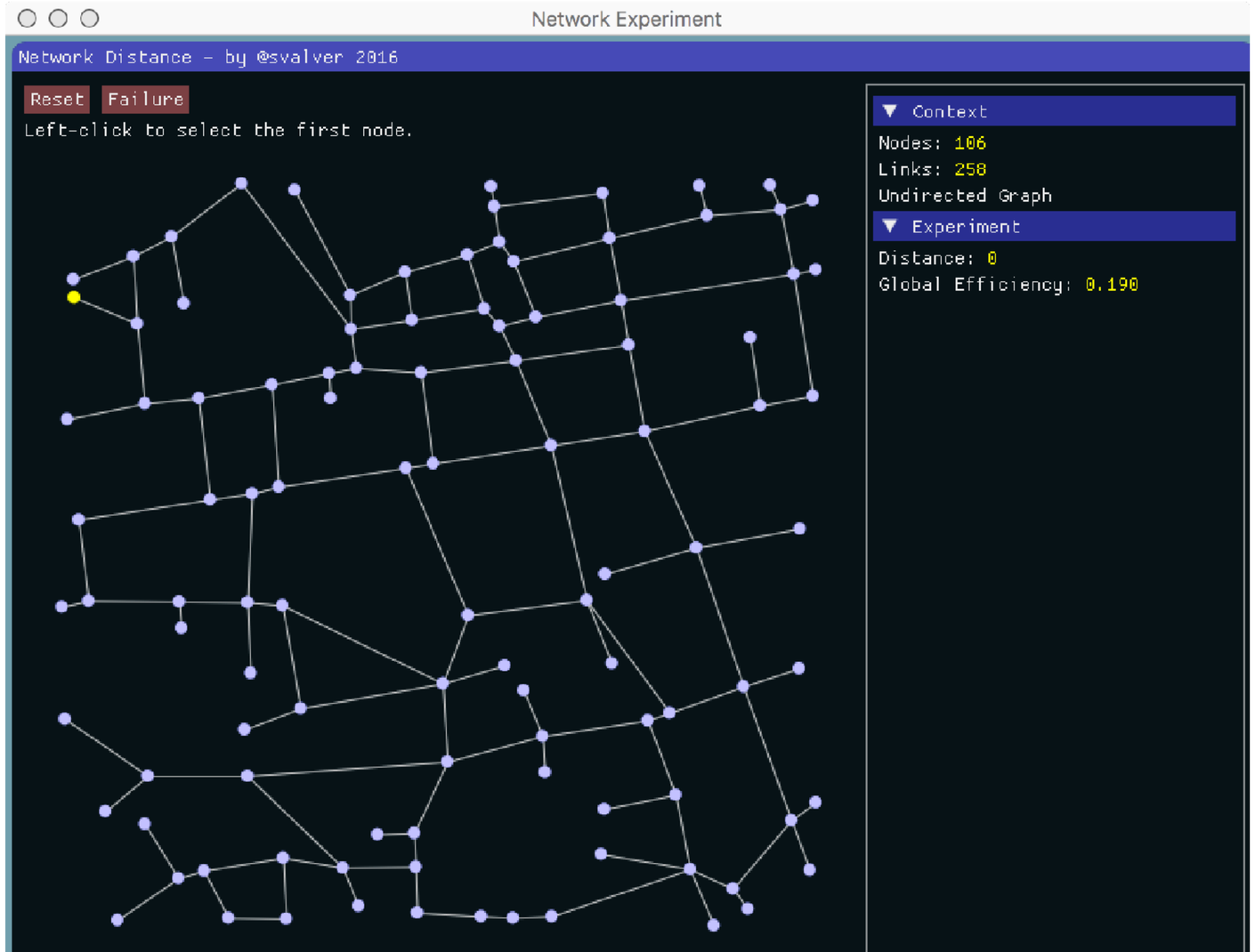
- Path Length
- Power of Matrices
- Geodesic Path
- Diameter
- Components
- Global Efficiency

Path Length



Activity: Shortest Paths

https://tinyurl.com/587wsvwj



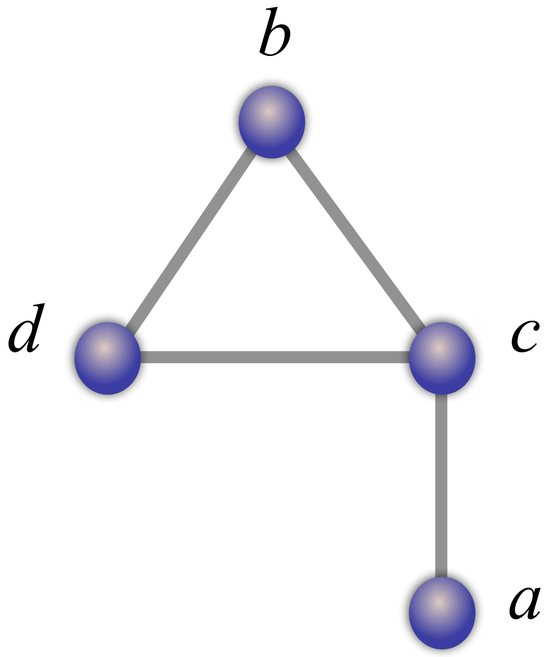
Click on a pair of nodes to see the shortest path connecting them.

Click the ‘Failure’ button repeatedly to remove nodes at random.

Describe the dynamical evolution of the shortest path under random failures.

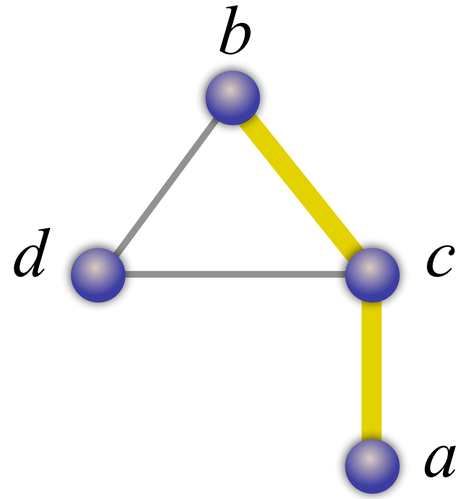
Network distance: path length

Length *of a path is the number of edges traversed along a path (not the nodes).*



$$A = \begin{array}{c} \\ a \\ b \\ c \\ d \\ \\ a \quad b \quad c \quad d \end{array} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

Network Distance



Power Matrices

$$A^2 = AA$$

$$A^2 = \begin{matrix} a \\ \\ \\ \\ \end{matrix} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix} \begin{matrix} \\ \\ \\ \\ \end{matrix} \begin{matrix} b \\ \\ \\ \\ \end{matrix} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix} = \begin{matrix} \\ \\ \\ \\ \end{matrix} \begin{matrix} N^2_{ab} \\ \\ \\ \\ \end{matrix} \begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 1 \\ 0 & 1 & 3 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}$$

Network Distance

Number of walks of given length

Number of length-*2* walks from *i* to *j*:

$$N_{ij}^{(2)} = \sum_{k=1}^{N} A_{ik} A_{kj} = [A^2]_{ij}$$

Number of length-*3* walks from *i* to *j*:

$$N_{ij}^{(3)} = \sum_{k=1}^{N} \sum_{l=1}^{N} A_{ik} A_{kl} A_{lj} = [A^3]_{ij}$$

Number of length-*r* walks from *i* to *j*:

$$N_{ij}^{(r)} = [A^r]_{ij}$$

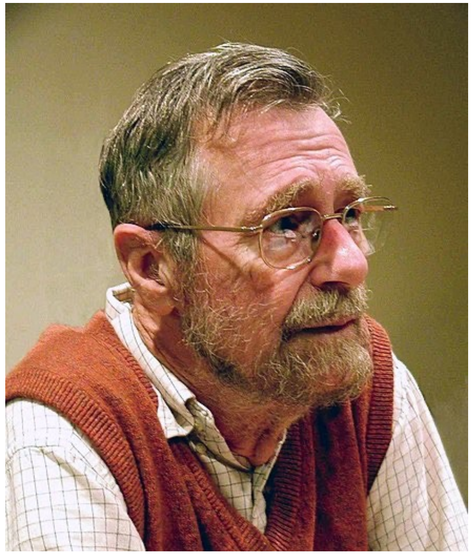
Network Distance

A geodesic path (or **shortest path**) is a path through a network between two vertices such that no shorter path exists.

The **shortest path distance *d(i,j)*** is the length of the shortest path, i.e., the minimal value of ***r*** such that:

$$\left[A^r\right]_{ij} > 0$$

In practice, there are more efficient ways of calculating shortest distances in a graph (e.g., **Dijkstra's Algorithm**).

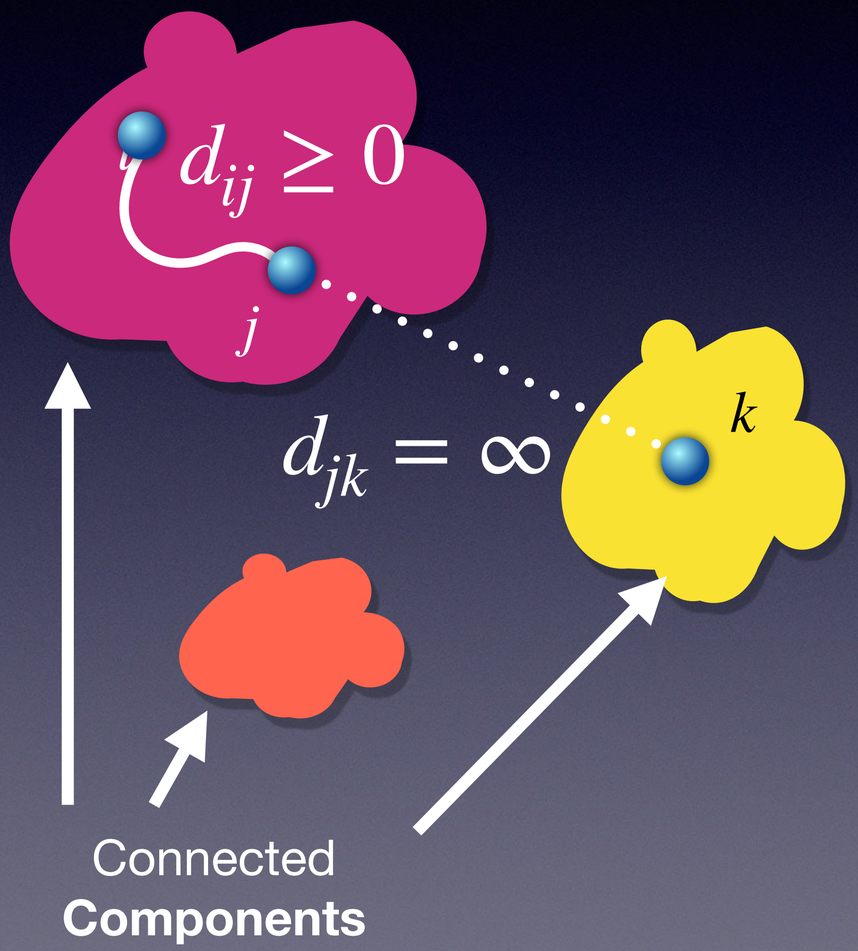


Edsger W. Dijkstra
(1930-2002)
Turing Award (1972)

Network Efficiency

$$d_{ij} = \min\{r \geq 1 : (A^r)_{ij} > 0\}$$

$$E_{glob} = \frac{1}{N(N-1)} \sum_{i \neq j} \frac{1}{d_{ij}}, \frac{1}{\infty} = 0$$

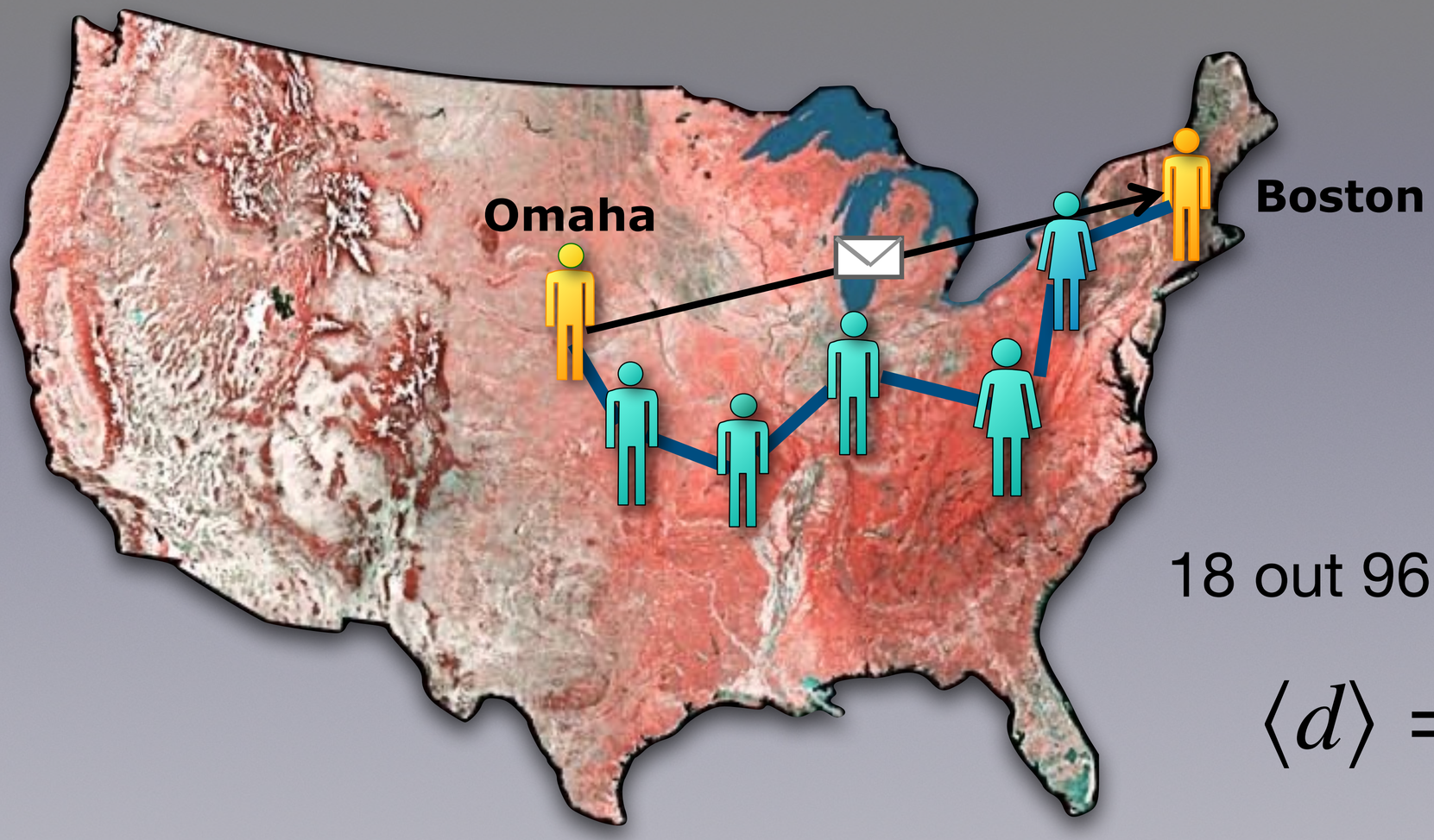


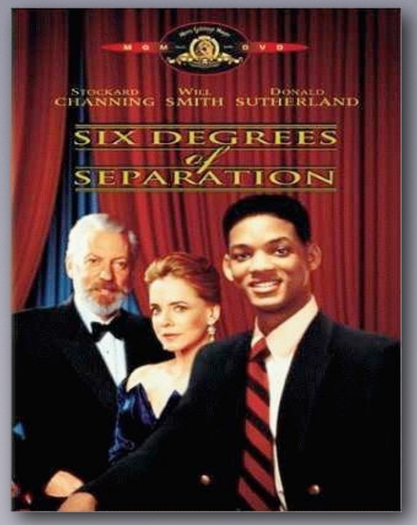
$$A = \begin{bmatrix} \blacksquare & 0 \\ 0 & \blacksquare \; \blacksquare \end{bmatrix}$$

Block diagonal form

Network Distance

Is this network large or small?





18 out 96 received

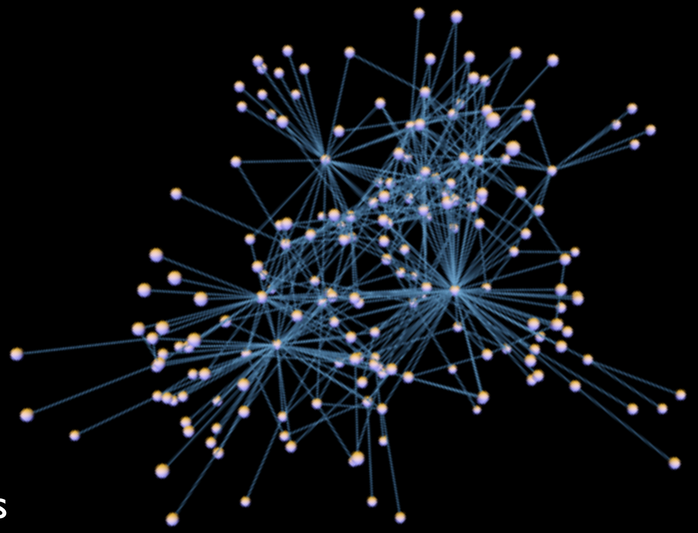
$$\langle d \rangle = 5.9$$

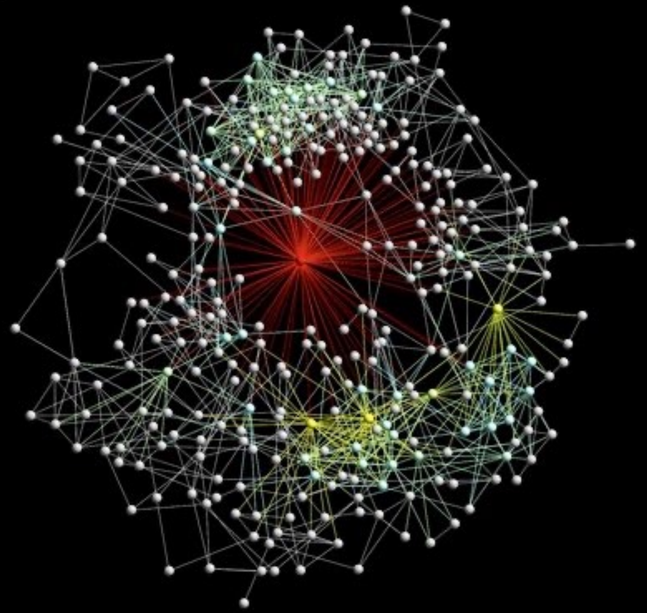
Stanley Milgram (1967)

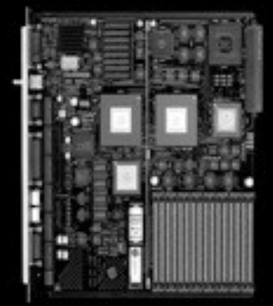
Between Order and Randomness

Why Many Networks are Small and yet Clustered?

Linguistic Networks

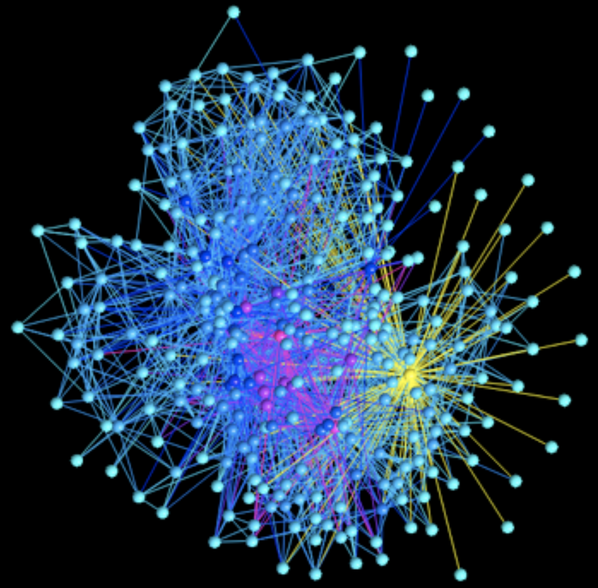






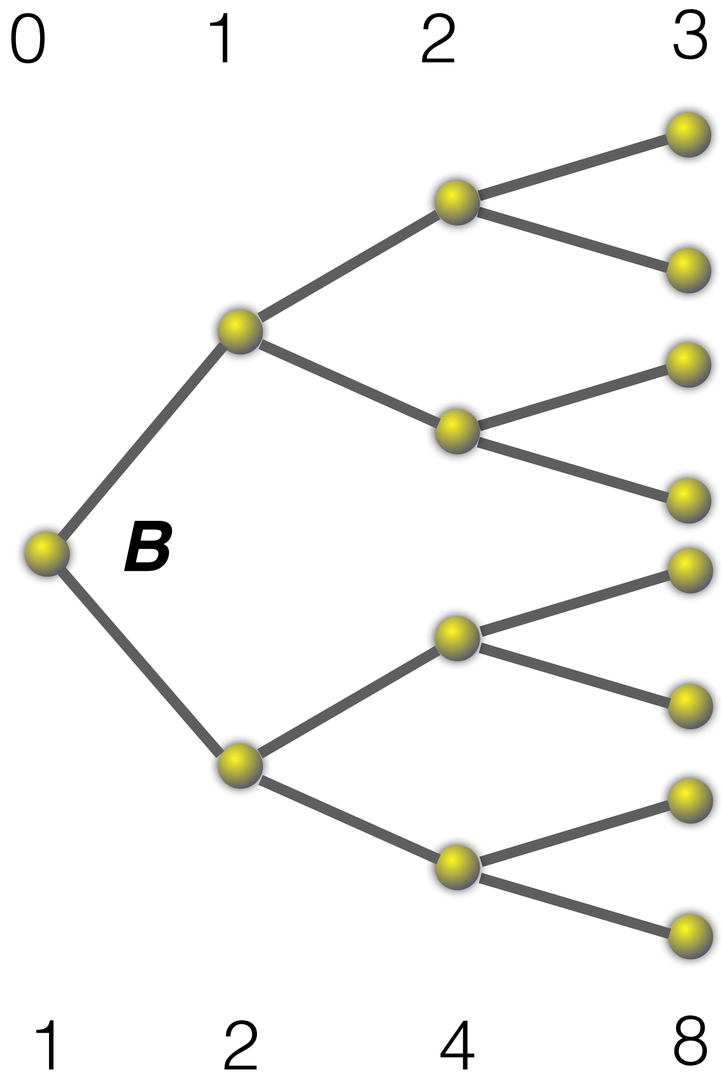
Electronic Circuits

Brain of a worm (*C. Elegans*)



Power grids

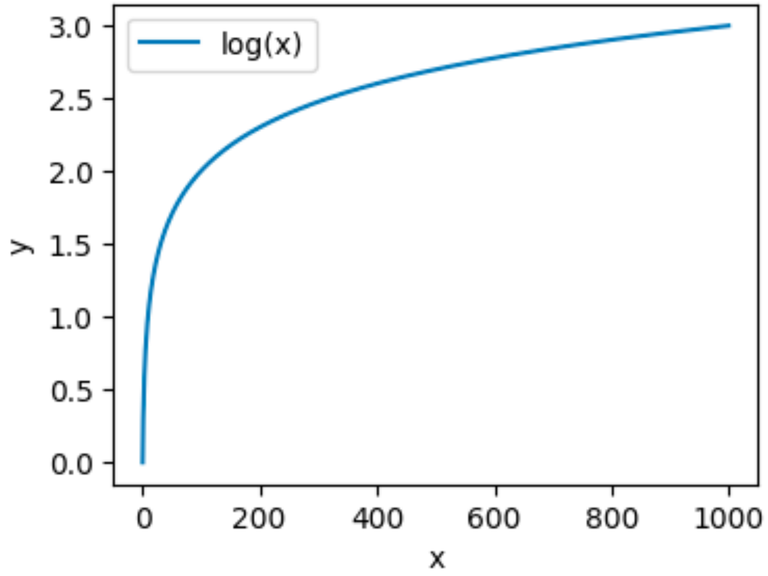
Average Path Length



$$N_d = B^d$$

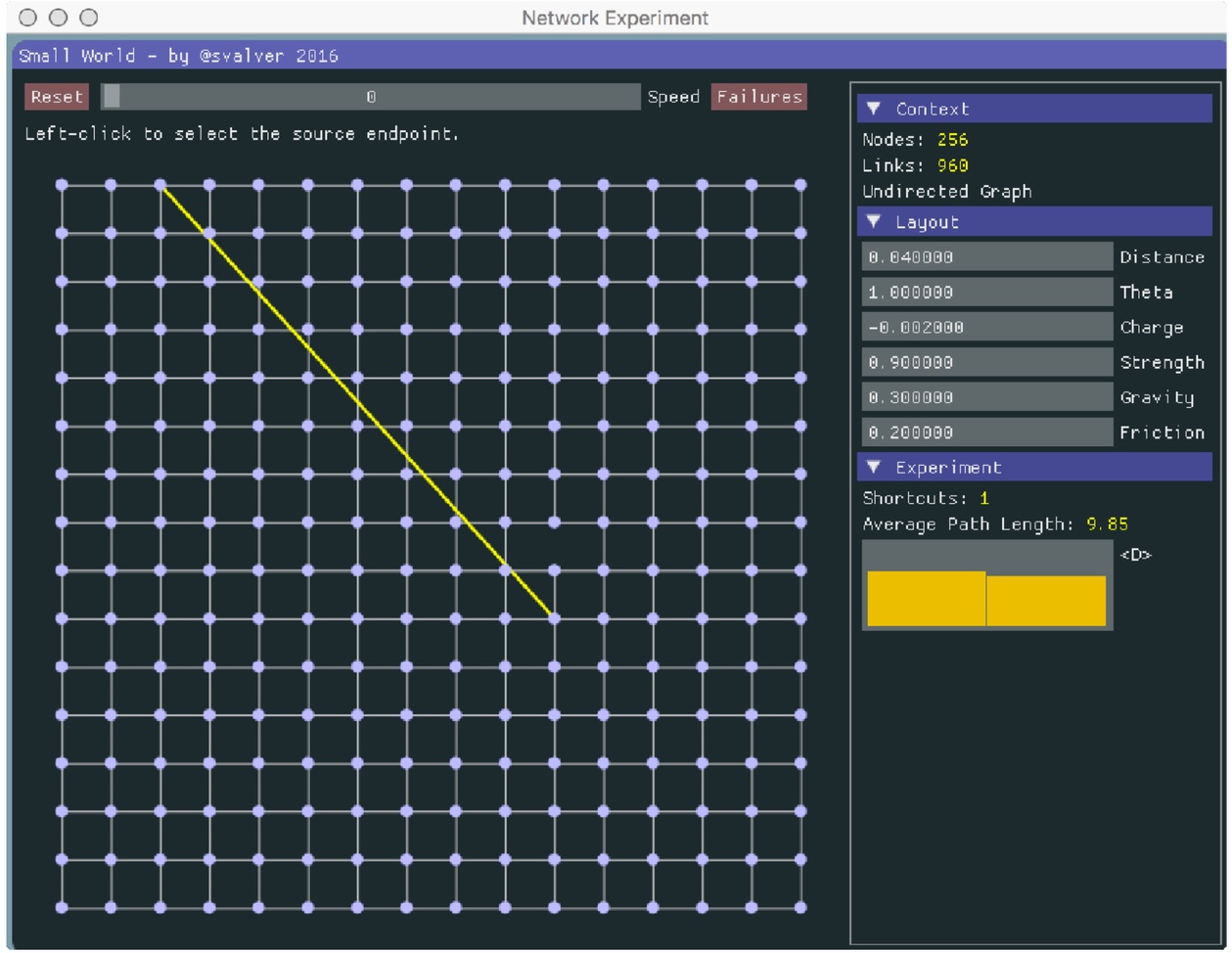
$$\log(N) = d\log(B)$$

$$\langle d \rangle \approx \frac{\log(N)}{\log(B)}$$

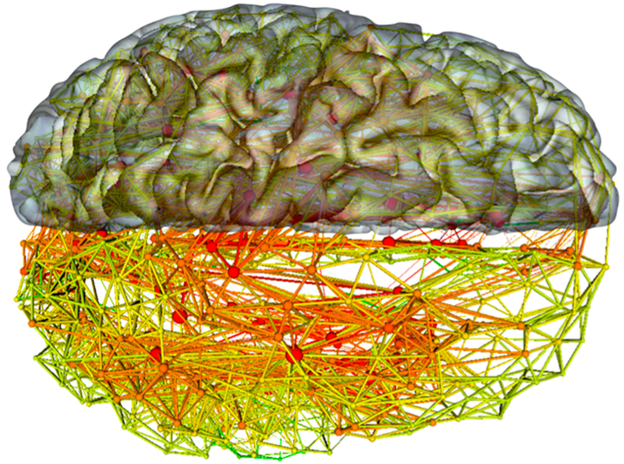


Activity: Small Worlds

https://tinyurl.com/yv5u4kpu



11. Which shortcuts reduce the average distance ?

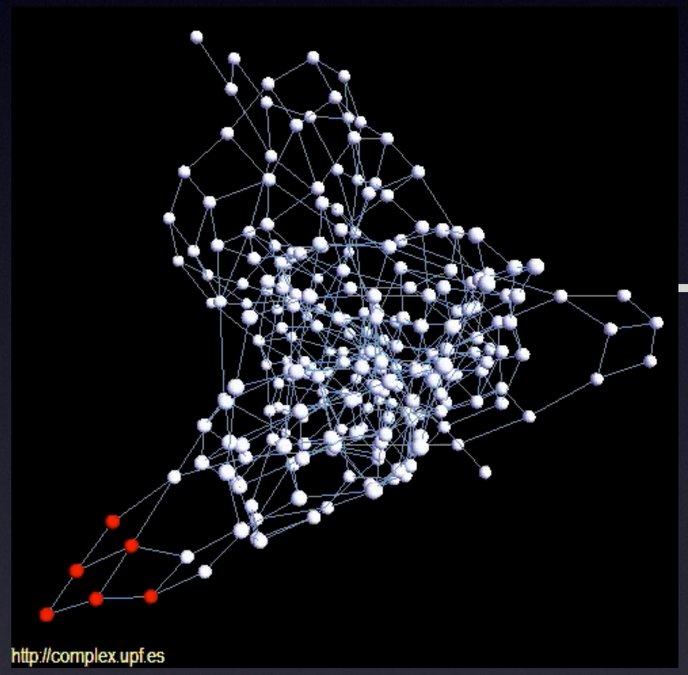


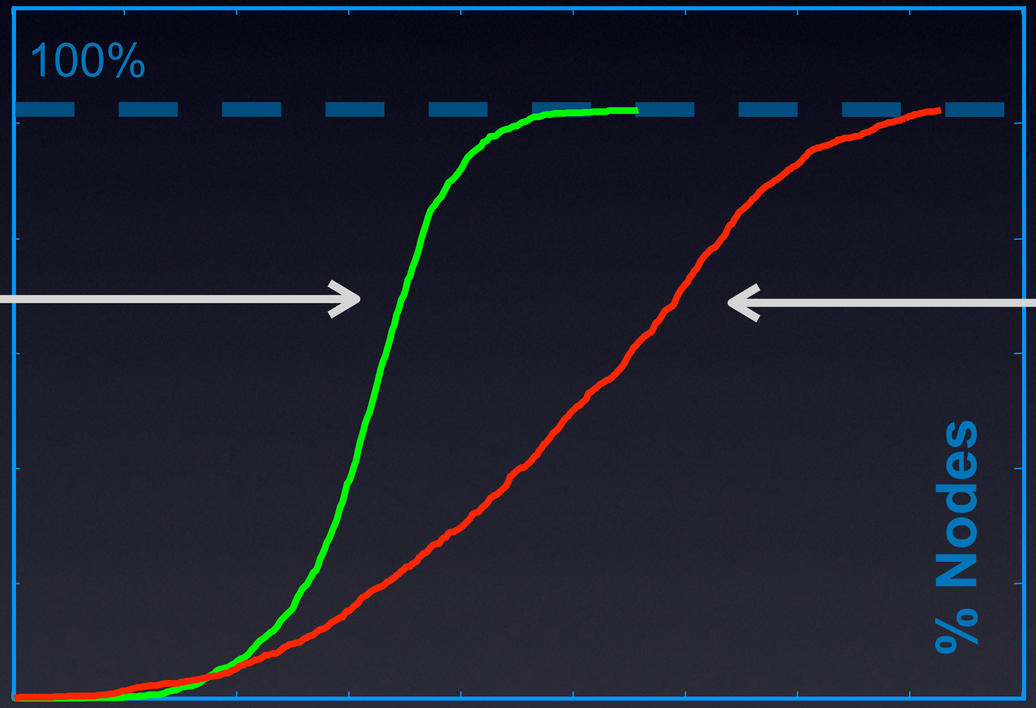
12. After completing 10 experiments, plot the (shortcuts, mean path length) curve. Can the distinction between good and poor networks be made?

Diffusion Processes

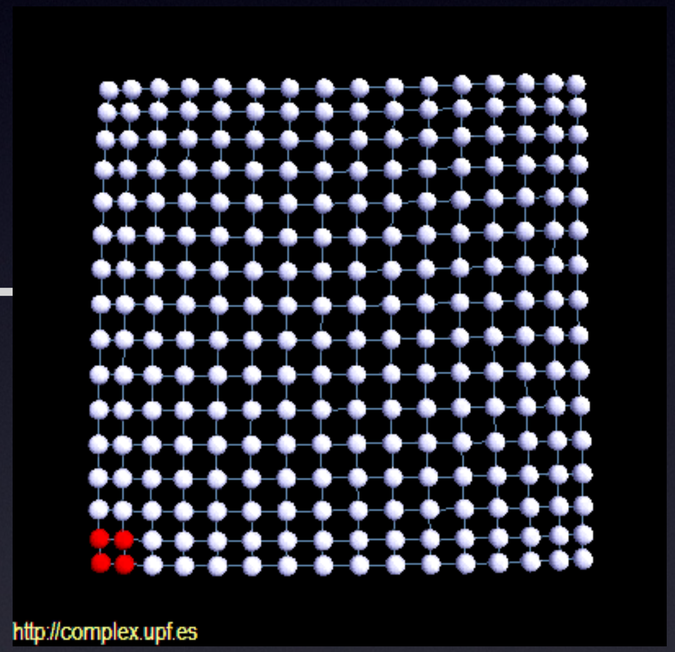
By defining a few long-distance links, diffusion may be accelerated

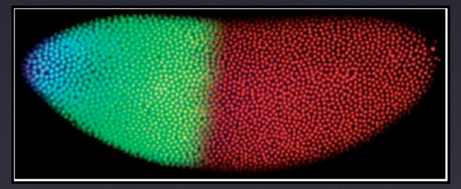
Small-World





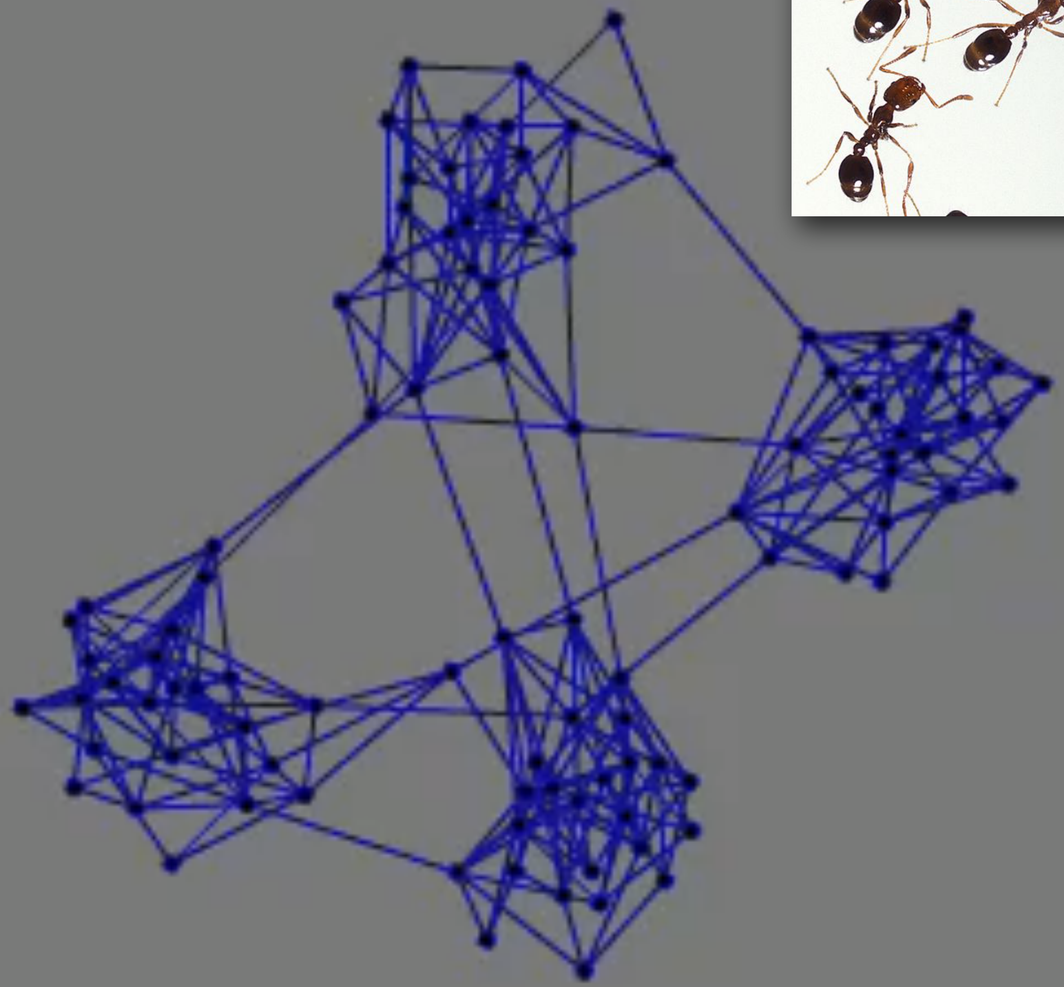
Lattice



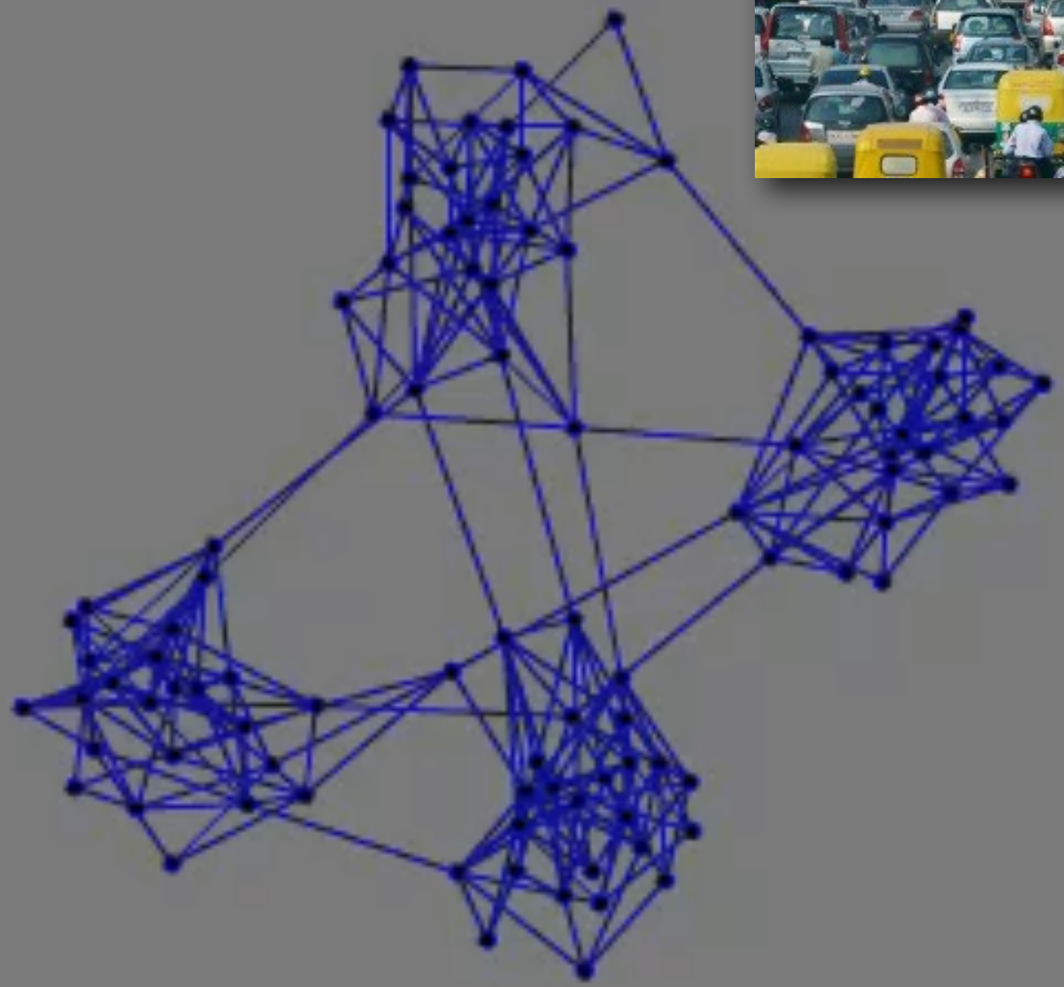


Structure-Function Relationship

Random Walk

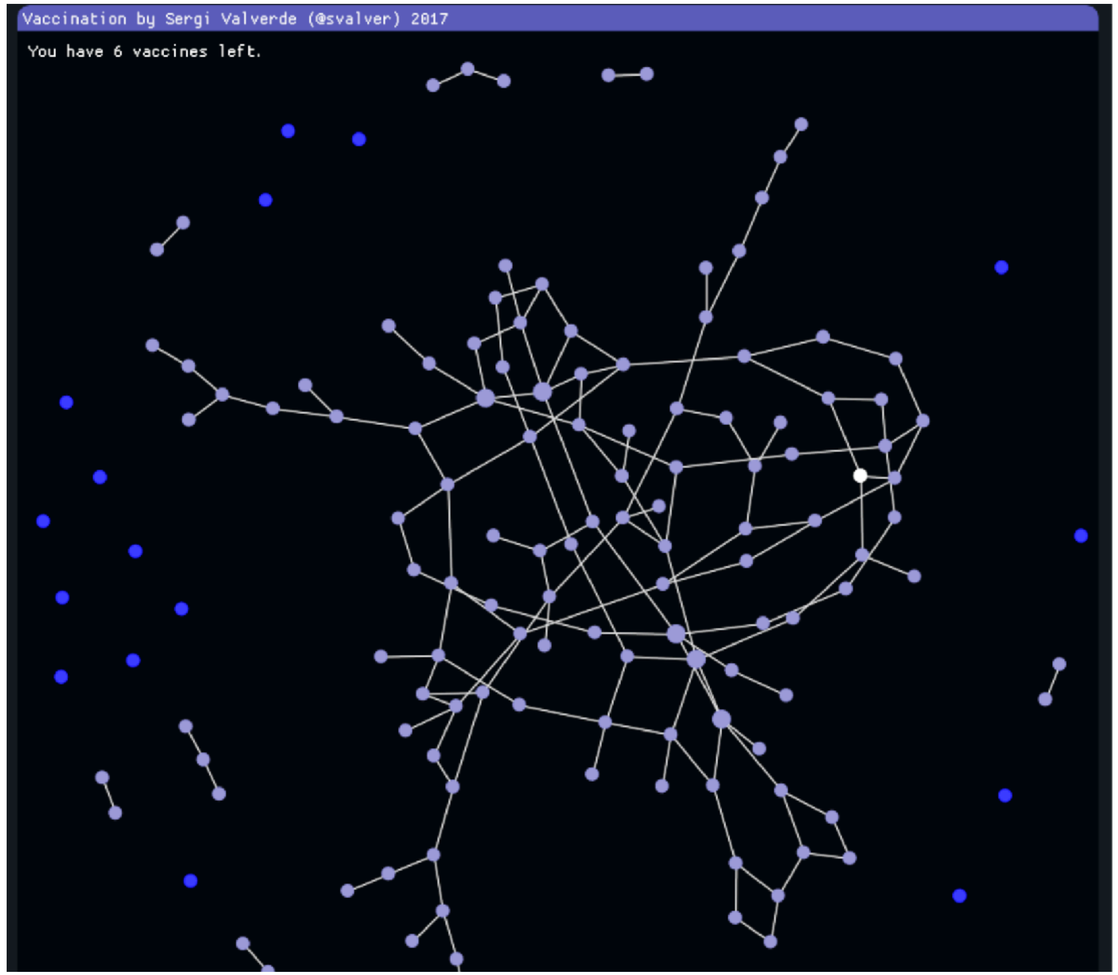


Shortest Path



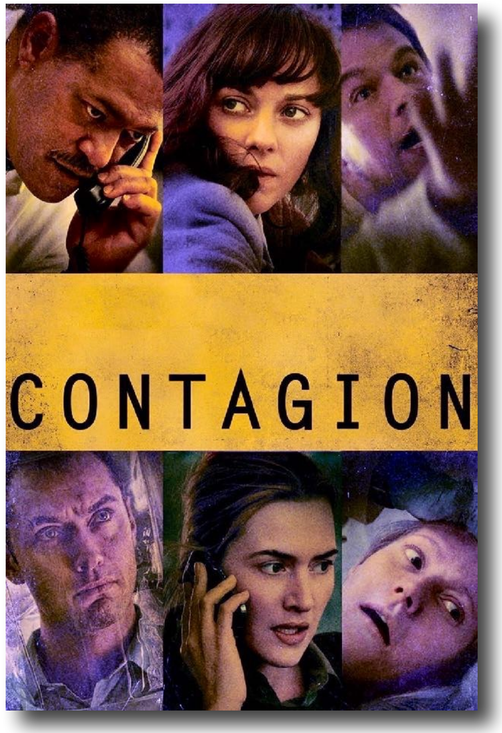
Vaccination Game

https://tinyurl.com/c42yx3pc



Can you control an epidemic?

Take action to prevent the spread of illness in various urban settings. After a small amount of vaccinations have been distributed, the epidemic continues to spread, and the players must act quickly to isolate everybody who could be sick.

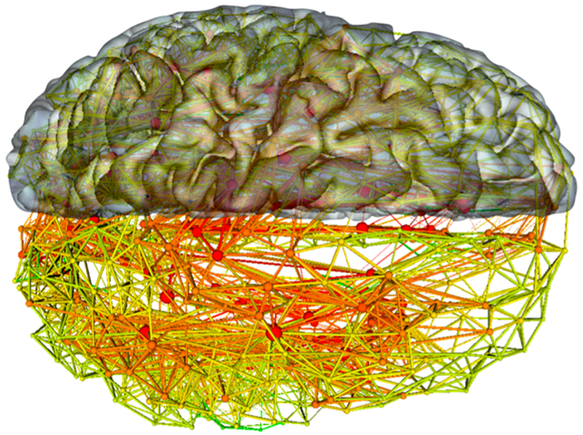


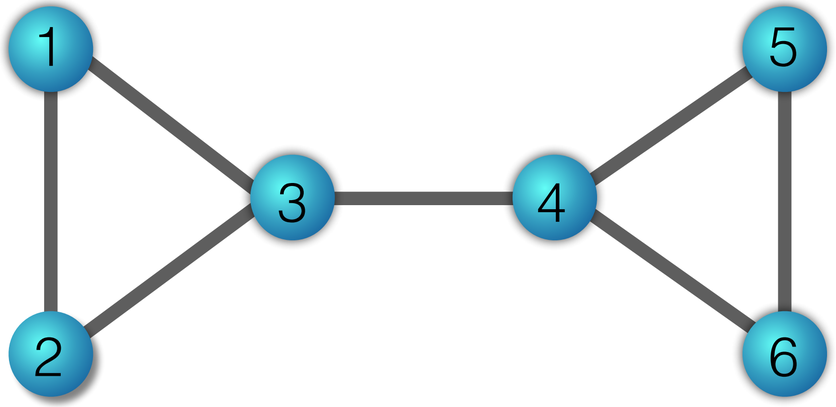
NOTE: This game was designed in 2017.

Modularity
Evolution & Tinkering

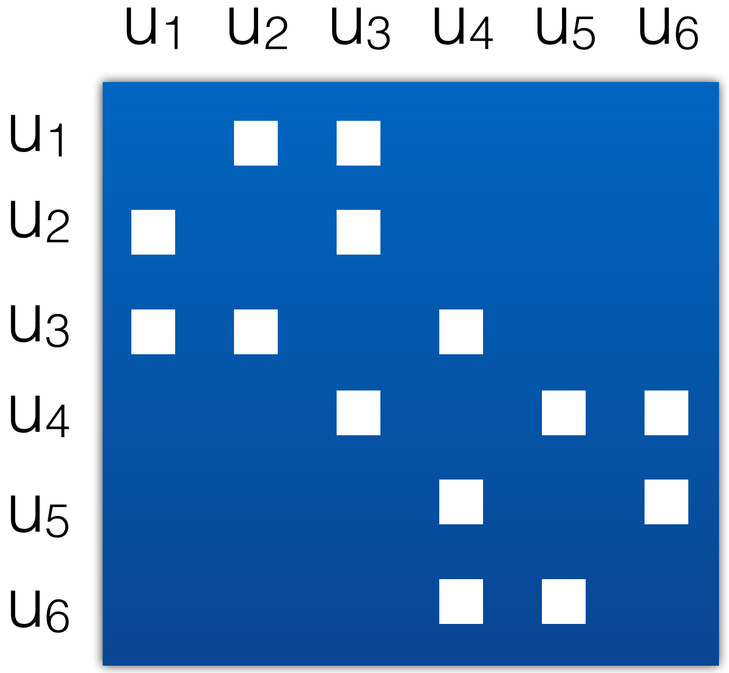
Definition

Modularity quantifies the degree to which nodes are clustered together and dependent on one another.





Network



Adjacency Matrix

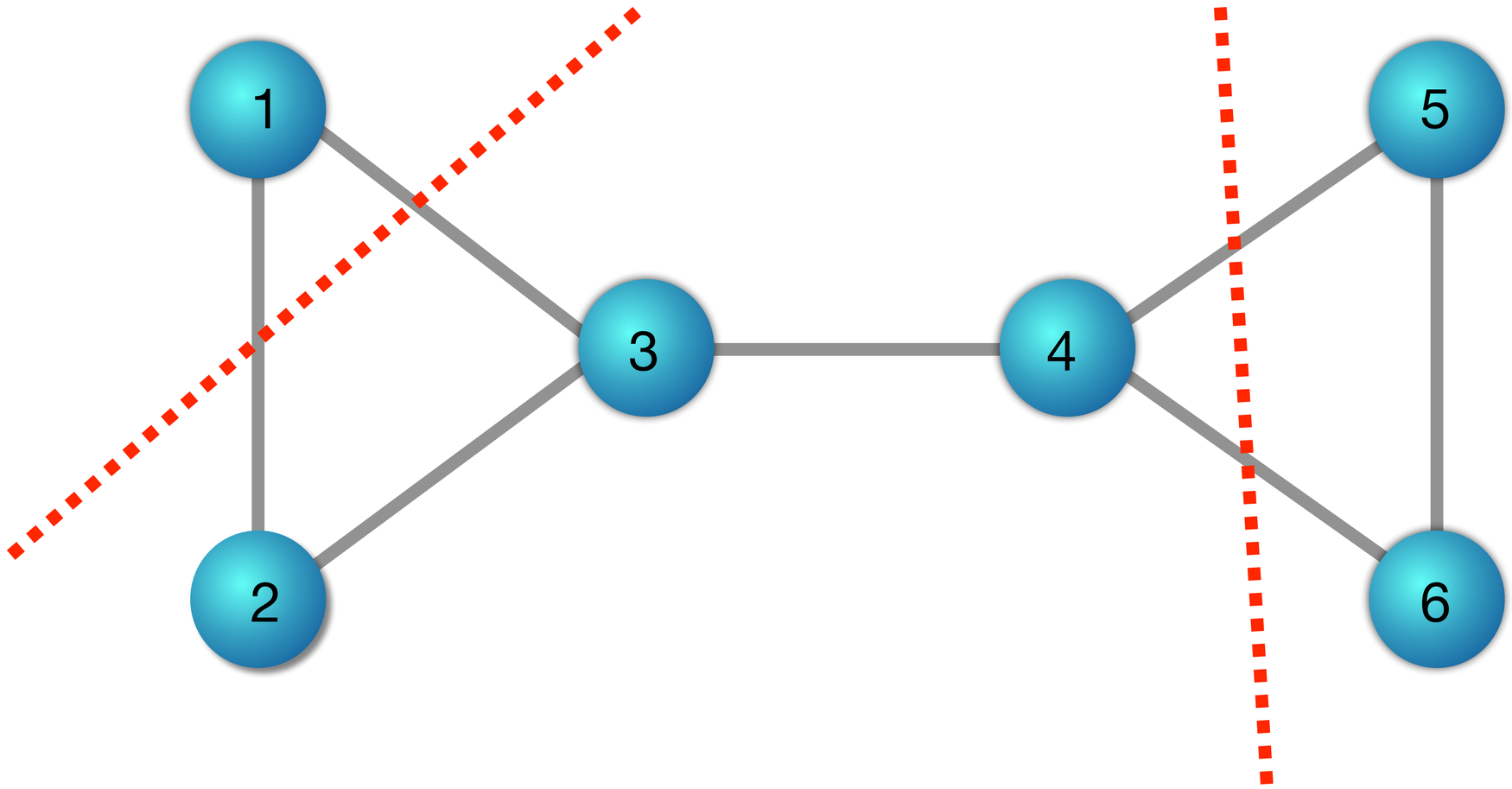
Newman & Girvan **Phys Rev E** 69, 026113 (2004)

(1) Divide up the network

(2) Calculate the modularity value (*Q*)

(3) Repeat until a solution is found

(1) Divide up the network



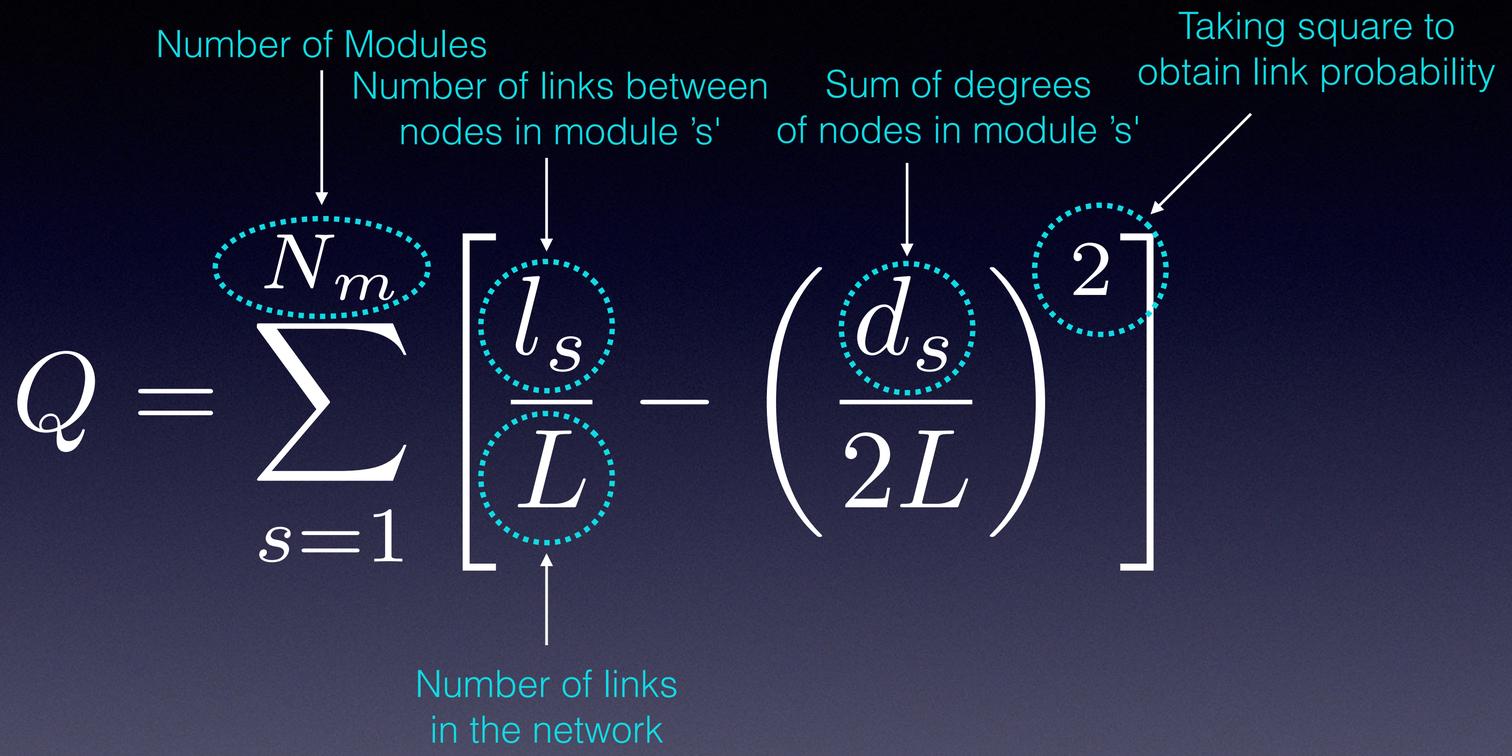
(2) Calculate the **modularity** value (Q)

$$Q = \sum$$

For each of the modules

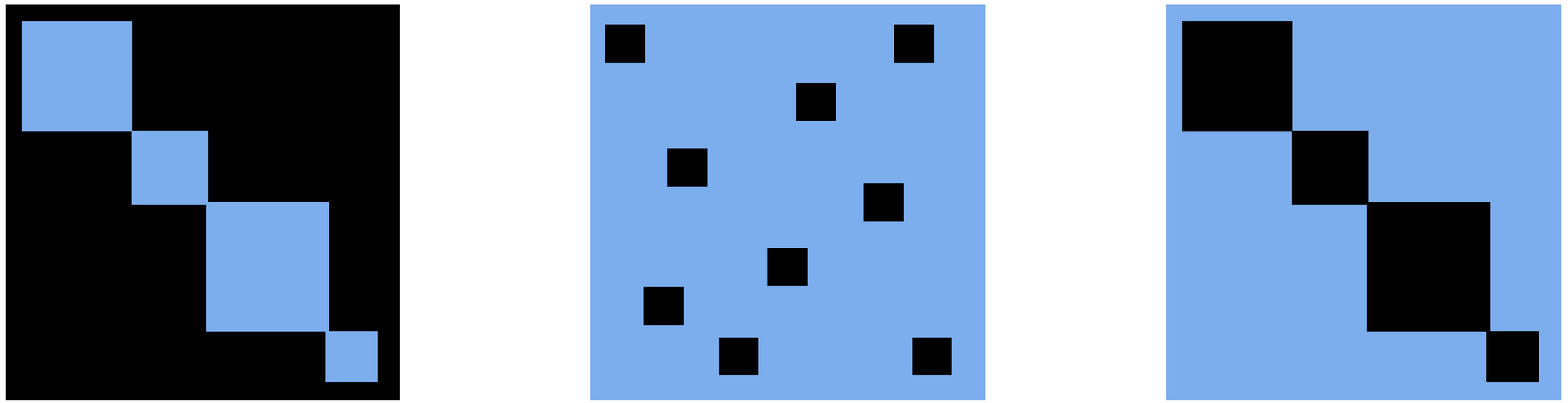
[**Observed** fraction of links in group − **Expected** fraction of links in group]

(2) Calculate the **modularity** value (Q)



Girvan and Newman **PNAS** 99:7821 (2002)

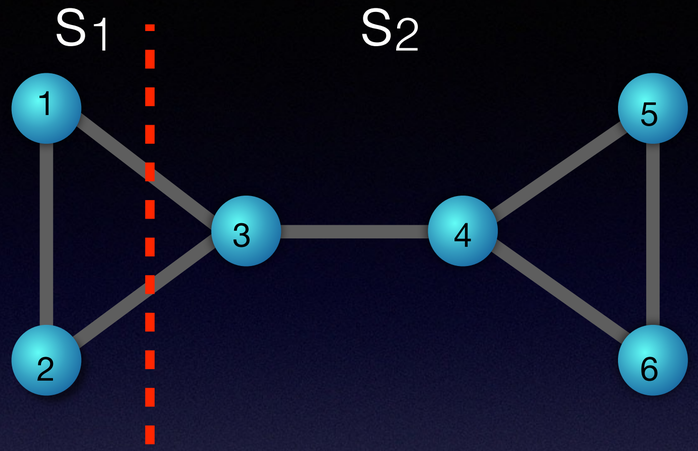
$Q < 0$ $Q = 0$ $0 < Q \leq 1$



ANTI-MODULAR RANDOM MODULAR

$$Q_{\text{mod}} = \frac{1}{2L} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2L} \right) \mathbf{1}[g_i = g_j] = \sum_{s} \left[\frac{l_s}{L} - \left(\frac{d_s}{2L} \right)^2 \right]$$

Example (1/2)



$$Q = \sum_{s=1}^{N_m} \left[\frac{l_s}{L} - \left(\frac{d_s}{2L} \right)^2 \right]$$

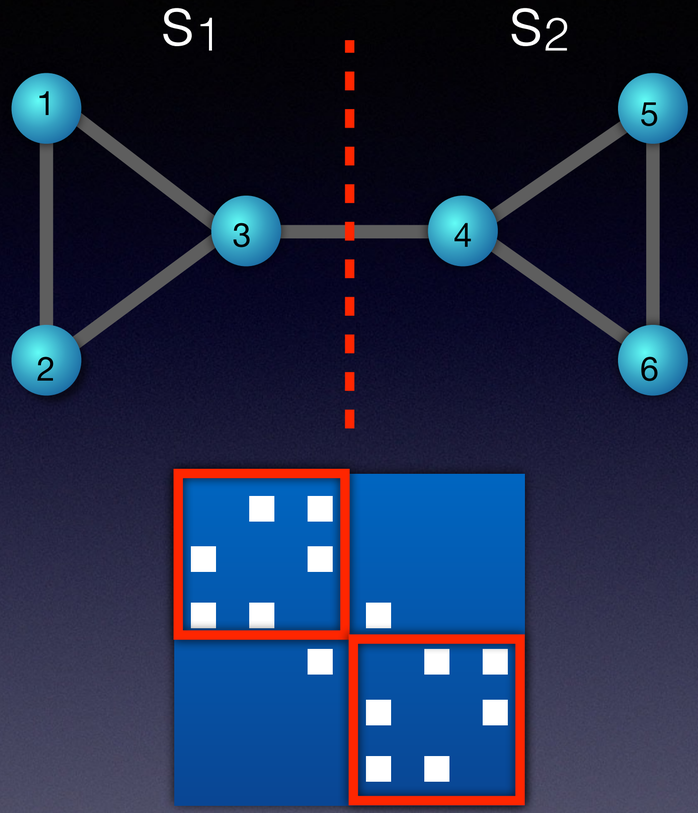
$$Q_{s_1} = \frac{1}{7} - \left(\frac{4}{14} \right)^2 = 0.06$$

$$Q_{s_2} = \frac{4}{7} - \left(\frac{10}{14} \right)^2 = 0.06$$

$$Q = Q_{s_1} + Q_{s_2} = 0.12$$

$$L = 7 \ , (l_1, d_1) = (1,4) \ , (l_2, d_2) = (4,10)$$

Example (2/2)



$$Q = \sum_{s=1}^{N_m} \left[\frac{l_s}{L} - \left(\frac{d_s}{2L} \right)^2 \right]$$

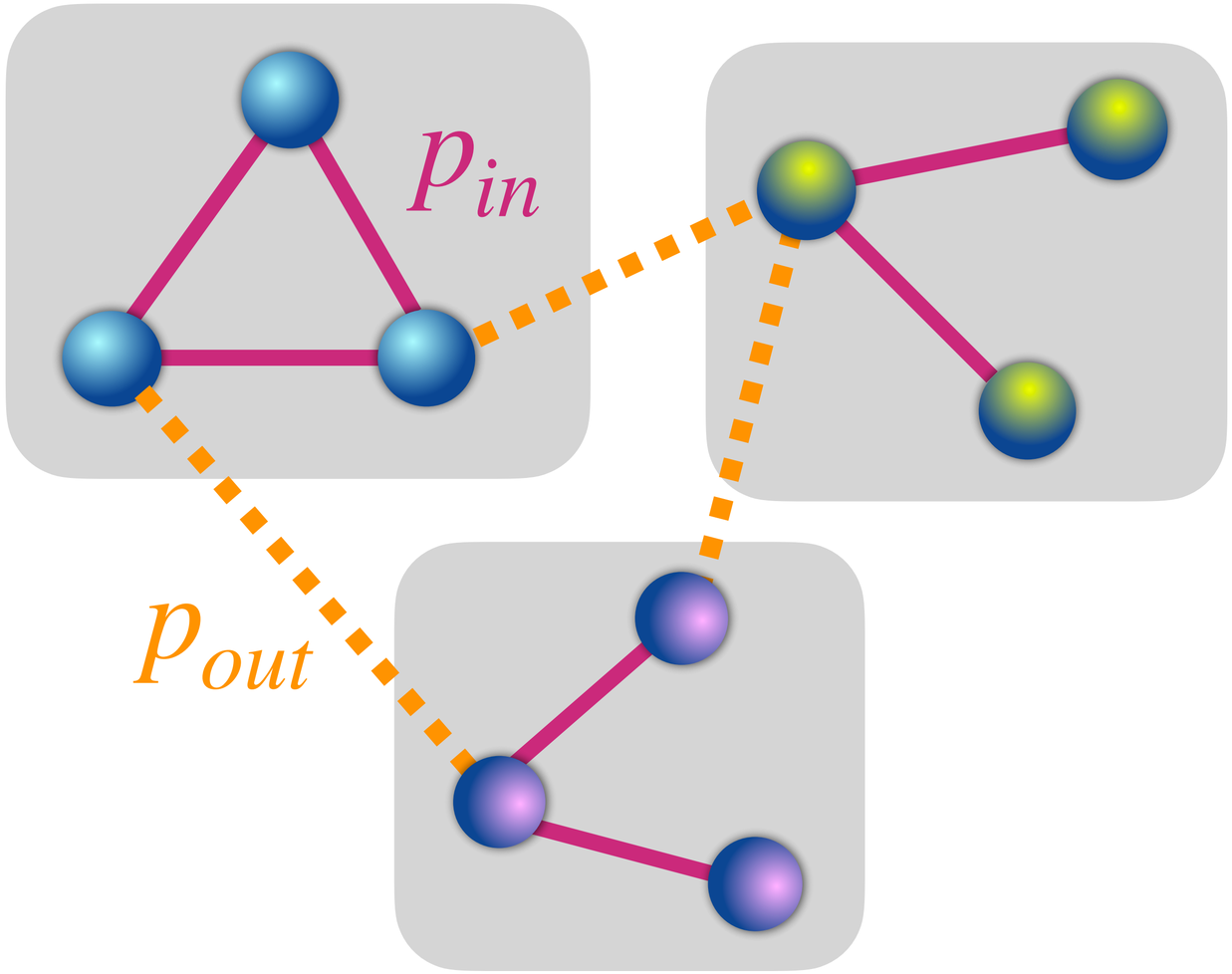
$$Q_{s_1} = \frac{3}{7} - \left(\frac{7}{14} \right)^2 = 0.18$$

$$Q_{s_2} = Q_{s_1} = 0.18$$

$$Q = Q_{s_1} + Q_{s_2} = 0.36 > 0.12$$

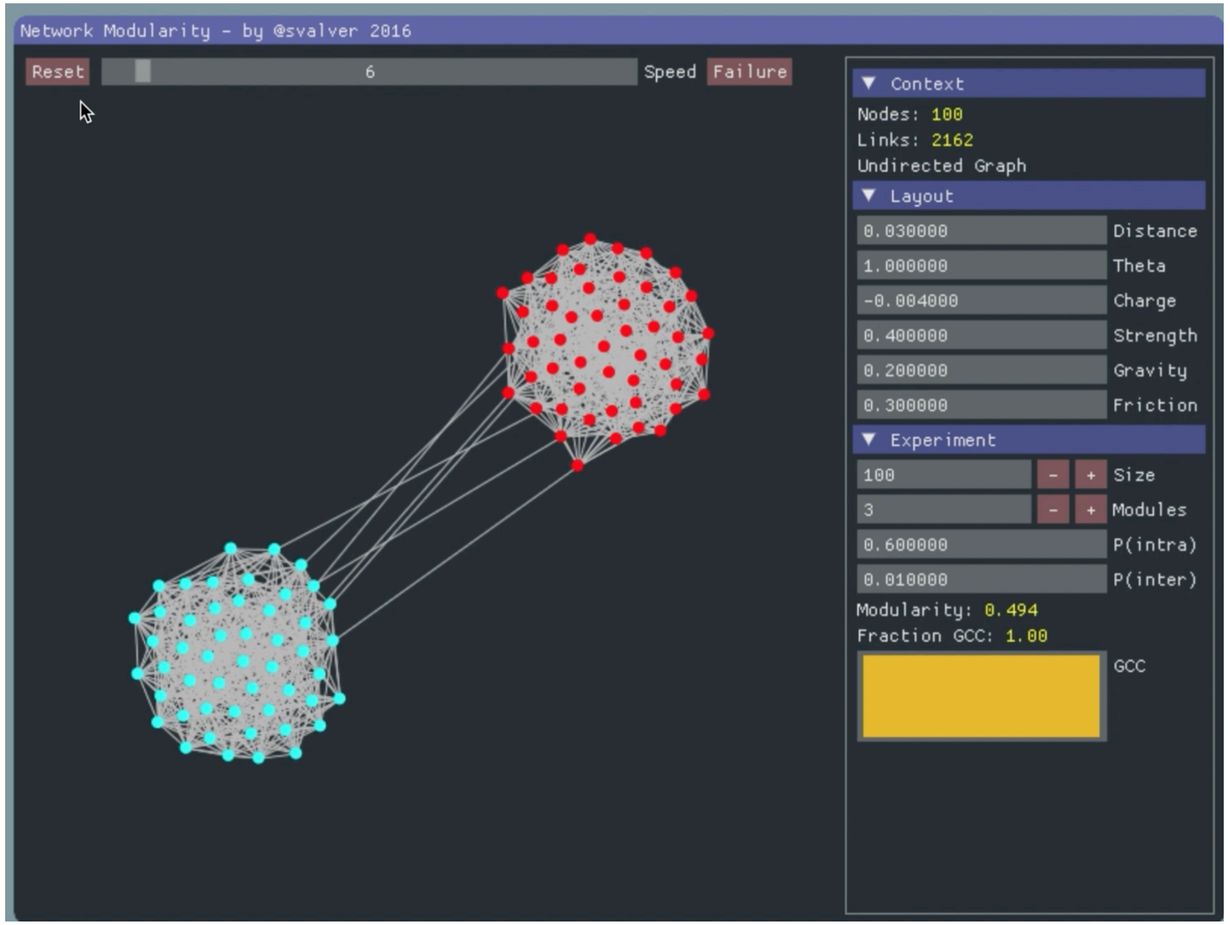
Random Modular Networks

$$RMG(p_{in}, p_{out})$$

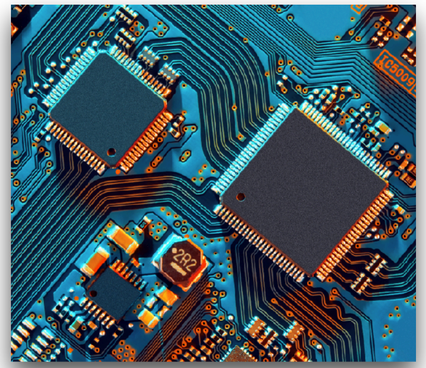


Activity: Random Modular Networks

https://tinyurl.com/4a7syzuk



13. Can you use this model to generate a random graph? How?

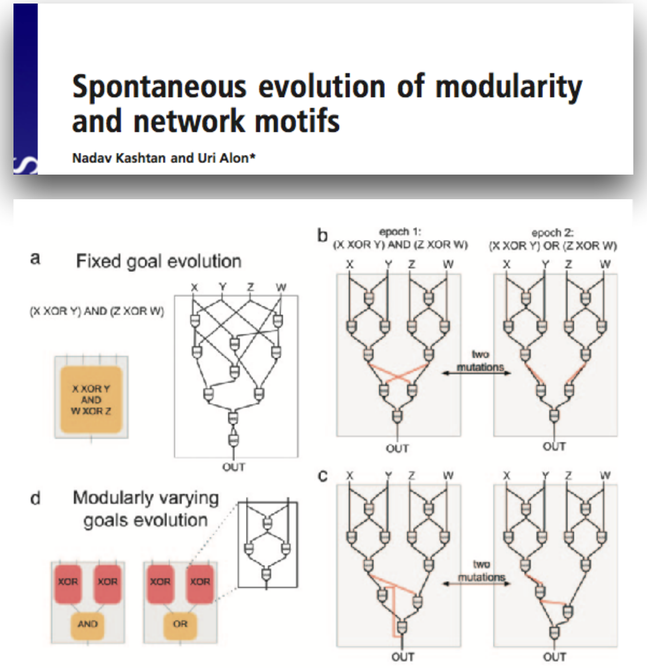


14. Which network has more links, RMG(a,b) *or* RMG(b,a)*? Which one is more modular? Why?*

Evolution of modularity

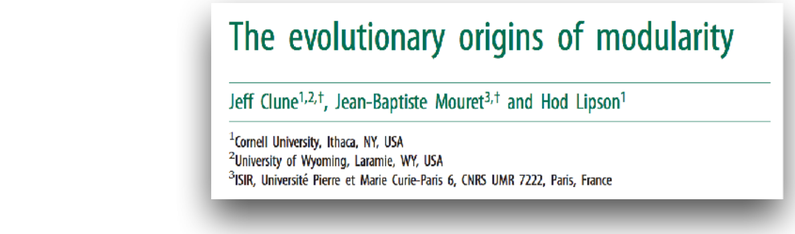
Understanding the contributions of multiples forces in the evolutionary origins of modularity

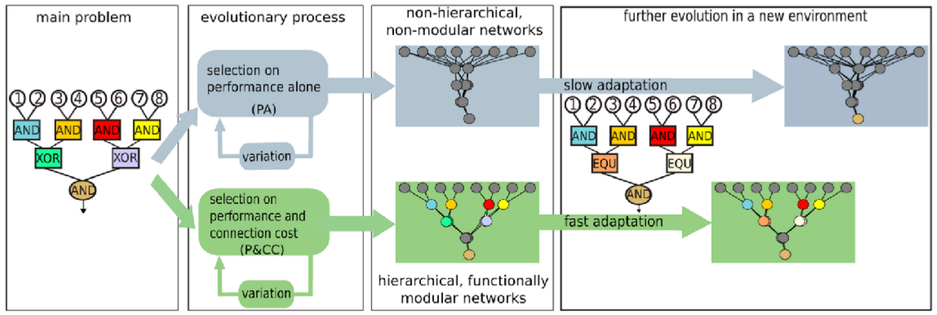
I) Selection



It has been suggested that networks evolved under “modularly varying goals” must be modular. However, it is unclear how many biological environments change in a modular way and if they change frequently enough.

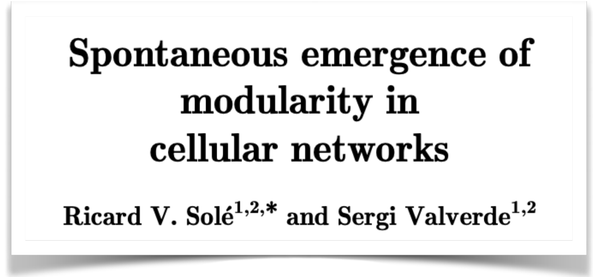
II) Connection costs

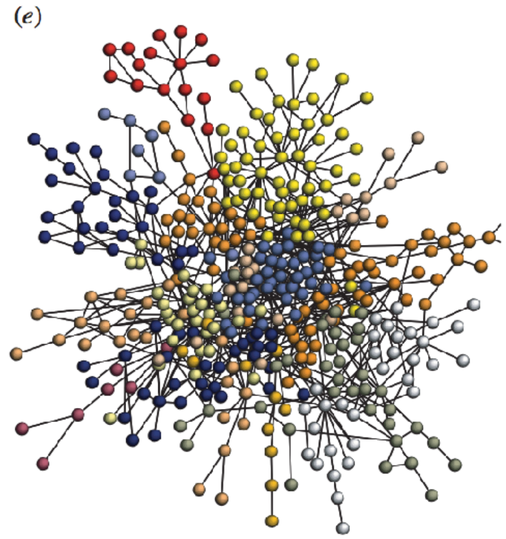




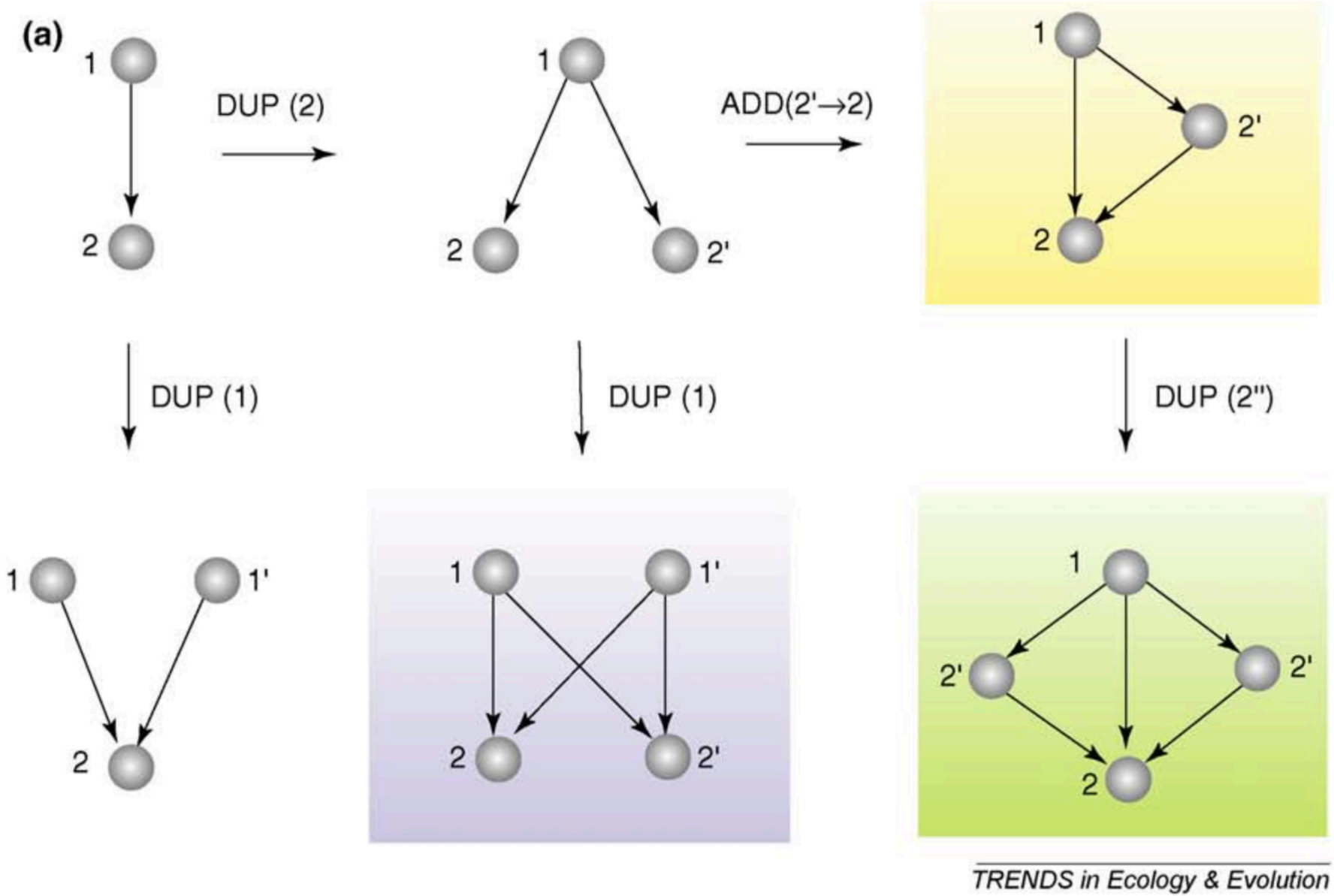
Most hypotheses of the emergence of modularity assume indirect selection for evolvability, but a direct selection pressure to reduce the cost of links causes the emergence of modular networks.

III) Duplication

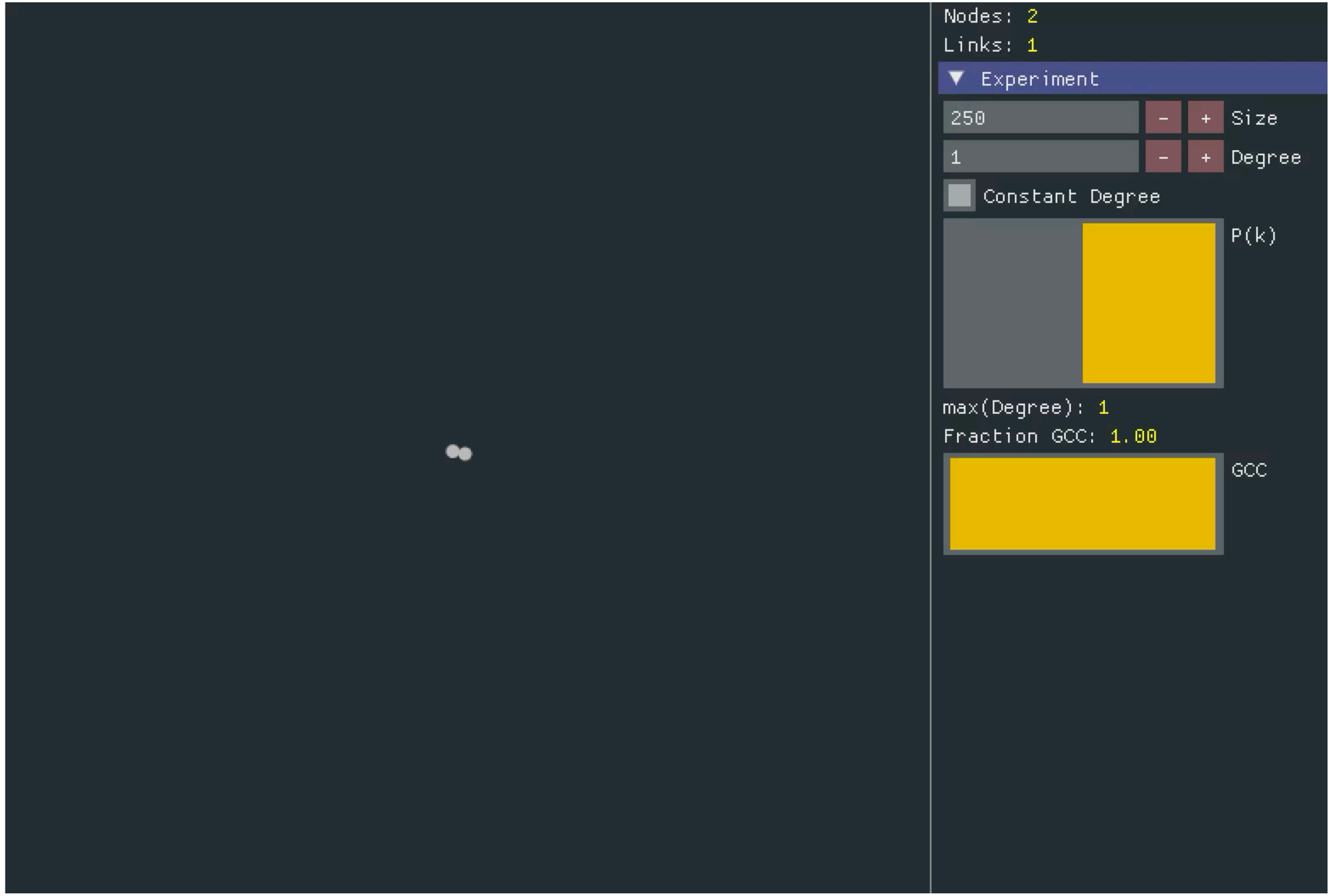




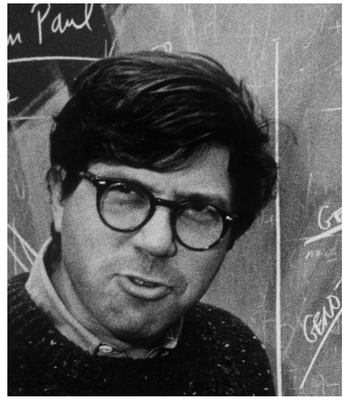
Structural diversity from duplication



Tinkered evolution of networks



Stephen Jay Gould



Richard Lewontin

Evolving complexity: how tinkering shapes cells, software and ecological networks

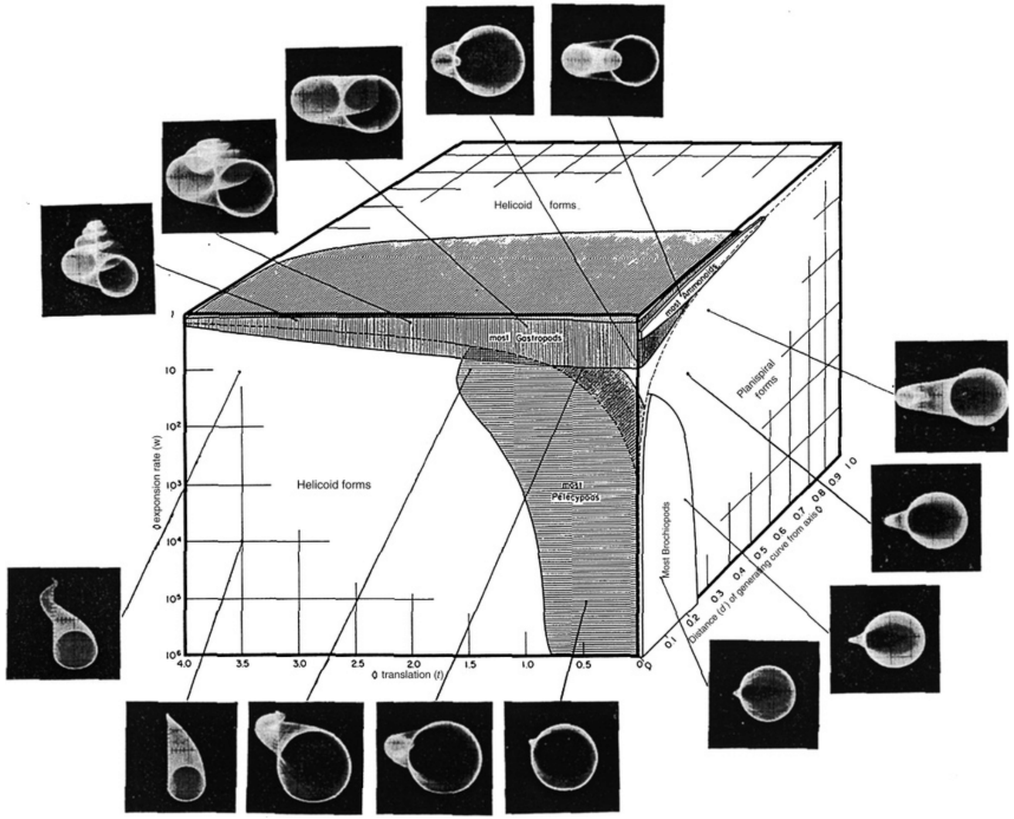
Ricard Solé[1,2,3,4] and Sergi Valverde[4,5]

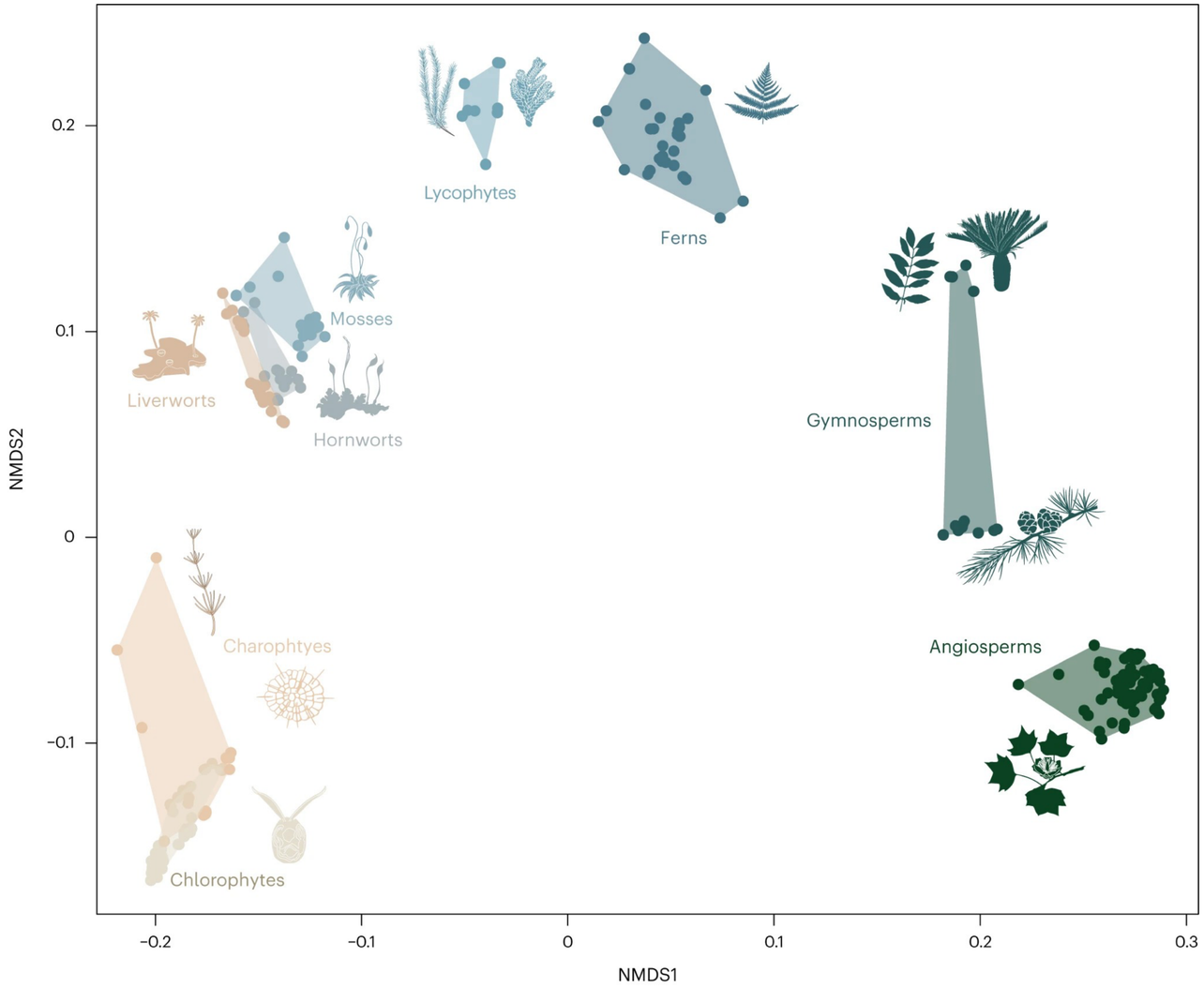
Valverde and Solé, **Physical Review E** (2005)

Solé and Valverde, **Trends Eco Evol** (2006)

Morphospaces

A coordinate system for network properties





Indeterminism & degeneracy

Uncertain future

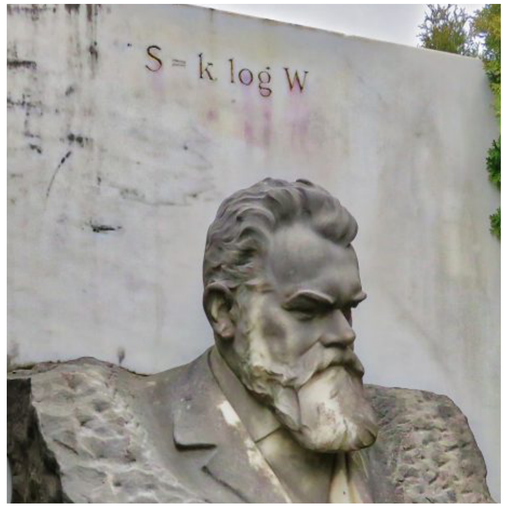
Uncertain past

Entropy

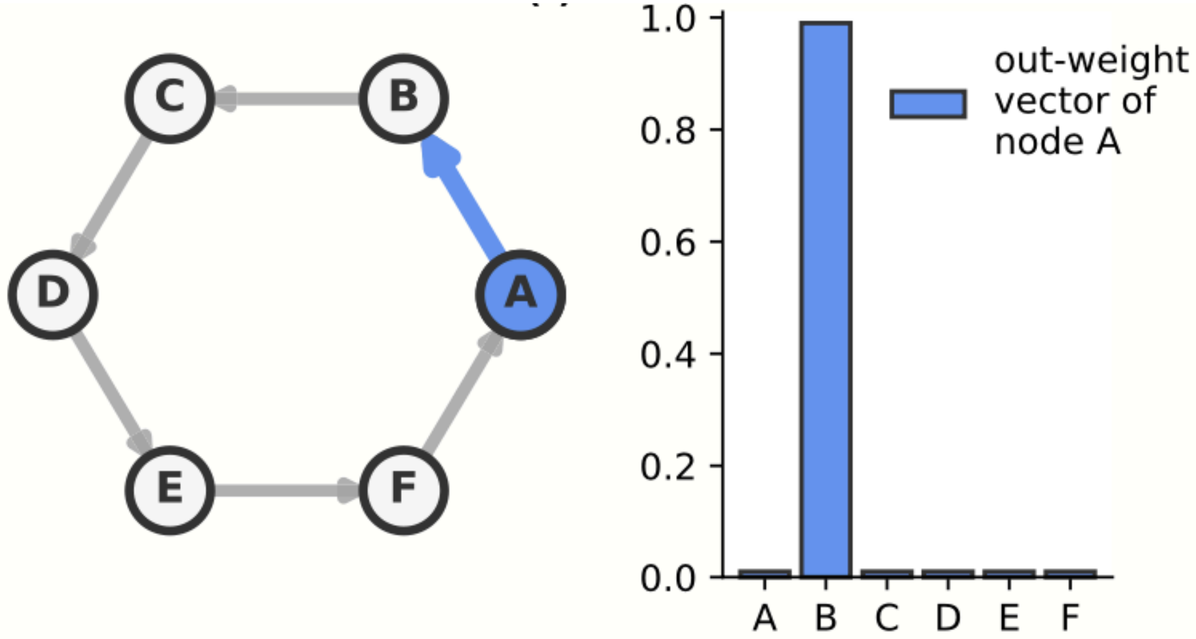
$$H(p) = -\sum_j p_j \log_2 p_j$$

0 bits | **1 bit** | **2 bits**

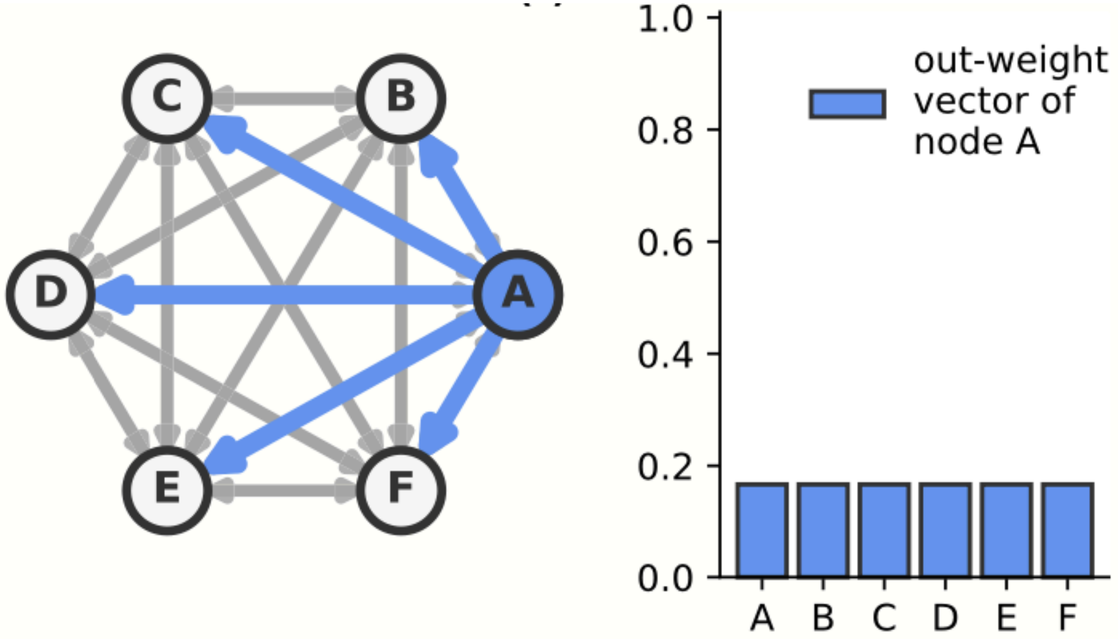
(1,0) | (1/2,1/2) | (1/4,1/4,1/4,1/4)



$$\text{Determinism} = log_2(N) - \left\langle H\left(W_i^{out}\right)\right\rangle$$



higher determinism

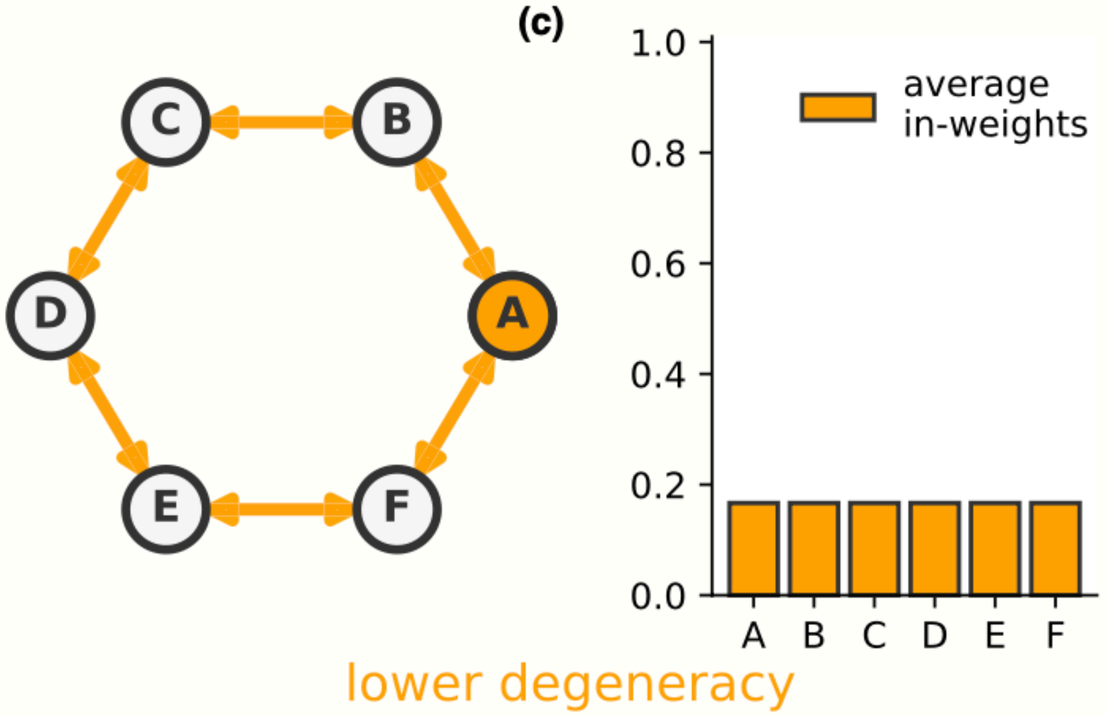


lower determinism

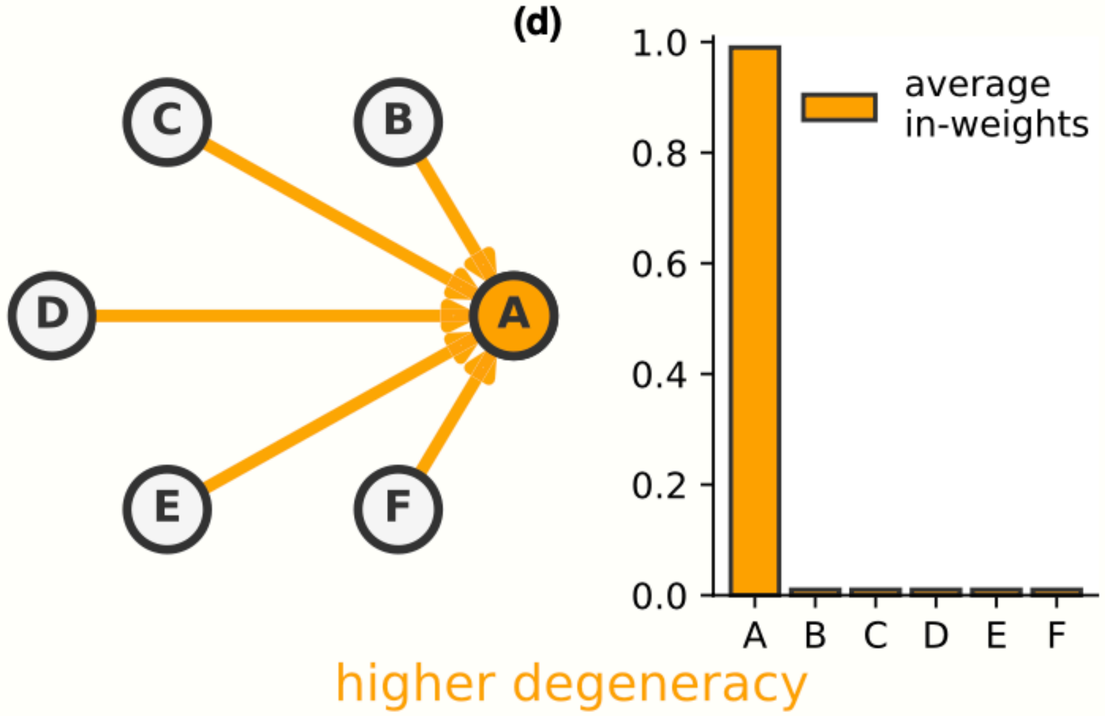
Hoel, E., Albantakis, L., & Tononi, G. (2013)

Degeneracy

$$\text{Degeneracy} = log_2(N) - H\left(\left\langle W_i^{in} \right\rangle\right)$$



lower degeneracy



higher degeneracy

Hoel, E., Albantakis, L., & Tononi, G. (2013)

Effective Information

$$EI = Degeneracy - Determinism$$

$$EI = log_2(N) - H\left(\left\langle W_i^{in} \right\rangle\right) - log_2(N) + \left\langle H\left(W_i^{out}\right)\right\rangle$$

Hoel, E., Albantakis, L., & Tononi, G. (2013)

Effective Information

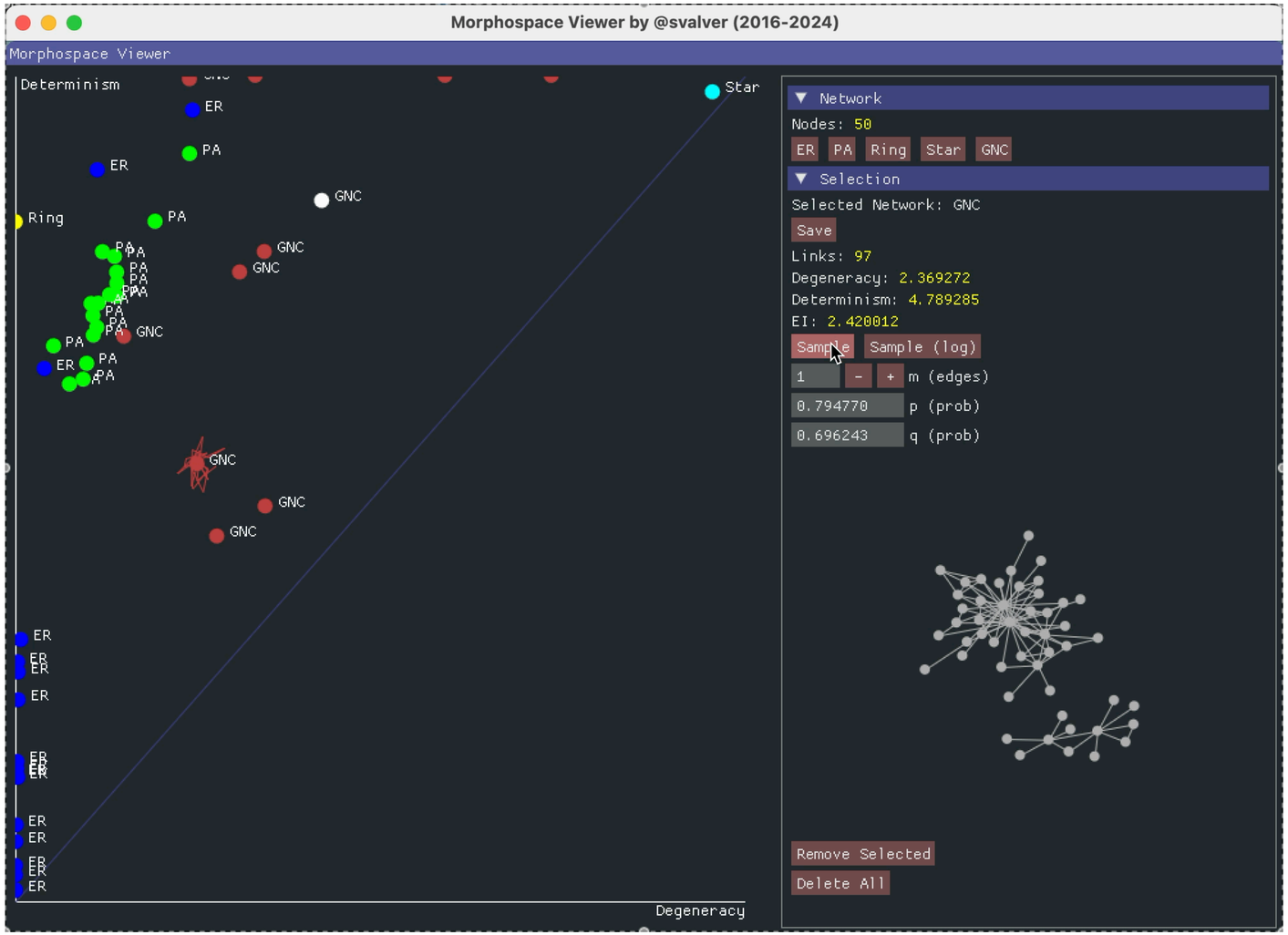
$$EI = Degeneracy - Determinism$$

$$EI = -H\left(\left\langle W_i^{in} \right\rangle\right) + \left\langle H\left(W_i^{out}\right) \right\rangle$$

Hoel, E., Albantakis, L., & Tononi, G. (2013)

Interactive Morphospace Exploration

https://tinyurl.com/5cvjz42b



15) Explore how different networks are positioned within this morphospace. Rank them according to filled morphospace.

16) Can you adjust model parameters to cross the diagonal? Why / Why not?

Summary

Networks are the language of complexity.

Many real networks are sparse.

Tradeoffs between robustness & efficiency.

Structure evidences multiple evolutionary mechanisms.

Complexity emerges from simplicity.

“The future cannot be predicted, but futures can be invented”

–Dennis Gabor (Hungarian physicist)